Analytic Number Theory

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Introduction

Analytic number theory is a branch of mathematics that uses analytical techniques (mostly from complex analysis) to address number-theoretical problems. There are many different areas of analytic number theory, such as prime number theory, Diophantine equations, modular functions, Dirichlet L-series, their analytic continuations and functional equations and the geometry of numbers.

Often analytic number theory is divided into *multiplicative number theory* and *additive number theory*. The first one deals with the distribution of prime numbers and with factorization and divisors. The focus is usually on developing approximate formulas for counting these objects in various contexts. The prime number theorem is a key result in this subject. Also, Dirichlet's theorem on primes in arithmetic progressions belongs to it.

Additive number theory is concerned with the additive structure of the integers, such as Goldbach's conjecture that every even number greater than 2 is the sum of two primes. One of the main results in additive number theory is the solution to Waring's problem.

Here is a list of some mathematicians having contributed to analytic number theory:

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Viggo Brun (1885-1978),
Pafnuty Lvovich Chebyshev (1821-1894),
Johann Peter Gustav Lejeune Dirichlet (1805-1859),
Paul Erdos (1913-1996),
Leonhard Euler (1707-1783),
Pierre de Fermat (1601-1665),
Carl Friedrich Gauss (1777-1855),
Jacques Salomon Hadamard (1865-1963),
Godfrey Harold Hardy (1877-1947),
Carl Gustav Jacob Jacobi (1804-1851),
Edmund Georg Hermann Landau (1877-1938),
Charles Jean Gustave Nicolas Baron de la Vallee Poussin (1866-1962),
Adrien-Marie Legendre (1752-1833),
Hans von Mangoldt (1854-1925),
Franz Mertens (1840-1927),
Srinivasa Aiyangar Ramanujan (1887-1920)
Georg Friedrich Bernhard Riemann (1826-1866),
Atle Selberg (1917-2007),
Carl Ludwig Siegel (1896-1981)
Ivan Matveevich Vinogradov (1891-1983).
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Let us give a few examples of open problems about prime numbers.

Conjecture 0.0.1. *Let* $H_n = \sum_{j=1}^{n} \frac{1}{j}$ *be the* n*-th harmonic number and* $\sigma(n) = \sum_{d|n} d$ *the sum of divisors function. Then*

$$\sigma(n) < H_n + \exp(H_n)\log(H_n)$$

for all $n \geq 2$*.*

Here the relationship to prime numbers is given by

$$\sigma(n) = \prod_{p^a \| n} (1 + p + p^2 + \cdots + p^a).$$

It might be surprising that this conjecture is equivalent to the famous *Riemann hypothesis*, about the non-trivial zeros of the Riemann zeta function $\zeta(s) = \sum_{n\geq 1} n^{-s}$, which converges for $\Re(s) > 1$, and has an analytic continuation to the complex plane with one singularity, a simple pole with residue 1 at $s = 1$.

Conjecture 0.0.2 (Riemann Hypothesis 1859). *The Riemann zeta function has its zeros only at the negative even integers and at complex numbers* s *with* $\Re(s) = \frac{1}{2}$*.*

The *non-trivial zeros* are those which lie in the critical strip with $0 < \Re(s) < 1$. Many consider it to be the most important unsolved problem in pure mathematics. It is of great interest in number theory in general, and in particular in prime number theory because it implies results about the distribution of prime numbers. We have the Euler product

$$\zeta(s) = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-s}}$$

for $\Re(s) > 1$.

Conjecture 0.0.3 (Strong Goldbach 1742). *Every even integer* $n \geq 4$ *is the sum of two primes.*

This has been verified for all even numbers $\leq 4 \cdot 10^{18}$ by computer. Often there are many possible decompositions of an even integer into a sum of two primes, e.g.,

$$\begin{aligned} 100 &= 3 + 97 \\ &= 11 + 89 \\ &= 17 + 83 \\ &= 29 + 71 \\ &= 41 + 59 \\ &= 47 + 53 \end{aligned}$$

In fact, the number $N_2(n)$ of Goldbach decompositions of an even integer n is conjectured to satisfy the asymptotics

$$N_2(n) \sim \left(2 \cdot C_2 \cdot \prod_{\substack{p|n \\ p>2}} \frac{p-1}{p-2} \right) \cdot \frac{n}{\log^2(n)},$$

where

$$C_2 = \prod_{p>2} \frac{p(p-2)}{(p-1)^2} \sim 0.66016181584686957$$

is the *twin prime constant.*

CONJECTURE 0.0.4 (Weak Goldbach 1742). *Every odd integer $n \geq 7$ is the sum of three primes.*

Goldbach's strong conjecture implies the weak conjecture, for if every even number greater than 4 is the sum of two odd primes, adding 3 to each even number greater than 4 will produce the odd numbers greater than 7, and 7 itself is equal to $2 + 2 + 3$. In 2013, *Harald Helfgott* released a **proof** of Goldbach's weak conjecture. The proof was accepted for publication in the Annals of Mathematics Studies series in 2015, and has been undergoing further review and revision since; fully-refereed chapters in close to final form are being made public in the process. Helfgott improved major and minor arc estimates sufficiently to unconditionally prove the weak Goldbach conjecture (it was known to follow from the generalized Riemann hypothesis).

CONJECTURE 0.0.5 (Twin prime conjecture 1849). *There are infinitely many twin primes, i.e., pairs of primes (p, q) with $q = p + 2$.*

This has been first conjectured by Alphonse de Polignac. The first pairs are

$$\begin{aligned}(p,q) = {} & (3,5),(5,7),(11,13),(17,19),(19,31),\\ & (41,43),(59,61),(71,73),(101,103),(107,109),\\ & (137,139),(149,151),(179,181),(191,193),(197,199).\end{aligned}$$

Denote by

$$\pi_2(X) = \sum_{\substack{p \leq x \\ p,p+2 \in \mathbb{P}}} 1$$

the number of twin prime pairs. Then Hardy and Littlewood conjectured that

$$\pi_2(x) \sim 2 \cdot C_2 \cdot \frac{x}{\log^2(x)}.$$

Yitang Zhang proved in 2013 that there exists an integer N that is less than 70 million, where there are infinitely many pairs of primes that differ by N. Terence Tao and others reduced the bound to only 246. The project had the name *small gaps between primes.*

CONJECTURE 0.0.6 (Legendre 1808). *There is always a prime number between n^2 and $(n+1)^2$ for $n \geq 1$.*

The conjecture has been verified up to $n = 10^{10}$.

CONJECTURE 0.0.7 (Landau 1912). *There are infinitely many prime numbers of the form $n^2 + 1$.*

The conjecture has been verified up to $n = 10^{20}$. Hardy and Littlewood conjectured, that the number of primes $q < x$ of the form $q = n^2 + 1$ is given by

$$\pi_q(x) \sim C_q \cdot \frac{\sqrt{x}}{\log(x)},$$

where

$$C_q = \prod_{p \geq 3} \left(1 - \frac{(-1)^{\frac{p-1}{2}}}{p-1}\right) \sim 1.3728134628182460091 1.$$

There is a more general version of this conjecture by *Bunyakovsky*, for polynomials $f(x)$ in $\mathbb{Z}[x]$ satisfying certain conditions, that $f(n)$ is a prime infinitely often. Here $f(x) = x^2 + 1$ gives Landau's conjecture, and $f(x) = ax + b$ with $\gcd(a, b) = 1$ gives Dirichlet's theorem on primes in arithmetic progressions. This we will prove later.

For integers $a \geq 2$ and $n \geq 2$ it is easy to prove that if $a^n - 1$ is prime, then $a = 2$ and $n = p$ is prime. So we are left with primes of the form $M_p = 2^p - 1$ for a given prime p. These primes are called *Mersenne primes.* The first ones are $2^2 - 1 = 3$, $2^3 - 1 = 7$, $2^5 - 1 = 31$, and $2^7 - 1 = 127$. However, $2^{11} - 1 = 23 \cdot 89$ is not prime.

Conjecture 0.0.8 (Mersenne 1644). *There are infinitely many Mersenne primes.*

Only 52 Mersenne primes have been found until today. The numbers grow very quickly. The largest one, found in 2024, is M_p for $p = 136279841$.

A prime number p is a Sophie Germain prime if $2p + 1$ is also prime.

Conjecture 0.0.9. *There are infinitely many Sophie Germain primes.*

The first ones are given by

$$2, 3, 5, 11, 23, 29, 41, 53, 83, 89, 113, 131, 173, 179, 191.$$

There are many more interesting conjectures concerning prime numbers, and also some strange conjecture. For example, call a prime number p *palindromic* if it remains the same when its digits are reversed. The first palindromic primes are given as follows

$2, 3, 5, 7, 11, 101, 131, 151, 181, 191, 313, 353, 373, 383, 727, 757, 787, 797,$
$919, 929, 10301, 10501, 10601, 11311, 11411, \ldots$

The sum $\sum_p \frac{1}{p}$ over all palindromic primes converges, and is ~ 1.32398.

Conjecture 0.0.10. *There are infinitely many palindromic primes.*

Here are two pyramids of palindromic primes, created by G. L. Honaker and M. Kenn:

```
                    2
                  30203
                133020331
              1713302033171
            12171330203317121
          151217133020331712151
        1815121713302033171215181
      16181512171330203317121518161
    331618151217133020331712151816133
  9333161815121713302033171215181613339
11933316181512171330203317121518161333911
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5
97579
389757983
3138975798313
15313897579831351
741531389757983135147
9074153138975798313514709
73907415313897579831351470937
907390741531389757983135147093709
3690739074153138975798313514709370963
38369073907415313897579831351470937096383
393836907390741531389757983135147093709638393
7039383690739074153138975798313514709370963839307
71703938369073907415313897579831351470937096383930717
347170393836907390741531389757983135147093709638393071743
9534717039383690739074153138975798313514709370963839307174359
93953471703938369073907415313897579831351470937096383930717435939
799395347170393836907390741531389757983135147093709638393071743593997
3679939534717039383690739074153138975798313514709370963839307174359399763
14367993953471703938369073907415313897579831351470937096383930717435939976341
761436799395347170393836907390741531389757983135147093709638393071743593997634167
1776143679939534717039383690739074153138975798313514709370963839307174359399763416771
70177614367993953471703938369073907415313897579831351470937096383930717435939976341677107

CHAPTER 1

Elementary Prime number theory

In this chapter we will discuss results on prime numbers and their distribution, which can be proved by elementary means not using complex analysis.

1.1. Euclid and the prime numbers

Let us start with some basic definitions and notations.

DEFINITION 1.1.1. Let R be an integral domain, i.e., a commutative ring with 1 without zero divisors. An element $p \in R$ is called *prime*, if it is not a unit and if $p \mid ab$ implies $p \mid a$ or $p \mid b$. It is called *irreducible*, if it is nonzero and not a unit, and if $p = ab$ implies that either a or b is a unit.

Every prime element is irreducible. The converse need not be true. However, over a PID also every irreducible element is prime. Since $R = \mathbb{Z}$ is a PID, we obtain the usual definition of a prime number in $\mathbb{Z}$, namely that it is irreducible, and every decomposition $p = ab$ implies that a or b is ± 1. It is customary to restrict a prime in $\mathbb{Z}$ to be a positive integer. The set of (positive) primes in $\mathbb{Z}$ is denoted by $\mathbb{P}$. An integer n, which is not prime is called *composed.*

Gauß proved the Fundamental theorem of arithmetic:

THEOREM 1.1.2 (Fundamental theorem of arithmetic). *For every positive integer n there exist unique non-negative integers $e(p)$ such that*

$$n = \prod_{p \in \mathbb{P}} p^{e(p)}$$

In particular only finitely many of the integers $e(p)$ are nonzero.

We consider here $n = 1$ as the empty product. The existence of such a representation is proved by induction. The uniqueness is based on the property that $p \mid ab$ implies $p \mid a$ or $p \mid b$ for primes p.

REMARK 1.1.3. The first rigorous proof of the fundamental theorem was given only in 1801 by Gauß , in his *Disquisitiones arithmeticae.* The result says that $\mathbb{Z}$ is a UFD. It is well known that many other rings arising in number theory are not UFD's. The most accessible examples are the rings of integers $\mathcal{O}_d$ of a quadratic number field $\mathbb{Q}[\sqrt{d}]$ for a squarefree $d \in \mathbb{Z}$. We have $\mathcal{O}_d = \mathbb{Z}[\alpha]$, with $\alpha = \sqrt{d}$ for $d \equiv 2, 3 \mod 4$ and $\alpha = \frac{1+\sqrt{d}}{2}$ for $d \equiv 1 \mod 4$.

Rings of integers are UFD's if and only if they are PID's, i.e., if they have class number 1. The standard example of such a ring without a unique prime factor decomposition is given by $R = \mathbb{Z}[\sqrt{-5}]$. For example, the elements $2, 3, 1 \pm \sqrt{-5}$ are all prime, but

$$6 = 2 \cdot 3 = (1 + \sqrt{-5}) \cdot (1 - \sqrt{-5})$$

represents two different decompositions of $n = 6$ in R. And indeed, this ring has class number two, and hence is not a PID.

For imaginary quadratic fields $\mathcal{O}_d$ with squarefree $d < 0$ there is a classification of those which are PID's. This is the case if and only if

$$d = -1, -2, -3, -7, -11, -19, -43, -67, -163.$$

For the other d the ring is not even a UFD. Besides the standard example $d = -5$ one might consider $d = -26$, i.e., $R = \mathbb{Z}[\sqrt{-26}]$. Here the elements $3, 1 \pm \sqrt{-26}$ are prime and non-associated, and

$$27 = 3 \cdot 3 \cdot 3 = (1 - \sqrt{-26}) \cdot (1 + \sqrt{-26})$$

gives two different factorizations of 27 into primes.

For real-quadratic number fields, with $d > 0$, there is no such classification. Gauß conjectured that there are *infinitely many* factorial rings $\mathcal{O}_d$, with class number 1. Consider the list of $1 \leq d \leq 70$, for which the rings $\mathcal{O}_d$ have class number > 1, in fact class number 2:

$$d = 10, 15, 26, 30, 34, 35, 39, 42, 51, 55, 58, 65, 66, 70.$$

Let us return to the study of prime numbers.

Definition 1.1.4. Let $x \geq 0$ be a real number. Denote by $\pi(x)$ the number of primes p with $p \leq x$, i.e.,

$$\pi(x) = \sum_{p \leq x} 1.$$

Proposition 1.1.5 (Euclid). *There are infinitely many prime numbers, i.e. we have $\pi(x) \to \infty$ for $x \to \infty$.*

Proof. Assume that there are only finitely many primes, listed as $p_1, p_2, \ldots, p_r$. Then consider the integer

$$N := \prod_{i=1}^{r} p_i + 1.$$

By the fundamental theorem of arithmetic, Proposition 1.1.2, N is a finite product of prime powers. Hence N is divisible by one of the primes p_k from our list. Since p_k obviously divides $p_1 \cdots p_r$, it also must divide the difference, so that $p_k \mid 1$. This is a contradiction. $\square$

Remark 1.1.6. It is possible to shorten this beautiful proof by Euclid by considering the number

$$N = n! + 1.$$

Indeed, this number is not divisible by any d with $2 \leq d \leq n$. Hence it has only prime factors $p > n$. Thus we always find a prime, which is bigger than all primes from any finite list of given primes.

We will give more proofs for Euclid's result. Some of them show more than the infinity of primes, i.e., they show that the series

$$\sum_{p \in \mathbb{P}} \frac{1}{p}$$

diverges.

II. The proof by Goldbach, 1730*:* denote by $F_n = 2^{2^n} + 1$ the n-th Fermat number. We will show that each two different Fermat numbers are coprime, i.e., that $\gcd(F_m, F_n) = 1$ for all $n \neq m$. For this we use the well known formula

$$F_0 F_1 \cdots F_{n-1} = F_n - 2,$$

which can be easily proven by induction. Let $k \in \mathbb{N}$ be with $k \mid F_n$ and $k \mid F_m$. Then the formula implies $k \mid 2$, and hence $k = 1$ or $k = 2$. However, since all F_n are odd, it follows that $k = 1$. Hence all prime divisors of F_n are distinct. Since we have infinitely many distinct Fermat numbers, we also have infinitely many prime numbers.

Note that the proof also implies that

$$p_n \leq 2^{2^{n-1}}$$

for the n-th prime p_n. Of course, we also can prove this by induction on n. Clearly $p_1 = 2$ and $2 \leq 2^{2^1} = 4$. Assuming $p_i \leq 2^{2^i}$ for all $i \leq n$, we conclude, because $p_1 \cdot p_2 \cdots p_n + 1$ is not divisible by any p_i, that

$$\begin{aligned} p_{n+1} &\leq \left(\prod_{i=1}^n p_i \right) + 1 \\ &\leq \left(\prod_{i=1}^n 2^{2^i} \right) + 1 \\ &= 2^{\sum_{i=1}^n 2^i} + 1 = 2^{2^{n+1}-2} + 1 \leq 2^{2^{n+1}}. \end{aligned}$$

For the next proof we need the following lemma.

Lemma 1.1.7. *Let $p_1, p_2, \ldots, p_r$ be different primes. Then we have*

$$\prod_{k=1}^r \left(1 - \frac{1}{p_k}\right)^{-1} = \sum_{n \in \mathcal{N}(p_1,\ldots,p_r)} \frac{1}{n}, \tag{1.1}$$

where $\mathcal{N}(p_1, \ldots, p_r)$ denotes the set of positive integers $n \in \mathbb{N}$ which satisfy

$$p \mid n \Rightarrow p = p_i \text{ for some } i \leq r.$$

Proof. Each factor in the product can be written as a geometric series

$$\frac{1}{1 - \frac{1}{p}} = 1 + \frac{1}{p} + \frac{1}{p^2} + \frac{1}{p^3} + \cdots$$

This implies

$$\begin{aligned} \prod_{k=1}^r \left(1 - \frac{1}{p_k}\right)^{-1} &= \prod_{k=1}^r \left(\sum_{\ell=0}^\infty \frac{1}{p_k^\ell} \right) \\ &= \sum_{\ell_1=0}^\infty \sum_{\ell_2=0}^\infty \cdots \sum_{\ell_r=0}^\infty \frac{1}{p_1^{\ell_1} \cdots p_r^{\ell_r}} \\ &= \sum_{n \in \mathcal{N}(p_1,\ldots,p_r)} \frac{1}{n} \end{aligned}$$

$\square$

III. The proof by Euler, 1737*:* We assume that there are only finitely many primes $p_1, \dots, p_r$. Then, by the fundamental theorem of arithmetic we have $\mathcal{N}(p_1, \dots, p_r) = \mathbb{N}$. Because of (1.1) we have

$$\prod_{k=1}^{r} \left(1 - \frac{1}{p_k}\right)^{-1} = \sum_{n \in \mathbb{N}} \frac{1}{n}.$$

The left side here is finite by assumption. Hence the harmonic series converges. This is clearly a contradiction.

We want to refine Euler's proof as follows.

DEFINITION 1.1.8. Euler's constant γ is defined by

$$\gamma = \lim_{n \to \infty} \left(\sum_{k=1}^{n} \frac{1}{k} - \log(n) \right)$$

It is not difficult to see that the limit exists, and that we have

$$\gamma \sim 0.57721566490153286.$$

LEMMA 1.1.9. *For $n \geq 1$ we have*

$$\log(n) + \gamma + \frac{1}{2n+1} \leq \sum_{k=1}^{n} \frac{1}{k} \leq \log(n) + \gamma + \frac{1}{2n-1}.$$

PROOF. Consider the sequence $a_n = H_n - \log(n)$. By Jensen's Inequality we have

$$\begin{aligned} a_n - a_{n+1} &= \log\left(1 + \frac{1}{n}\right) - \frac{1}{n+1} \\ &= \int_n^{n+1} \frac{1}{x} \, \mathrm{d}x - \frac{1}{n+1} \\ &\geq \left(\int_n^{n+1} x \, \mathrm{d}x \right)^{-1} - \frac{1}{n+1} \\ &= \frac{1}{n + \frac{1}{2}} - \frac{1}{n+1} \\ &= \frac{1}{(2n+1)(n+1)} \\ &\geq \frac{1}{(2n+1)\left(n + \frac{3}{2}\right)} \\ &= \frac{1}{2n+1} - \frac{1}{2n+3}. \end{aligned}$$

Thus, a_n is a decreasing sequence and the limit is γ, namely Euler's Constant. Summing this yields

$$H_n - \log(n) \geq \gamma + \frac{1}{2n+1}.$$

We obtain the upper bound on $H_n - \log(n)$ as follows.

$$\begin{aligned} a_n - a_{n+1} &= \log\left(1 + \frac{1}{n}\right) - \frac{1}{n+1} \\ &= \int_0^{1/n} \frac{x}{(1+x)^2}\,\mathrm{d}x \\ &\leq \int_0^{1/n} x\,\mathrm{d}x \\ &= \frac{1}{2n^2} \\ &\leq \frac{1}{2n-1} - \frac{1}{2n+1}. \end{aligned}$$

Summing this yields

$$H_n - \log(n) \leq \gamma + \frac{1}{2n-1}.$$

□

As functions we often just write

$$\sum_{n \leq x} \frac{1}{n} = \log(x) + \gamma + O\left(\frac{1}{x}\right) = \log(x) + O(1)$$

using Landau's O-notation.

Definition 1.1.10. Let $A \subseteq \mathbb{R}$ and $f\colon A \to \mathbb{R}$, $g\colon A \to \mathbb{R}_+$ be two functions. Then we write $f(x) = O(g(x))$, if there exists a constant $c > 0$ such that $|f(x)| \leq c \cdot g(x)$ for all $x \in A$. We write $f(x) = o(g(x))$, if $\lim_{x \to \infty} \frac{f(x)}{g(x)} = 0$.

One may extend this definition also to complex functions, as long as the image of g lies in $\mathbb{R}_+$. Note that the *Vinogradov notation* is $f(x) \ll g(x)$ for $f(x) = O(g(x))$.

Example 1.1.11. *The notation $f(x) = O(1)$ just says that f is bounded. For example, $\sin(x) = O(1)$.*

Another example is $e^{-x} = O(x^{-n})$ for all $n \geq 1$.

IV. The refined proof: Let $x \geq 2$ and $p_1, \ldots, p_r$ the primes p with $p \leq x$. By the fundamental theorem, $\mathcal{N}(p_1, \ldots, p_r)$ contains the set $\{1, 2, \ldots [x]\}$. By Lemma 1.1.9 we obtain, in the same way as (1.1),

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right)^{-1} \geq \sum_{n \leq x} \frac{1}{n} > \log(x). \tag{1.2}$$

Here we have $\log(x) \to \infty$ for $x \to \infty$. Hence the product will diverge for $x \to \infty$.

This refined proof gives more than Euler's proof:

Proposition 1.1.12. *For $x \geq 2$ we have*

$$\sum_{p\le x}\frac{1}{p} > \log(\log(x)) - 1.$$

In particular, the series $\sum_{p\in\mathbb{P}}\frac{1}{p}$ *diverges.*

For the proof we need another lemma.

LEMMA 1.1.13. *For all* $t\in[0,\frac{1}{2}]$ *we have* $\frac{1}{1-t}\le e^{t+t^2}$.

PROOF. Let $f(t) = (1-t)\exp(t+t^2)$. The claim is that $f(t)\ge 1$ for all $t\in[0,\frac{1}{2}]$. We have $f(0)=1$, and the function is monotoneously increasing on this interval, because of $f'(t) = t(1-2t)\exp(t+t^2)\ge 0$ for all $t\in[0,\frac{1}{2}]$. This implies the claim. $\square$

Proof of Proposition 1.1.12: Using (1.2) and the above lemma with $t=\frac{1}{p}\in(0,\frac{1}{2}]$ we obtain

$$\log(x) < \prod_{p\le x}\left(1-\frac{1}{p}\right)^{-1} < \prod_{p\le x}\exp\left(\frac{1}{p}+\frac{1}{p^2}\right).$$

Because the logarithm is monotoneous, this implies

$$\log(\log(x)) < \sum_{p\le x}\left(\frac{1}{p}+\frac{1}{p^2}\right) < \sum_{p\le x}\frac{1}{p} + \sum_{p\in\mathbb{P}}\frac{1}{p^2}.$$

The second series is less than one, because we have

$$\sum_{p\in\mathbb{P}}\frac{1}{p^2} < \sum_{n=2}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6} - 1 < 1.$$

Actually, the value of this series is approximately 0.45224742. $\square$

REMARK 1.1.14. One can also show the following for $x\ge 2$, see [**16**]:

$$\sum_{p\le x}\frac{1}{p} = \log(\log(x)) + c + O\left(\frac{1}{\log x}\right).$$

Here the constant $c>0$ is given by

$$c = \gamma - \sum_{p\in\mathbb{P}}\left(\log\left(\frac{1}{1-\frac{1}{p}}\right) - \frac{1}{p}\right) \sim 0.2614972128.$$

The following result by Euler is perhaps the birth of analytic number theory. It generalizes the fundamental theorem of arithmetic.

PROPOSITION 1.1.15 (Euler). *For* $\sigma > 1$ *we have*

$$\sum_{n=1}^{\infty}\frac{1}{n^\sigma} = \prod_{p\in\mathbb{P}}\left(1-\frac{1}{p^\sigma}\right)^{-1}.$$

PROOF. Let $\sigma>0$. Then we have, again in the same way as for (1.1),

$$\prod_{p\le x}\left(1-\frac{1}{p^\sigma}\right)^{-1} = \sum_{\substack{n\in\mathbb{N}\\ p\mid n\Rightarrow p\le x}}\frac{1}{n^\sigma}$$

For $\sigma > 1$ we obtain the claim with $x \to \infty$. $\square$

This implies again Euclid's result by considering the limit $\sigma \to 1$. The series in Euler's result is of great importance. Riemann had the idea to consider it also for complex variables.

Definition 1.1.16. For $s = \sigma + it$, $\sigma > 1$ we define the *Riemann zeta function* by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Proposition 1.1.17. *The series is absolutely convergent for all $s = \sigma + it$ with $\sigma > 1$. It is uniformly convergent in each half plane $\sigma \geq \sigma_0 > 1$. In particular, $\zeta(s)$ is holomorphic in the half plane $\Re(s) > 1$.*

Proof. We have

$$\begin{aligned} |n^s| = n^\sigma &= |e^{-s\log(n)}| \\ &= e^{-\Re(s\log(n))} = e^{-\sigma\log(n)} \\ &= n^{-s}. \end{aligned}$$

Hence the convergence is uniform in every closed half plane $\{\Re(s) \geq \sigma_0\}$. The absolute convergence follows form the following estimate.

$$\left|\sum_{n=1}^{\infty} \frac{1}{n^s}\right| \leq \sum_{n=1}^{\infty} \frac{1}{n^\sigma} \leq \sum_{n=1}^{\infty} \frac{1}{n^{\sigma_0}} < 1 + \int_1^\infty \frac{dt}{t^{\sigma_0}} = 1 + \frac{1}{\sigma_0 - 1}$$

By the Weierstrass Theorem, $\zeta(s)$ is holomorphic as a limit function of a uniformly convergent sequence of holomorphic functions. $\square$

Euler's result, Proposition 1.1.15, is also valid for all $s \in \mathbb{C}$ with $\Re(s) > 1$.

Proposition 1.1.18 (Euler product). *For $s \in \mathbb{C}$ with $\Re(s) > 1$ we have*

$$\zeta(s) = \prod_{p \in \mathbb{P}} \left(1 - \frac{1}{p^s}\right)^{-1}.$$

Actually, Euler's result for real numbers already implies that the sum of reciprocal primes diverges.

Another proof of Proposition 1.1.12: for $\sigma > 1$ we have

$$\begin{aligned} \log(\zeta(\sigma)) &= -\sum_p \log\left(1 - \frac{1}{p^\sigma}\right) = -\sum_p \sum_{m=1}^{\infty} -\frac{1}{mp^{m\sigma}} \\ &= \sum_p \frac{1}{p^\sigma} + \sum_p \sum_{m=2}^{\infty} \frac{1}{mp^{m\sigma}}. \end{aligned}$$

This will imply the result if we can show that the double sum is bounded. In fact, then we have

$$\log(\zeta(\sigma)) = \sum_p \frac{1}{p^\sigma} + O(1),$$

where the left side diverges for $\sigma \to 1$. It follows that $\sum_{p\in\mathbb{P}} \frac{1}{p} = \infty$. Because of $1 - \frac{1}{p^{2\sigma}} \geq \frac{1}{2}$ and $\sigma > 1$ we have

$$\begin{aligned}\sum_p \sum_{m=2}^{\infty} \frac{1}{mp^{m\sigma}} &< \sum_p \sum_{m=2}^{\infty} \frac{1}{p^{m\sigma}} \\ &= \sum_p \frac{1}{p^{2\sigma}} \cdot \frac{1}{1-\frac{1}{p^{2\sigma}}} \\ &< 2\sum_p \frac{1}{p^{2\sigma}} \\ &\leq 2\sum_p \frac{1}{p^2}.\end{aligned}$$

$\square$

1.2. Arithmetic functions

Arithmetic functions naturally arise in prime number theory. To give an example, consider the function $\omega\colon \mathbb{N} \to \mathbb{N}$, defined by

$$\omega(n) = \sum_{p \mid n} 1.$$

This function counts the positive prime divisors of n. Of course, $\omega(p) = 1$ for prime numbers p.

DEFINITION 1.2.1. An *arithmetic function* is a function $f\colon \mathbb{N} \to \mathbb{C}$. It is called *multiplicative*, if $f(1) = 1$ and if we have for all $n, m \in \mathbb{N}$ with $\gcd(n, m) = 1$ that

$$f(mn) = f(m)f(n).$$

It is called *strongly multiplicative*, if this holds for *all* $n, m \in \mathbb{N}$.

We require $f(1) = 1$ to exclude the zero function. If f is strongly multiplicative, then $f(p^\alpha) = f(p)^\alpha$ for all primes p.

EXAMPLE 1.2.2. *The arithmetic function ω is not multiplicative. The function*

$$\tau(n) = \sum_{d \mid n} 1$$

is multiplicative, but not strongly multiplicative. The functions $f_s(n) = n^s$, for $s \in \mathbb{C}$, are all strongly multiplicative.

Indeed, if ω were multiplicative, and p, q two different primes, we had

$$1 = \omega(p)\omega(q) = \omega(pq) = 2.$$

The function τ counts the positive divisors of n. We will show that

$$\tau(n) = \prod_{p^\alpha \| n} (\alpha + 1)$$

for all $n \geq 1$. Recall that $p^\alpha \| n$ means that $p^\alpha \mid n$, but $p^{\alpha+1} \nmid n$. This formula shows that τ is multiplicative.

DEFINITION 1.2.3. Let $k \geq 0$. The functions σ_k are defined by

$$\sigma_k(n) = \sum_{d \mid n} d^k.$$

The Möbius μ-function is defined by

$$\mu(n) = \begin{cases} (-1)^{\omega(n)} & \text{if } n \text{ is squarefree,} \\ 0 & \text{otherwise.} \end{cases}$$

Euler's totient function φ is defined by

$$\varphi(n) = \sum_{\substack{1 \leq k \leq n \\ (k,n)=1}} 1.$$

The von Mangold function Λ is defined by

$$\Lambda(n) = \begin{cases} \log(p) & \text{if } n = p^m \text{ for some } m \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

We also use the notations $\sigma_0(n) = \tau(n)$ and $\sigma_1(n) = \sigma(n)$.

PROPOSITION 1.2.4. *The functions τ and σ are multiplicative, but not strongly multiplicative. For $n = p_1^{\alpha_1} \cdots p_\ell^{\alpha_\ell}$ we have,*

$$\tau(n) = \prod_{i=1}^{\ell} (\alpha_i + 1),$$
$$\sigma(n) = \prod_{i=1}^{\ell} \frac{p_i^{\alpha_i+1} - 1}{p_i - 1}.$$

PROOF. We have $d \mid n$ if and only if $d = p_1^{\beta_1} \cdots p_\ell^{\beta_\ell}$ with $0 \leq \beta_i \leq \alpha_i$ for all $1 \leq i \leq \ell$. Hence the positive divisors d are in bijective correspondence to the n-tuples $(\beta_1, \dots, \beta_\ell)$ with $0 \leq \beta_i \leq \alpha_i$. Since there are exactly $\alpha_i + 1$ choices for the β_i, one has exactly $(\alpha_1 + 1) \cdots (\alpha_\ell + 1)$ such tuples.

For proving the second formula one writes

$$\sigma(n) = \sum_{d \mid n} d = \sum_{(\beta_1, \dots, \beta_\ell)} p_1^{\beta_1} \cdots p_\ell^{\beta_\ell},$$

where the sum extends over all tuples $(\beta_1, \dots, \beta_\ell)$ with $0 \leq \beta_i \leq \alpha_i$. This sum is evaluated as follows.

$$\left(\sum_{\beta_1=0}^{\alpha_1} p_1^{\beta_1} \right) \left(\sum_{\beta_2=0}^{\alpha_2} p_2^{\beta_2} \right) \cdots \left(\sum_{\beta_\ell=0}^{\alpha_\ell} p_\ell^{\beta_\ell} \right) = \frac{p_1^{\alpha_1+1} - 1}{p_1 - 1} \cdot \frac{p_2^{\alpha_2+1} - 1}{p_2 - 1} \cdots \frac{p_\ell^{\alpha_\ell+1} - 1}{p_\ell - 1}.$$

□

REMARK 1.2.5. The formula for $\sigma(n)$ generalizes for all functions $\sigma_k(n)$. We have

$$\sigma_k(n) = \prod_{i=1}^{\ell} \frac{p_i^{k(\alpha_i+1)} - 1}{p_i^k - 1}.$$

Hence all functions $\sigma_k(n)$ are multiplicative.

PROPOSITION 1.2.6. *the Möbius $\mu(n)$ function is multiplicative and we have*

$$\sum_{d \mid n} \mu(d) = \left\lfloor \frac{1}{n} \right\rfloor = \begin{cases} 1 & \textit{if } n = 1, \\ 0 & \textit{otherwise.} \end{cases}$$

PROOF. We have $\mu(1) = 1$ and $\mu(p_1 \cdots p_\ell) = (-1)^\ell$ for different prime numbers $p_1, \dots, p_\ell$. Furthermore, $\mu(p^\alpha) = 0$ for $\alpha \geq 2$. Let n, m be two coprime integers. If either n or m has a factor p^2, so does the product nm, and we have $\mu(mn) = \mu(m)\mu(n) = 0$. Otherwise we have, because of $\gcd(n, m) = 1$, that

$$\mu(mn) = (-1)^{\omega(m)+\omega(n)} = (-1)^{\omega(m)}(-1)^{\omega(m)} = \mu(m)\mu(n).$$

For the formula we may assume that $n \geq 2$. Let $n = p_1^{\alpha_1} \cdots p_\ell^{\alpha_\ell}$. Then the only nonzero terms in the sum $\sum_{d|n} \mu(d)$ arise by $d = 1$ or by products of r different primes from the set $\{p_1, \ldots, p_\ell\}$. There are $\binom{\ell}{r}$ many such terms, and hence

$$\begin{aligned}\sum_{d|n} \mu(d) &= \mu(1) + (\mu(p_1) + \cdots + \mu(p_\ell)) \\ &\quad + \mu(p_1p_2) + \mu(p_1p_3) + \cdots + \mu(p_{\ell-1}p_\ell) + \cdots + \mu(p_1p_2\cdots p_\ell) \\ &= 1 + \binom{\ell}{1}(-1)^1 + \binom{\ell}{2}(-1)^2 + \cdots + \binom{\ell}{\ell}(-1)^\ell \\ &= (1-1)^\ell = 0.\end{aligned}$$

□

Corollary 1.2.7. *We have*

$$\sum_{d|n} \mu^2(d) = 2^{\omega(n)}.$$

Proof. Replacing $\mu(d)$ by $\mu^2(d) = |\mu(d)|$ in the above proof, we obtain

$$\sum_{d|n} \mu^2(d) = \sum_{\ell=0}^{\omega(n)} \binom{\omega(n)}{\ell} = (1+1)^{\omega(n)} = 2^{\omega(n)}.$$

□

Let us consider now Euler's totient function φ. We have $|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n)$ and

$$(\mathbb{Z}/n\mathbb{Z})^\times \times (\mathbb{Z}/m\mathbb{Z})^\times \cong (\mathbb{Z}/nm\mathbb{Z})^\times,$$

by the Chinese remainder theorem, for all coprime m, n. This implies

$$\varphi(mn) = \varphi(n)\varphi(m)$$

for all m, n with $\gcd(m, n) = 1$, i.e., φ is multiplicative.

Proposition 1.2.8. *Euler's totient function $\varphi(n)$ is multiplicative and we have*

$$\sum_{d|n} \varphi(d) = n.$$

Proof. Let $S = \{1, 2, \ldots, n\}$. We distribute the numbers in S into disjoint sets as follows. For $d \mid n$ let

$$A(d) = \{k \in \mathbb{N} \mid 1 \leq k \leq n,\ \gcd(k, n) = d\}.$$

This set contains those elements of S, whose gcd with n is equal to d. Define $f(d) = |A(d)|$. Then we have

$$\bigcup_{d|n} A(d) = S, \text{ hence } \sum_{d|n} f(d) = n.$$

Now $\gcd(k, n) = d$ is equivalent to $\gcd(\frac{k}{d}, \frac{n}{d}) = 1$, where we have $0 < k \leq n$ if and only if $0 < \frac{k}{d} \leq \frac{n}{d}$. With $q = \frac{k}{d}$ we obtain a bijective correspondence between the elements of $A(d)$

and the elements in $\{q \in \mathbb{N} \mid 0 < q \le \frac{n}{d}, (q, \frac{n}{d}) = 1\}$. The number of such q's is given by $\varphi(\frac{n}{d})$. This implies $f(d) = \varphi(\frac{n}{d})$ and hence

$$\sum_{d|n} \varphi\left(\frac{n}{d}\right) = n.$$

This is equivalent to the claim of the proposition, because with d also $\frac{n}{d}$ runs through all divisors of n. $\square$

Note that $\varphi(p) = p - 1$ for all primes, and that $\varphi(n) \le n - \sqrt{n}$ for composite n. On the other hand, $\varphi(n) \ge \sqrt{n}$ for all $n \ge 7$. A much better lower bound, for all $n \ge 3$, is given by

$$\varphi(n) > \frac{n}{e^\gamma \log(\log(n)) + \frac{3}{\log(\log(n))}}.$$

The functions $\varphi(n)$ and $\mu(n)$ are connected as follows:

Proposition 1.2.9. *For $n \ge 1$ we have*

$$\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d}.$$

Proof. By Proposition 1.2.6 we have

$$\varphi(n) = \sum_{k=1}^{n} \left\lfloor \frac{1}{(n,k)} \right\rfloor = \sum_{k=1}^{n} \sum_{d|(n,k)} \mu(d) = \sum_{k=1}^{n} \sum_{\substack{d|n \\ d|k}} \mu(d)$$
$$= \sum_{d|n} \sum_{q=1}^{n/d} \mu(d) = \sum_{d|n} \mu(d) \left(\sum_{q=1}^{n/d} 1 \right) = \sum_{d|n} \mu(d) \frac{n}{d}.$$

Here we have written the summation index k as $k = qd$. For fixed $d \mid n$ we have to sum over such $1 \le k \le n$, which are multiples of d. Here we have $1 \le k \le n$ if and only if $1 \le q \le \frac{n}{d}$. $\square$

Proposition 1.2.10 (Product formula). *For $n \ge 2$ we have*

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

Proof. Denote by $p_1, \ldots p_r$ the distinct prime divisors of n. Then we have

$$\begin{aligned} \prod_{p|n} \left(1 - \frac{1}{p}\right) &= \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right) \\ &= 1 - \sum \frac{1}{p_i} + \sum \frac{1}{p_i p_j} - \sum \frac{1}{p_i p_j p_k} + \cdots + \sum \frac{(-1)^r}{p_1 p_2 \cdots p_r} \\ &= \sum_{d|n} \frac{\mu(d)}{d} \\ &= \frac{\varphi(n)}{n}. \end{aligned}$$

The last equality follows by Proposition 1.2.9. The notation $\sum \frac{1}{p_i p_j}$ means, that one has to consider all possible products $p_i p_j$ of two different prime divisors of n for the sum. Each summand is of the form $\frac{\mu(d)}{d}$, where $d \mid n$, with either $d = 1$ or d is a product of distinct prime divisors. □

Note that this product formula implies again that φ is multiplicative. It implies also some more properties.

Proposition 1.2.11. *The following properties hold for φ.*

(1) $\varphi(p^n) = p^n - p^{n-1}$, *for* $p \in \mathbb{P}$ *and* $n \geq 1$.
(2) $\varphi(n^2) = n\varphi(n)$ *for* $n \geq 1$.
(3) $\varphi(mn) = \varphi(m)\varphi(n)\frac{d}{\varphi(d)}$, *where* $d = \gcd(m, n)$.
(4) $m \mid n \Rightarrow \varphi(m) \mid \varphi(n)$.
(5) $\varphi(n)$ *is even for all* $n \geq 3$.

Proof. The product formula implies $\varphi(p^n) = p^n\left(1 - \frac{1}{p}\right) = p^n - p^{n-1}$, so (1).

(2) follows directly from (3).

For (3), consider

$$\frac{\varphi(n)}{n} = \prod_{p \mid n}\left(1 - \frac{1}{p}\right).$$

Each prime divisor $p \mid mn$ either divides m or n. Those dividing n and m also divide $\gcd(m, n)$. So we have

$$\frac{\varphi(mn)}{mn} = \prod_{p \mid mn}\left(1 - \frac{1}{p}\right) = \frac{\prod_{p \mid m}\left(1 - \frac{1}{p}\right)\prod_{p \mid n}\left(1 - \frac{1}{p}\right)}{\prod_{p \mid (m,n)}\left(1 - \frac{1}{p}\right)}$$

$$= \frac{\varphi(m)}{m} \cdot \frac{\varphi(n)}{n} \cdot \left(\frac{\varphi(d)}{d}\right)^{-1},$$

and hence the claim.

To (4): because of $m \mid n$ we have $n = mc$ with $1 \leq c \leq n$. For $c = n$ we have $m = 1$ and hence $\varphi(m) \mid \varphi(n)$. So we may assume that $c < n$. By (3) we have

$$\varphi(n) = \varphi(mc) = \varphi(m)\varphi(c)\frac{d}{\varphi(d)} = d\varphi(m)\frac{\varphi(c)}{\varphi(d)}$$

with $d = (m, c)$. We will prove (4) by induction over n. For $n = 1$ we have $m = c = 1$ and the claim is true. Assume now (4) for all $1 \leq k < n$. Then in particular $\varphi(d) \mid \varphi(c)$ because of $d \mid c$. Hence the above equations says that $\varphi(n)$ is an integer multiple of $\varphi(m)$i.e., that $\varphi(m) \mid \varphi(n)$.

To (5): for $n = 2^r$, $r \geq 2$ the number $\varphi(n) = 2^r - 2^{r-1}$ is even. If n has at least one odd prime divisor then the claim follows from

$$\varphi(n) = n\prod_{p \mid n}\frac{p-1}{p} = \frac{n}{\prod_{p \mid n} p} \cdot \prod_{p \mid n}(p-1),$$

because $\prod_{p|n}(p-1)$ then is even, and the fraction is an integer. Actually we see that each prime divisor $p > 2$ contributes a factor of 2 for this product. Therefore we even have

$$2^r \mid \varphi(n),$$

if n has exactly r different odd prime divisors. $\square$

PROPOSITION 1.2.12. *For $n \geq 1$ we have*

$$\sum_{d|n} \Lambda(d) = \log(n).$$

PROOF. For $n = 1$ the claim is true. So let $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r} \geq 2$. Taking the logarithm yields $\log(n) = \sum_{i=1}^r \alpha_i \log(p_i)$. The nonzero terms in $\sum_{d|n} \Lambda(d)$ arise by those divisors d, which are of the form p_i^m, with $m = 1, 2, \ldots, \alpha_i$ and $i = 1, 2, \ldots, r$. Therefore we have

$$\sum_{d|n} \Lambda(d) = \sum_{i=1}^{r} \sum_{m=1}^{\alpha_i} \Lambda(p_i^m) = \sum_{i=1}^{r} \sum_{m=1}^{\alpha_i} \log(p_i) = \sum_{i=1}^{r} \alpha_i \log(p_i) = \log(n).$$

$\square$

DEFINITION 1.2.13. The Liouville λ function is defined by

$$\lambda(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^{\sum_{i=1}^r \alpha_i} & \text{if } n = p_1^{\alpha_1} \cdots p_r^{\alpha_r} \geq 2. \end{cases}$$

It is obvious that λ is strongly multiplicative.

REMARK 1.2.14. The introduced arithmetic functions yields interesting identities involving the Riemann zeta function, by forming a Dirichlet series $\sum_{n=1}^\infty a_n n^{-s}$ with them:

$$\sum_{n=1}^{\infty} \lambda(n) n^{-s} = \frac{\zeta(2s)}{\zeta(s)}, \quad \sigma > 1$$
$$\sum_{n=1}^{\infty} \mu(n) n^{-s} = \frac{1}{\zeta(s)}, \quad \sigma > 1$$
$$\sum_{n=1}^{\infty} \sigma_\alpha(n) n^{-s} = \zeta(s)\zeta(s-\alpha), \quad \sigma > \max\{1, 1 + \Re(\alpha)\}$$
$$\sum_{n=1}^{\infty} \tau(n) n^{-s} = \zeta(s)^2, \quad \sigma > 1$$
$$\sum_{n=1}^{\infty} \varphi(n) n^{-s} = \frac{\zeta(s-1)}{\zeta(s)}, \quad \sigma > 2$$
$$\sum_{n=1}^{\infty} \Lambda(n) n^{-s} = -\frac{\zeta'(s)}{\zeta(s)}, \quad \sigma > 1.$$

1.3. The Dirichlet product

The Dirichlet product of two arithmetic functions has its origin in the natural product of two Dirichlet series. Let f b an arithmetic function. The *formal Dirichlet series* of f is defined by

$$D_f(s) = \sum_{n=1}^{\infty} f(n)n^{-s}.$$

the sum and the product of two such series are defined naturally. Indeed, with

$$h(n) = \sum_{dd'=n} f(d)g(d')$$

we have

$$\begin{aligned} D_f(s) + D_g(s) &= \sum_{n=1}^{\infty}(f(n) + g(n))n^{-s}, \\ D_f(s)D_g(s) &= \sum_{n=1}^{\infty} h(n)n^{-s}. \end{aligned}$$

The formula for the product can be justified by a formal computation:

$$\begin{aligned} \sum_{m=1}^{\infty} f(m)m^{-s} \sum_{k=1}^{\infty} g(k)k^{-s} &= \sum_{m,k=1}^{\infty} f(m)g(k)(mk)^{-s} \\ &= \sum_{n=1}^{\infty} n^{-s} \left(\sum_{km=n} f(m)g(k) \right). \end{aligned}$$

For two arithmetic functions f, g we will define a sum and a new product $f + g, f * g$ in such a way, that we have

$$\begin{aligned} D_{f+g}(s) &= D_f(s) + D_g(s), \\ D_{f*g}(s) &= D_f(s)D_g(s). \end{aligned}$$

This amounts to the following definition.

Definition 1.3.1. For two arithmetic functions f, g we define

$$\begin{aligned} (f + g)(n) &= f(n) + g(n), \\ (f * g)(n) &= \sum_{d|n} f\left(\frac{n}{d}\right)g(d) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right). \end{aligned}$$

The star product is called the *Dirichlet product*.

This product is commutative, because we have

$$(f * g)(n) = \sum_{\substack{ab=n \\ a,b\in\mathbb{N}}} f(a)g(b) = (g * f)(n).$$

DEFINITION 1.3.2. For $n \geq 1$ we define the arithmetic functions I, ε and id as follows:

$$I(n) = \left\lfloor \frac{1}{n} \right\rfloor, \quad \varepsilon(n) = 1, \quad \mathrm{id}(n) = n.$$

Note that $I(n) = 0$ for all $n \geq 2$.

PROPOSITION 1.3.3. *The set of arithmetic functions forms a UFD, and hence an integral domain, with respect to the addition and the Dirichlet product. It is denoted by $\mathcal{D}$. The unit element is given by I, i.e.,*

$$I * f = f * I = f.$$

PROOF. The ring multiplication is indeed associative:

$$\begin{aligned}
((f * g) * h)(n) &= \sum_{\ell c=n} (f * g)(\ell)h(c) = \sum_{lc=n} h(c) \sum_{ab=\ell} f(a)g(b) \\
&= \sum_{abc=n} f(a)g(b)h(c), \\
(f * (g * h))(n) &= \sum_{ad=n} f(a)(g * h)(d) = \sum_{ad=n} f(a) \sum_{bc=d} g(b)h(c) \\
&= \sum_{abc=n} f(a)g(b)h(c).
\end{aligned}$$

Furthermore, I is a unit element because of

$$(f * I)(n) = \sum_{d|n} f(d) I\left(\frac{n}{d}\right) = \sum_{d|n} f(d) \left\lfloor \frac{d}{n} \right\rfloor = f(n).$$

Here $\left\lfloor \frac{d}{n} \right\rfloor = 0$ for $d < n$. The commutativity implies $f = f * I = I * f$. It is not difficult to prove the other axioms, see the proof given in [**3**]. ☐

What are the units of this ring, i.e., which elements are invertible? We will show that $f \in \mathcal{D}$ is invertible if and only if $f(1) \neq 0$.

PROPOSITION 1.3.4. *The unit group $U(\mathcal{D})$ consists of the arithmetic functions f with $f(1) \neq 0$. For such f one can compute the Dirichlet inverse $g(n) = f^{-1}(n)$ recursively as follows:*

$$\begin{aligned}
g(1) &= f(1)^{-1} \\
g(n) &= -f(1)^{-1} \sum_{\substack{d|n \\ d<n}} f\left(\frac{n}{d}\right) g(d), \quad n \geq 2.
\end{aligned}$$

PROOF. The equation $f * g = I$ is equivalent to a family of equations for every $n \geq 1$, namely

$$\sum_{d \mid n} f\Big(\frac{n}{d}\Big) g(d) = I(n).$$

If $f(1) = 0$, then the equation for $n = 1$ has no solution and hence f is not invertible. Otherwise the equations exactly yield the given recursion. Clearly the inverse is unique if it exists. Suppose that $f * g = g * f = I$ and $f * h = h * f = I$ for some $g, h \in \mathcal{D}$, then $f * h = f * g = I$ and hence

$$h = h * I = h * (f * g) = (h * f) * g = I * g = g.$$

$\square$

EXAMPLE 1.3.5. *The μ-function has a Dirichlet inverse because of $\mu(1) = 1$. This inverse is the ε-function, i.e., we have*

$$\mu * \varepsilon = \varepsilon * \mu = I.$$

Indeed, by Proposition 1.2.6 we have

$$(\varepsilon * \mu)(n) = \sum_{d \mid n} \mu(d) = I(n).$$

The Dirichlet product enables us to rewrite our results on divisor sums in a compact form:

$$n = \sum_{d \mid n} \varphi(d) \Longleftrightarrow \mathrm{id} = \varepsilon * \varphi,$$

$$\varphi(n) = \sum_{d \mid n} \mu(d) \frac{n}{d} \Longleftrightarrow \varphi = \mu * \mathrm{id},$$

$$\log(n) = \sum_{d \mid n} \Lambda(d) \Longleftrightarrow \log = \Lambda * \varepsilon.$$

It also leads to easier proofs of such identities. For example, $\varepsilon * \varphi = \mathrm{id}$ implies that $\varphi = \varepsilon^{-1} * \mathrm{id} = \mu * \mathrm{id}$, showing the second identity, see also Proposition 1.2.9. Furthermore $\log = \Lambda * \varepsilon$ implies $\Lambda = \log * \varepsilon^{-1} = \log * \mu$. Hence the Λ-function is a Dirichlet product of two well-known arithmetic functions.

Multiplicative functions are units of $\mathcal{D}$, since they satisfy $f(1) = 1$. They even form a subgroup of $U(\mathcal{D})$.

PROPOSITION 1.3.6. *The set of multiplicative arithmetic functions forms a subgroup of $U(\mathcal{D})$.*

PROOF. Let $f, g \in U(\mathcal{D})$ be multiplicative. We will show that $h := f * g$ is again multiplicative. Let m, n be two integers with $\gcd(m, n) = 1$. Then we have by definition

$$h(mn) = \sum_{c \mid mn} f(c) g\Big(\frac{mn}{c}\Big).$$

Each divisor $c \mid mn$ can be written in the form $c = ab$, with $a \mid m$ and $b \mid n$. Because of $\gcd(m, n) = 1$ we have

$$\gcd(a, b) = \gcd\Big(\frac{m}{a}, \frac{n}{b}\Big) = 1.$$

So there is a bijective correspondence between the set of products ab and the divisors $c \mid mn$. Hence we have

$$\begin{aligned} h(mn) &= \sum_{\substack{a\mid m \\ b\mid n}} f(ab)g\left(\frac{mn}{ab}\right) = \sum_{\substack{a\mid m \\ b\mid n}} f(a)f(b)g\left(\frac{m}{a}\right)g\left(\frac{n}{b}\right) \\ &= \sum_{a\mid m} f(a)g\left(\frac{m}{a}\right) \cdot \sum_{b\mid n} f(b)g\left(\frac{n}{b}\right). = h(n)h(m). \end{aligned}$$

In the same way one shows that the Dirichlet inverse of f is again multiplicative, see [**16**] or [**1**]. $\square$

Example 1.3.7. *Since ε is multiplicative and $\tau = \varepsilon * \varepsilon$, we again obtain that τ is multiplicative.*

Indeed, we have

$$\tau(n) = \sum_{d\mid n} 1 = \sum_{d\mid n} \varepsilon(d)\varepsilon\left(\frac{n}{d}\right).$$

More generally we have the following result.

Example 1.3.8. *All divisor sum functions σ_k are multiplicative.*

Indeed, we have

$$\sigma_k(n) = \sum_{d\mid n} d^k = (j * \varepsilon)(n),$$

where $\varepsilon(n) = 1$ and $j(n) = n^k$ are multiplicative.

Proposition 1.3.9. *Let f be strongly multiplicative. Then the Dirichlet inverse of f is given by $f^{-1} = \mu \cdot f$, i.e., by $f^{-1}(n) = \mu(n)f(n)$.*

Proof. We write $f \cdot g$ for the above multiplication, to distinguish it from $f * g$. Let $g = \mu \cdot f$. Since f is strongly multiplicative we have

$$(g * f)(n) = \sum_{d\mid n} \mu(d)f(d)f\left(\frac{n}{d}\right) = f(n)\sum_{d\mid n} \mu(d) = f(n)I(n) = I(n),$$

where we have used Proposition 1.2.6 and $f(1) = 1$, $I(n) = 0$ for $n \geq 2$. Then we obtain $g * f = I$, and hence $g = f^{-1}$. $\square$

Example 1.3.10. *The Dirichlet inverse of the φ-function is given by*

$$\varphi^{-1}(n) = \sum_{d\mid n} d\mu(d),$$

i.e., by $\varphi^{-1} = \varepsilon * (\mu \cdot \mathrm{id})$.

Indeed, because of $\varphi = \mu * \mathrm{id}$ we have $\varphi^{-1} = \mathrm{id}^{-1} * \mu^{-1}$. Since id is strongly multiplicative, $\mathrm{id}^{-1} = \mu \cdot \mathrm{id}$ follows from the above proposition. Hence we have $\varphi^{-1} = \mu^{-1} * \mu \cdot \mathrm{id} = \varepsilon * (\mu \cdot \mathrm{id})$.

Example 1.3.11. *The Dirichlet inverse of the λ-function is given by $\lambda^{-1}(n) = \mu(n)\lambda(n) = \mu^2(n) = |\mu(n)|$.*

This follow again from Proposition 1.3.9, because λ is strongly multiplicative.

PROPOSITION 1.3.12. *If f is multiplicative, we have*

$$\sum_{d|n} \mu(d) f(d) = \prod_{p|n} (1 - f(p)).$$

PROOF. Let

$$g(n) = \sum_{d|n} \mu(d) f(d).$$

By Proposition 1.3.6, also g is multiplicative. Hence we only need to compute $g(p^\alpha)$:

$$g(p^\alpha) = \sum_{d|p^\alpha} \mu(d) f(d) = \mu(1) f(1) + \mu(p) f(p) = 1 - f(p).$$

Here we have used that $\mu(p^\alpha) = 0$ for $\alpha \geq 2$. $\square$

COROLLARY 1.3.13. *We have*

$$\varphi^{-1}(n) = \prod_{p|n} (1 - p).$$

PROOF. Because of $\varphi^{-1}(n) = \sum_{d|n} d\mu(d)$ the corollary follows from Proposition 1.3.12 with $f(d) = d$. $\square$

DEFINITION 1.3.14. The *Dirichlet derivative* f' of an arithmetic function f is given by the arithmetic function

$$f'(n) = f(n) \log(n).$$

For example, $I'(n) = 0$, since $I(n) \log(n) = 0$ for $n \geq 1$. Another example is $\varepsilon'(n) = \log(n)$. The formula

$$\sum_{d|n} \Lambda(d) = \log(n)$$

can be rewritten then by $\Lambda * \varepsilon = \varepsilon'$. The following result justifies the name "derivation".

PROPOSITION 1.3.15. *For arithmetic functions f, g we have*

$$\begin{aligned}
(f+g)' &= f' + g' \\
(f * g)' &= f' * g + f * g' \\
(f^{-1})' &= -f' * (f * f)^{-1}, \quad \textit{for } f(1) \neq 0.
\end{aligned}$$

PROOF. We have

$$(f+g)'(n) = (f+g)(n) \log(n) = f(n) \log(n) + g(n) \log(n) = f'(n) + g'(n).$$

because of $\log(n) = \log(d) + \log\left(\frac{n}{d}\right)$ we have

$$\begin{aligned}
(f * g)'(n) &= \sum_{d|n} f(d) g\left(\frac{n}{d}\right) \log(n) \\
&= \sum_{d|n} f(d) \log(d) g\left(\frac{n}{d}\right) + \sum_{d|n} f(d) \log\left(\frac{n}{d}\right) g\left(\frac{n}{d}\right) \\
&= (f' * g)(n) + (f * g')(n).
\end{aligned}$$

This implies

$$0 = I' = (f * f^{-1})' = f' * f^{-1} + f * (f^{-1})'.$$

Multiplying both sides with f^{-1} yields

$$\begin{aligned} 0 &= f^{-1} * f' * f^{-1} + (f^{-1})' \\ &= f' * f^{-1} * f^{-1} + (f^{-1})' \\ &= f' * (f * f)^{-1} + (f^{-1})'. \end{aligned}$$

□

An application of the Dirichlet derivative is a formula by Selberg, which is used in the so called "elementary proof" of the prime number theorem by Selberg and Erdös.

PROPOSITION 1.3.16 (Selberg). *For $n \geq 1$ we have*

$$\Lambda(n)\log(n) + \sum_{d|n} \Lambda(d)\Lambda\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d)\log^2\left(\frac{n}{d}\right).$$

PROOF. We form the Dirichlet derivative of the above identity $\Lambda * \varepsilon = \varepsilon'$. This yields

$$\begin{aligned} \varepsilon'' &= (\Lambda * \varepsilon)' = \Lambda' * \varepsilon + \Lambda * \varepsilon' \\ &= \Lambda' * \varepsilon + \Lambda * (\Lambda * \varepsilon) \end{aligned}$$

Now $\varepsilon^{-1} = \mu$ implies

$$\varepsilon'' * \mu = \Lambda' + \Lambda * \Lambda.$$

This shows the claim. □

REMARK 1.3.17. Let us state the *prime number theorem*. It is the asymptotic statement

$$\pi(x) \sim \frac{x}{\log(x)}.$$

In the elementary proof, the Λ-function is used, together with its summatory function

$$\psi(x) = \sum_{n \leq x} \Lambda(n).$$

In fact, the prime number theorem is equivalent to the statement $\psi(x) \sim x$.

PROPOSITION 1.3.18 (First Möbius inversion formula). *Let $f, g \in \mathcal{D}$. Then the following statements are equivalent:*

$$g(n) = \sum_{d|n} f(d)$$

$$f(n) = \sum_{d|n} g(d)\mu\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d)g\left(\frac{n}{d}\right).$$

PROOF. The first statement is equivalent to $g = f * \varepsilon$; the second one is equivalent to $f = g * \mu$. The claim follows from $\mu^{-1} = \varepsilon$. □

EXAMPLE 1.3.19. *For $n \geq 1$ we have*

$$\Lambda(n) = \sum_{d|n} \mu(d)\log\left(\frac{n}{d}\right) = -\sum_{d|n} \mu(d)\log(d).$$

In Proposition 1.2.12 w have shown that $\log = \Lambda * \varepsilon$. By the inversion formula this is equivalent to $\Lambda = \log * \mu$. Writing this explicitly gives the first part of the claim. Furthermore we have

$$\begin{aligned}\sum_{d|n} \mu(d) \log\left(\frac{n}{d}\right) &= \log(n) \sum_{d|n} \mu(d) - \sum_{d|n} \mu(d) \log(d) \\ &= \log(n) I(n) - \sum_{d|n} \mu(d) \log(d) \\ &= -\sum_{d|n} \mu(d) \log(d).\end{aligned}$$

It is possible to generalize the Dirichlet product.

Definition 1.3.20. Let F be a real or complex valued function, defined on $(0, \infty)$ with $F(x) = 0$ for $0 < x < 1$, and let $\alpha \in \mathcal{D}$. Then define

$$(\alpha \circ F)(x) = \sum_{n \leq x} \alpha(n) F\left(\frac{x}{n}\right).$$

If $F(x) = 0$ for all $x \notin \mathbb{Z}$, then $F_{|\mathbb{Z}}$ is arithmetic and we have $(\alpha \circ F)(n) = (\alpha * F)(n)$. However, the operation $\circ$ is neither commutative, nor associative. It does satisfy the following property, see [**1**]:

Proposition 1.3.21. *For $\alpha, \beta \in \mathcal{D}$ we have*

$$\alpha \circ (\beta \circ F) = (\alpha * \beta) \circ F.$$

Note that the function I is also a left unit with respect to the product $\circ$.

Proposition 1.3.22 (Second Möbius inversion formula). *Let $\alpha \in U(\mathcal{D})$. Then the following statements are equivalent:*

$$G(x) = \sum_{n \leq x} \alpha(n) F\left(\frac{x}{n}\right)$$

$$F(x) = \sum_{n \leq x} \alpha^{-1}(n) G\left(\frac{x}{n}\right).$$

If α is strongly multiplicative, then $\alpha^{-1}(n) = \mu(n)\alpha(n)$.

Proof. The first identity says that $G = \alpha \circ F$, the second one says that $F = \alpha^{-1} \circ G$. The first implies the second as follows:

$$\alpha^{-1} \circ G = \alpha^{-1} \circ (\alpha \circ F) = (\alpha^{-1} * \alpha) \circ F = I \circ F = F.$$

The second one implies the first one in the same way. □

Of course, the second Möbius inversion formula implies the first one by taking $\alpha = \varepsilon$.

Proposition 1.3.23. *Let $h = f * g$ and $H(x) = \sum_{n \leq x} h(n)$, $F(x) = \sum_{n \leq x} f(n)$, $G(x) = \sum_{n \leq x} g(n)$. Then we have*

$$H(x) = \sum_{n \leq x} f(n) G\left(\frac{x}{n}\right) = \sum_{n \leq x} g(n) F\left(\frac{x}{n}\right).$$

Proof. Define

$$U(x) = \begin{cases} 0 & \text{falls } 0 < x < 1, \\ 1 & \text{falls } x \geq 1. \end{cases}$$

then we have $F = f \circ U$, $G = g \circ U$ and $H = h \circ U$. By Proposition 1.3.21 it follows

$$\begin{aligned} f \circ G &= f \circ (g \circ U) = (f * g) \circ U = H, \\ g \circ F &= g \circ (f \circ U) = (g * f) \circ U = H. \end{aligned}$$

□

Corollary 1.3.24. *Let $F(x) = \sum_{n \leq x} f(n)$. Then we have*

$$\sum_{n \leq x} \sum_{d \mid n} f(d) = \sum_{n \leq x} f(n) \left[\frac{x}{n} \right] = \sum_{n \leq x} F\left(\frac{x}{n} \right).$$

Proof. This follows from Proposition 1.3.23 with $g(n) = 1$, i.e., with $G(x) = [x]$. □

The corollary has the following application.

Proposition 1.3.25. *For $x \geq 1$ we have*

$$\begin{aligned} \sum_{n \leq x} \mu(n) \left[\frac{x}{n} \right] &= 1, \\ \sum_{n \leq x} \Lambda(n) \left[\frac{x}{n} \right] &= \log([x]!). \end{aligned}$$

Proof. By Corollary 1.3.24 with $f(n) = \mu(n)$, respectively $f(n) = \Lambda(n)$, we obtain

$$\begin{aligned} \sum_{n \leq x} \mu(n) \left[\frac{x}{n} \right] &= \sum_{n \leq x} \sum_{d \mid n} \mu(d) = \sum_{n \leq x} \left\lfloor \frac{1}{n} \right\rfloor = 1, \\ \sum_{n \leq x} \Lambda(n) \left[\frac{x}{n} \right] &= \sum_{n \leq x} \sum_{d \mid n} \Lambda(d) = \sum_{n \leq x} \log n = \log([x]!). \end{aligned}$$

□

Remark 1.3.26. The first part of the above result suggests that we might have

$$\lim_{x \to \infty} \sum_{n \leq x} \frac{\mu(n)}{n} = 0.$$

In fact, this statement is equivalent to the prime number theorem.

Corollary 1.3.27. *For every $x \geq 1$ we have*

$$[x]! = \prod_{p \leq x} p^{\alpha(p)}, \quad \text{mit } \alpha(p) = \sum_{m=1}^{\infty} \left[\frac{x}{p^m} \right].$$

PROOF. Note that the sum is finite for $\alpha(p)$, since $[x/p^m] = 0$ for $p > x$. By using the identity for Λ from Proposition 1.3.25 we obtain

$$\log([x]!) = \sum_{n\le x} \Lambda(n) \left[\frac{x}{n}\right] = \sum_{p\le x} \sum_{m=1}^{\infty} \left[\frac{x}{p^m}\right] \log(p) = \sum_{p\le x} \alpha(p) \log(p).$$

Here we have used that only for $n = p^m$ the terms $\Lambda(n) = \Lambda(p^m) = \log(p)$ are nonzero. Moreover, $p^m \le x$ implies $p \le x$. $\square$

1.4. Euler's summation formula

Our aim is to determine the asymptotics of certain sums involving arithmetic functions. This often can be done by comparison with an integral. The following summation formula by Euler is an important result in this direction.

PROPOSITION 1.4.1 (Euler's summation formula). *Let f be a real function of class C^1 on the interval $[y,x]$, $0<y<x$, i.e., f' is continuous on $[y,x]$. Then we have*

$$\begin{aligned}\sum_{y<n\le x} f(n) = \int_y^x f(t)dt + \int_y^x (t-[t])f'(t)dt\\ + f(x)([x]-x) - f(y)([y]-y).\end{aligned}$$

This formula can be generalized as follows.

PROPOSITION 1.4.2 (Abel's summation formula). *Let $a(n)$ be an arithmetic function, and*

$$A(x) = \begin{cases} \sum_{n\le x} a(n) & \text{for } x\ge 1,\\ 0 & \text{for } x<1.\end{cases}$$

Let $f\in C^1([y,x])$ for $0<y<x$. Then we have

$$\sum_{y<n\le x} a(n)f(n) = A(x)f(x) - A(y)f(y) - \int_y^x A(t)f'(t)dt.$$

PROOF. First note that $A(t)$ in constant on $[n,n+1)$. Let $k=[x]$ and $m=[y]$. Then we have $A(x)=A(k)$ and $A(y)=A(m)$. Since $A(t)$ is constant on $[k,x]$, we have

$$\begin{aligned}A(x)f(x) - \int_k^x A(t)f'(t)dt &= A(x)f(x) - A(x)\int_k^x f'(t)dt\\ &= A(x)f(x) - A(x)(f(x)-f(k)) = A(k)f(k).\end{aligned}$$

Similarly we have

$$A(y)f(y) - \int_{m+1}^y A(t)f'(t)dt = A(m)f(m+1).$$

Using $A(n)-A(n-1)=a(n)$ and the above formulas, which are used to replace $A(k)f(k)$ and $A(m)f(m+1)$, we have

$$\begin{aligned}
\sum_{y<n\leq x} a(n)f(n) &= \sum_{n=m+1}^{k} a(n)f(n) = \sum_{n=m+1}^{k} (A(n)-A(n-1))f(n) \\
&= \sum_{n=m+1}^{k} A(n)f(n) - \sum_{n=m}^{k-1} A(n)f(n+1) \\
&= A(k)f(k) - A(m)f(m+1) + \sum_{n=m+1}^{k-1} A(n)(f(n)-f(n+1)) \\
&= A(k)f(k) - A(m)f(m+1) - \sum_{n=m+1}^{k-1} A(n)\int_n^{n+1} f'(t)dt \\
&= A(k)f(k) - A(m)f(m+1) - \sum_{n=m+1}^{k-1} \int_n^{n+1} A(t)f'(t)dt \\
&= A(k)f(k) - A(m)f(m+1) - \int_{m+1}^{k} A(t)f'(t)dt \\
&= A(x)f(x) - A(y)f(y) - \int_y^{m+1} A(t)f'(t)dt \\
&\quad - \int_{m+1}^{k} A(t)f'(t)dt - \int_k^x A(t)f'(t)dt \\
&= A(x)f(x) - A(y)f(y) - \int_y^x A(t)f'(t)dt.
\end{aligned}$$

□

For $y<1$ we have $A(y)=0$. This implies the following corollary.

Corollary 1.4.3. *For $x\geq 1$ we have*

$$\sum_{n\leq x} a(n)f(n) = A(x)f(x) - \int_1^x A(t)f'(t)dt.$$

Proof of Proposition 1.4.1*:* Let $a(n)=1$ for $n\geq 1$, so that $A(x)=[x]$. Then Abel's summation formula yields

$$\sum_{y<n\leq x} f(n) = f(x)[x] - f(y)[y] - \int_y^x [t]f'(t)dt.$$

Combining this with the formula for partial integration

$$\int_y^x tf'(t)dt = xf(x) - yf(y) - \int_y^x f(t)dt,$$

yields the claim. □

Both summation formulas have many applications.

PROPOSITION 1.4.4. *Let $\alpha, \sigma \in \mathbb{R}$ and $x \geq 1$. Then we have*

$$\begin{aligned}
\sum_{n \leq x} \frac{1}{n} &= \log(x) + \gamma + O\left(\frac{1}{x}\right), \\
\sum_{n \leq x} \frac{1}{n^\sigma} &= \frac{x^{1-\sigma}}{1-\sigma} + \zeta(\sigma) + O(x^{-\sigma}), \quad \sigma \neq 1, \sigma > 0 \\
\sum_{n > x} \frac{1}{n^\sigma} &= O(x^{1-\sigma}), \quad \sigma > 1 \\
\sum_{n \leq x} n^\alpha &= \frac{x^{\alpha+1}}{\alpha+1} + O(x^\alpha), \quad \alpha \geq 0.
\end{aligned}$$

PROOF. For the first statement choose $f(t) = \frac{1}{t}$ and $y = 1$ in Euler's summation formula. Then it follows, with $f'(t) = -\frac{1}{t^2}$, that

$$\begin{aligned}
\sum_{n \leq x} \frac{1}{n} = 1 + \sum_{1 < n \leq x} \frac{1}{n} &= 1 + \int_1^x \frac{dt}{t} + \int_1^x \frac{t - [t]}{-t^2}\, dt + \frac{[x] - x}{x} - 0 \\
&= \log(x) + 1 - \int_1^x \frac{t - [t]}{t^2}\, dt + O\left(\frac{1}{x}\right) \\
&= \log(x) + \left(1 - \int_1^\infty \frac{t - [t]}{t^2}\, dt\right) + \int_x^\infty \frac{t - [t]}{t^2}\, dt + O\left(\frac{1}{x}\right).
\end{aligned}$$

Here $1 - \int_1^\infty \frac{t-[t]}{t^2}\, dt$ is a constant, since the integral if finite. In fact, it is dominated by $\int_1^\infty t^{-2} dt$. The integral after it is subsumed in the O-term, because of

$$0 \leq \int_x^\infty \frac{t - [t]}{t^2}\, dt \leq \int_x^\infty t^{-2} dt = \frac{1}{x}.$$

For $x \to \infty$ this yields

$$\gamma = \lim_{x \to \infty} \left(\sum_{n \leq x} \frac{1}{n} - \log(x) \right) = 1 - \int_1^\infty \frac{t - [t]}{t^2}\, dt.$$

This finishes the proof of the first statement.

For the second statement we choose $f(t) = t^{-\sigma}$ with $f'(t) = -\sigma t^{-\sigma-1}$, $y = 1$. Then Euler's summation formula yields

$$\sum_{n \leq x} \frac{1}{n^\sigma} = \int_1^x \frac{dt}{t^\sigma} - \sigma \int_1^x \frac{t - [t]}{t^{\sigma+1}}\, dt + \frac{[x] - x}{x^\sigma} + 1.$$

As above we may replace the upper index of the second integral by infinity, introducing the error term

$$\sigma \int_x^\infty \frac{t - [t]}{t^{\sigma+1}}\, dt = O(x^{-\sigma})$$

macht. Hence we have

$$\begin{aligned}\sum_{n\le x}\frac{1}{n^\sigma} &= \frac{x^{1-\sigma}}{1-\sigma}-\frac{1}{1-\sigma}+1-\sigma\int_1^\infty \frac{t-[t]}{t^{\sigma+1}}dt+O(x^{-\sigma})\\ &= \frac{x^{1-\sigma}}{1-\sigma}+C(\sigma)+O(x^{-\sigma}),\end{aligned}$$

where

$$C(\sigma)=1-\frac{1}{1-\sigma}-\sigma\int_1^\infty \frac{t-[t]}{t^{\sigma+1}}dt.$$

For $\sigma>1$ we consider the limit $x\to\infty$. The left side tends to $\zeta(\sigma)$, while both terms $x^{1-\sigma}$ and $x^{-\sigma}$ tend to zero. Hence we have

$$C(\sigma)=\zeta(\sigma),\quad \sigma>1.$$

For $0<\sigma<1$ we also have $x^{-\sigma}\to 0$ for $x\to\infty$ and $C(\sigma)=\zeta(\sigma)$, since we have a continuation of $\zeta(\sigma)$ in this strip by

$$\lim_{x\to\infty}\left(\sum_{n\le x}n^{-\sigma}-\frac{x^{1-\sigma}}{1-\sigma}\right)=C(\sigma)$$

This shows the second statement.

The third statement follows form the second, for $\sigma>1$:

$$\sum_{n>x}\frac{1}{n^\sigma}=\zeta(\sigma)-\sum_{n\le x}\frac{1}{n^\sigma}=\frac{x^{1-\sigma}}{\sigma-1}+O(x^{-\sigma})=O(x^{1-\sigma}),$$

because we have $x^{-\sigma}\le x^{1-\sigma}$ for $\sigma>1$.

For the last statement we choose $f(t)=t^\alpha$ with $f'(t)=\alpha t^{\alpha-1}$ and $\alpha\ge 0$. Then

$$\begin{aligned}\sum_{n\le x}n^\alpha &= \int_1^x t^\alpha dt+\alpha\int_1^x t^{\alpha-1}(t-[t])dt+1-(x-[x])x^\alpha\\ &= \frac{x^{\alpha+1}}{\alpha+1}-\frac{1}{\alpha+1}+O\left(\alpha\int_1^x t^{\alpha-1}dt\right)+O(x^\alpha)\\ &= \frac{x^{\alpha+1}}{\alpha+1}+O(x^\alpha).\end{aligned}$$

□

PROPOSITION 1.4.5. *For $x\ge 2$ we have*

$$\sum_{n\le x}\Lambda(n)\left[\frac{x}{n}\right]=\log([x]!)=x\log(x)-x+O(\log(x)).$$

PROOF. The first equality follows from Proposition 1.3.25. For the second one we choose $f(t)=\log(t)$ in Euler's summation formula. This yields

$$\begin{aligned}\sum_{n\le x}\log(n) &= \int_1^x \log(t)dt + \int_1^x \frac{t-[t]}{t}\,dt + ([x]-x)\log(x)\\ &= x\log(x) - x + 1 + \int_1^x \frac{t-[t]}{t}\,dt + O(\log(x))\\ &= x\log(x) - x + O(\log(x)).\end{aligned}$$

In the last step we have used that

$$\int_1^x \frac{t-[t]}{t}dt = O\left(\int_1^x \frac{1}{t}dt\right) = O(\log(x))$$

□

Corollary 1.4.6. *For $x \ge 2$ we have*

$$\sum_{p\le x}\left[\frac{x}{p}\right]\log(p) = x\log(x) + O(x).$$

Proof. As in corollary 1.3.27 we have

$$\sum_{n\le x}\Lambda(n)\left[\frac{x}{n}\right] = \sum_{p\le x}\sum_{m=1}^{\infty}\left[\frac{x}{p^m}\right]\log(p) = \sum_{p\le x}\left[\frac{x}{p}\right]\log(p) + \sum_{p\le x}\sum_{m=2}^{\infty}\left[\frac{x}{p^m}\right]\log(p).$$

This yields the claim with Proposition 1.4.5, if we can show that the last sum is $O(x)$. Indeed, we have

$$\begin{aligned}\sum_{p\le x}\sum_{m=2}^{\infty}\left[\frac{x}{p^m}\right]\log(p) &\le \sum_{p\le x}\log(p)\sum_{m=2}^{\infty}\frac{x}{p^m} = x\sum_{p\le x}\log(p)\sum_{m=2}^{\infty}p^{-m}\\ &= x\sum_{p\le x}\log(p)\cdot\frac{1}{p^2}\left(1-\frac{1}{p}\right)^{-1}\\ &= x\sum_{p\le x}\frac{\log(p)}{p(p-1)} \le x\sum_{n=2}^{\infty}\frac{\log(n)}{n(n-1)} = O(x),\end{aligned}$$

because the last series is bounded. □

Remark 1.4.7. In fact, the last sum is bounded as follows

$$\sum_{n=2}^{\infty}\frac{\log(n)}{n(n-1)} \le \sum_{r=1}^{\infty}\sum_{\substack{2^{r-1}<n\\ 2^r\ge n}}\frac{r\log(2)}{n(n-1)} = \sum_{r=1}^{\infty}\frac{r\log(2)}{2^r} = \log(4).$$

REMARK 1.4.8. We want to mention a few further results on the asymtotics of summatory functions for some arithmetic functions. For $x \geq 1$ we have

(1) $\displaystyle\sum_{n \leq x} \varphi(n) = \frac{3}{\pi^2} x^2 + O(x \log(x)).$

(2) $\displaystyle\sum_{n \leq x} \tau(n) = x \log(x) + (2\gamma - 1)x + O(\sqrt{x}).$

(3) $\displaystyle\sum_{n \leq x} \sigma(n) = \frac{\pi^2}{12} x^2 + O(x \log(x)).$

1.5. A Tauberian Theorem by Shapiro

We prove a theorem by H. N. Shapiro, connecting sums of the form $\sum_{n\le x} a(n)$ with sums of the form $\sum_{n\le x} a(n)[x/n]$, see [**15**]. As an application we obtain two theorems by Mertens.

THEOREM 1.5.1 (Shapiro). *Let $(a(n))$ be a sequence of non-negative real numbers such that, for $x \ge 1$,*

$$\sum_{n\le x} a(n)\left[\frac{x}{n}\right] = x\log(x) + O(x).$$

Then the following statements hold:

(1) $\displaystyle\sum_{n\le x} \frac{a(n)}{n} = \log(x) + O(1)$ *for $x \ge 1$.*

(2) $\displaystyle\sum_{n\le x} a(n) \le Bx$ *for $x \ge 1$ and a constant $B > 0$.*

(3) $\displaystyle\sum_{n\le x} a(n) \ge Ax$ *for $x \ge x_0$ and some $x_0 > 1$, and a constant $A > 0$.*

PROOF. Let

$$S(x) = \sum_{n\le x} a(n), \quad T(x) = \sum_{n\le x} a(n)\left[\frac{x}{n}\right], \quad A(x) = \sum_{n\le x} \frac{a(n)}{n}.$$

We first show, for $x \ge 1$, the inequality

$$S(x) - S\left(\frac{x}{2}\right) \le T(x) - 2T\left(\frac{x}{2}\right). \tag{1.3}$$

Because $y \ge 0$ implies $[2y] - 2[y] \ge 0$, we have

$$\begin{aligned} T(x) - 2T\left(\frac{x}{2}\right) &= \sum_{n\le x} a(n)\left[\frac{x}{n}\right] - 2\sum_{n\le x/2} a(n)\left[\frac{x}{2n}\right] \\ &= \sum_{n\le x/2} a(n)\left(\left[\frac{x}{n}\right] - 2\left[\frac{x}{2n}\right]\right) + \sum_{x/2<n\le x} a(n)\left[\frac{x}{n}\right] \\ &\ge \sum_{x/2<n\le x} a(n)\left[\frac{x}{n}\right] = \sum_{x/2<n\le x} a(n) = S(x) - S\left(\frac{x}{2}\right), \end{aligned}$$

because $[x/n] = 1$ for $x/2 < n \le x$. This will imply (2) as follows. By assumption we have $T(x) = x\log(x) + O(x)$. Hence we have

$$\begin{aligned} T(x) - 2T\left(\frac{x}{2}\right) &= x\log(x) - 2\cdot\frac{x}{2}\log\left(\frac{x}{2}\right) + O(x) \\ &= O(x). \end{aligned}$$

By (1.3) this implies $S(x) - S\left(\frac{x}{2}\right) = O(x)$. So there exists a constant $c > 0$ with

$$S(x) - S\left(\frac{x}{2}\right) \le cx, \quad x \ge 1.$$

Now replace x here successively by $\frac{x}{2}, \frac{x}{4}, \dots, \frac{x}{2^{n-1}}$ with $2^n > x$. We have $S\left(\frac{x}{2^n}\right) = 0$, since the argument then is less than one. This yields the inequalities

$$\begin{aligned} S(x) - S\left(\frac{x}{2}\right) &\le cx \\ S\left(\frac{x}{2}\right) - S\left(\frac{x}{4}\right) &\le c\frac{x}{2} \\ \vdots \qquad &\le \ \vdots \\ S\left(\frac{x}{2^{n-1}}\right) - S\left(\frac{x}{2^n}\right) &\le c\frac{x}{2^{n-1}}. \end{aligned}$$

Adding up all inequalities yields

$$S(x) - 0 \le cx\left(1 + \frac{1}{2} + \dots + \frac{1}{2^{n-1}}\right) < 2cx.$$

With $B := 2c$ this implies $\sum_{n \le x} a(n) = S(x) \le Bx$.

To (1). Writing $[x/n] = x/n + O(1)$we have, by using (2),

$$\begin{aligned} T(x) &= \sum_{n \le x} a(n)\left[\frac{x}{n}\right] = \sum_{n \le x}\left(\frac{x}{n} + O(1)\right)a(n) \\ &= x\sum_{n \le x}\frac{a(n)}{n} + O\left(\sum_{n \le x} a(n)\right) = x\sum_{n \le x}\frac{a(n)}{n} + O(x). \end{aligned}$$

hence we obtain

$$\sum_{n \le x}\frac{a(n)}{n} = \frac{1}{x}T(x) + O(1) = \frac{x\log(x)}{x} + O(1) = \log(x) + O(1).$$

Here the second equality follows by assumption.

To (3): We write (1) as $A(x) = \log(x) + R(x)$ with an error term $R(x)$, which satisfies $|R(x)| \le M$ for a constant $M > 0$. For $x \ge 1$ and $0 < \alpha < 1$ with $\alpha x \ge 1$ it follows

$$\begin{aligned} A(x) - A(\alpha x) &= \log(x) + R(x) - \log(\alpha x) - R(\alpha x) \\ &= -\log(\alpha) + R(x) - R(\alpha x) \\ &\ge -\log(\alpha) - |R(x)| - |R(\alpha x)| \ge -\log(\alpha) - 2M. \end{aligned}$$

Now choose $\alpha = \exp(-2M - 1)$. Then we have $\log(\alpha) = -2M - 1$, and therefore $-\log(\alpha) - 2M = 1$. This gives

$$A(x) - A(\alpha x) \geq 1, \quad x \geq \frac{1}{\alpha},$$

$$A(x) - A(\alpha x) = \sum_{\alpha x < n \leq x} \frac{a(n)}{n} \leq \frac{1}{\alpha x} \sum_{n \leq x} a(n) = \frac{S(x)}{\alpha x}.$$

It follows that $S(x) \geq \alpha x$ for $x \geq \frac{1}{\alpha}$. This yields (3) with $A = \alpha$ and $x_0 = 1/\alpha$. $\square$

PROPOSITION 1.5.2 (Mertens 1). *For all $x \geq 1$ we have*

$$\sum_{p \leq x} \frac{\log(p)}{p} = \log(x) + O(1).$$

PROOF. This follows from Shapiro's theorem with

$$a(n) = \begin{cases} \log(p) & \text{falls } p \in \mathbb{P}, \\ 0 & \text{otherwise.} \end{cases}$$

We have $a(n) \geq 0$ and all assumptions of Shapiro's theorem are satisfied, because by Corollary 1.4.6 we have

$$\sum_{n \leq x} a(n) \left[\frac{x}{n}\right] = \sum_{p \leq x} \left[\frac{x}{p}\right] \log(p) = x \log(x) + O(x).$$

Hence (1) of Theorem 1.5.1 yields

$$\sum_{p \leq x} \frac{\log(p)}{p} = \sum_{n \leq x} \frac{a(n)}{n} = \log(x) + O(1).$$

$\square$

PROPOSITION 1.5.3 (Mertens 2). *For all $x \geq 1$ we have*

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} = \log(x) + O(1).$$

PROOF. Choose $a(n) = \Lambda(n) \geq 0$ in Shapiro's theorem. By Proposition 1.4.5, the assumptions are satisfied:

$$\sum_{n \leq x} a(n) \left[\frac{x}{n}\right] = \sum_{n \leq x} \Lambda(n) \left[\frac{x}{n}\right] = x \log(x) + O(x).$$

Hence the claim again follows from (1) of Theorem 1.5.1. $\square$

REMARK 1.5.4. There is also a theorem Mertens 3. For all $x \geq 2$ we have,

$$\sum_{p \leq x} \frac{1}{p} = \log(\log(x)) + c + O\left(\frac{1}{\log x}\right),$$

see Remark 1.1.14. This is a bit weaker than the prime number theorem, where the error term would be $o\left(\frac{1}{\log(x)}\right)$. There is also a *Mertens formula*, namely, for $x \geq 2$,

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right) = \frac{e^{-\gamma}}{\log(x)} \left(1 + O\left(\frac{1}{\log x}\right)\right).$$

For a proof see [**16**], I.1.6. Finally, there is (was) also a *Mertens conjecture.* The Mertens function is defined as

$$M(n) = \sum_{1 \leq k \leq n} \mu(k).$$

The Mertens conjecture says that for all $n > 1$,

$$|M(n)| < \sqrt{n}.$$

The conjecture is true for all $n < 10^{16}$. In such a case, usually one believes that it is true always. However, in 1985, Andrew Odlyzko and Herman te Riele proved the Mertens conjecture false using the Lenstra–Lenstra–Lovász lattice basis reduction algorithm, see [**14**]. There exist infinitely many counterexamples, and the first one must lie in the interval

$$[10^{16}, 10^{8.512 \cdot 10^{18}}]$$

But until today no *explicit* counterexample is known. The Mertens conjecture, if true, would have implied the Riemann hypothesis. In fact, RH is equivalent to

$$\lim_{x \to \infty} \frac{M(x)}{x^{\frac{1}{2}+\varepsilon}} = 0$$

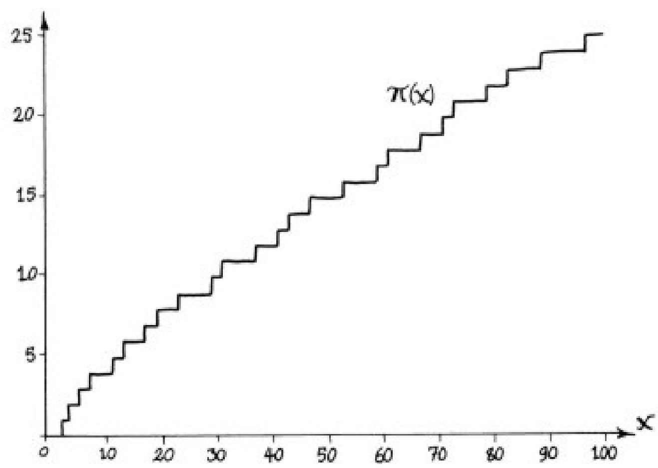
for all $\varepsilon > 0$.

1.6. Some results on the prime number distribution

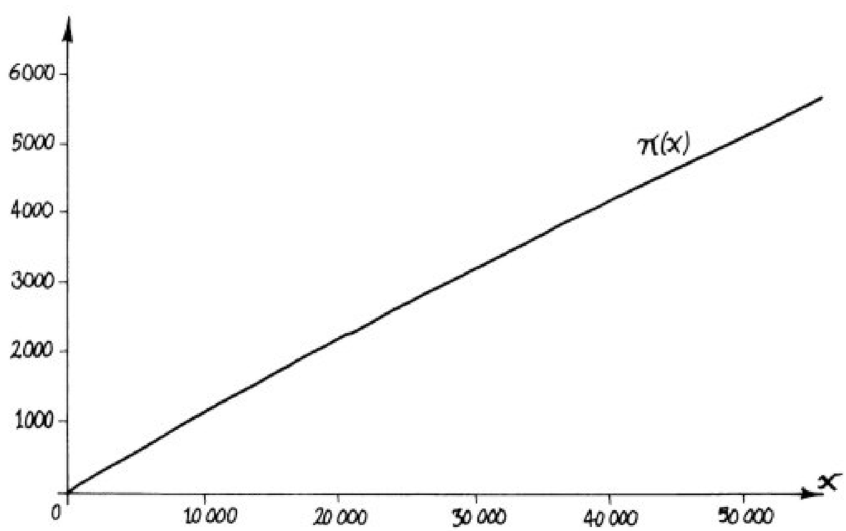
The distribution of prime numbers locally appears to be very irregular. The primes seem to be "randomly" distributed. For example, for what reason does the interval [10000000, 10000100] contain exactly the following two primes

$$10000019, \ 10000079$$

The following graph indicates, how irregular the values of the function $\pi(x)$ are distributed in the interval $[1, 100]$:



Globally, however, the picture is totally different:



Don Zagier writes in his essay on *the first* 50 *million prime numbers*: "For me, the smoothness with which this curve climbs is one of the most astonishing facts in mathematics." Gauß studied as a 15-year old boy prime number tables for hours. He already conjectured in 1792, that

$$\pi(x) \sim \int_2^x \frac{dt}{\log(t)}.$$

The integral is called *logarithmic integral* and we have

$$\mathrm{li}(x) = \int_2^x \frac{dt}{\log(t)} = \frac{x}{\log(x)} + O\left(\frac{x}{\log^2(x)}\right).$$

Legendre published the conjecture $\pi(x) \sim x/\log(x)$ in 1798. A proof was given much later, namely in 1896 by Hadamard, and independently by de la Vallée Poussin:

THEOREM 1.6.1 (Prime number theorem). *For $x \to \infty$ we have*

$$\pi(x) \sim \frac{x}{\log(x)} \sim \mathrm{li}(x).$$

Note that $\mathrm{li}(x)$ is a much better approximation to $\pi(x)$ than $\frac{x}{\log(x)}$. The difficulty in the proof is to show that the limit exists. Once the limit

$$\lim_{x\to\infty} \pi(x)\frac{\log(x)}{x}$$

is known to exist, it is easy to show that it must be equal to 1. This has been proved by Chebyshev. For the existence of the limit there is no easy proof. Most proofs are based on complex anaylysis. We will present a short proof, based on an article by Don Zagier [**17**] at the end of the lecture.

Here is a table for some values of $\pi(x)$, in comparison to $[\mathrm{li}(x)]$:

x	$\pi(x)$	$[\mathrm{li}(x)]$
10^1	4	5
10^2	25	29
10^3	168	177
10^4	1229	1245
10^5	9592	9629
10^6	78498	78627
10^7	664579	664917
10^8	5761455	5762208
10^9	50847534	50849234
10^{10}	455052511	455055614
10^{11}	4118054813	4118066400
10^{12}	37607912018	37607950280

The table suggests that we should have $\mathrm{li}(x) - \pi(x) > 0$ for all $x \geq 1$. However, this is false.

THEOREM 1.6.2 (Littlewood, 1914). *There are infinitely many values of $x \geq 1$ with $\pi(x) > \mathrm{li}(x)$.*

A *Skewes number* is the smallest natural number x for which we have $\pi(x) > \mathrm{li}(x)$. The upper bounds for a Skewes number are gigantic. In 2011, Stoll and Demichel gave the best upper bound so far, namely

$$1.39716 \times 10^{316}.$$

Riemann found an *exact* formula for $\pi(x)$. He considered the function

$$R(x) = 1 + \sum_{n=1}^{\infty} \frac{1}{n\zeta(n+1)} \frac{\log^n(x)}{n!}.$$

This series converges rather quickly and Riemann found that

$$\pi(x) = R(x) - \sum_{\rho \in \mathcal{N}} R(x^\rho),$$

where $\mathcal{N}$ is the set of roots of $\zeta(s)$. This sum does not converge absolutely, so that one has to sum it up in the right order, i.e., by the absolute value of $\mathrm{Im}(\rho)$. The formula was proven by von Mangoldt in 1895. Let us repeat the statement of the Riemann hypothesis from the introduction.

Conjecture 1.6.3 (Riemann). *The Riemann zeta function $\zeta(s)$ has no root in the strip $0 < \mathrm{Re}(s) < 1$, which does not lie on the critical line $\mathrm{Re}(s) = \frac{1}{2}$.*

The "first" zeros of $\zeta(s)$ are

$$\begin{aligned}
\rho_1 &= \frac{1}{2} + 14.134725i \\
\rho_2 &= \frac{1}{2} + 21.022040i \\
\rho_3 &= \frac{1}{2} + 25.010856i \\
\rho_4 &= \frac{1}{2} + 30.424878i \\
\rho_5 &= \frac{1}{2} + 32.9345057i.
\end{aligned}$$

Of course, with ρ also $\overline{\rho}$ is a root. The Riemann hypothesis is equivalent to the statement, that for every $\varepsilon > 0$ there exists a constant $C_\varepsilon > 0$ such that

$$|\pi(x) - \mathrm{li}(x)| \le C_\varepsilon x^{\frac{1}{2}+\varepsilon}.$$

We come back now to more elementary results concerning the prime number distribution.

PROPOSITION 1.6.4. *For every $n \geq 2$ we have*

$$\frac{1}{6} \cdot \frac{n}{\log(n)} < \pi(n) < 6 \cdot \frac{n}{\log(n)}.$$

PROOF. We start with the inequalities

$$2^n \leq \binom{2n}{n} < 4^n. \tag{1.4}$$

The left side follows by induction, and the right side as follows:

$$4^n = (1+1)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} > \binom{2n}{n}.$$

Taking logarithms in (1.4) yields

$$n\log(2) \leq \log(2n!) - 2\log(n!) < n\log(4). \tag{1.5}$$

We already know from the proof of Corollary 1.3.27, that

$$\log(n!) = \sum_{p \leq n} \alpha(p)\log(p), \quad \text{with} \quad \alpha(p) = \sum_{m=1}^{\left[\frac{\log(n)}{\log(p)}\right]} \left[\frac{n}{p^m}\right].$$

Hence we have

$$\log(2n!) - 2\log(n!) = \sum_{p \leq 2n} \sum_{m=1}^{\left[\frac{\log(2n)}{\log(p)}\right]} \left(\left[\frac{2n}{p^m}\right] - 2\left[\frac{n}{p^m}\right]\right)\log(p). \tag{1.6}$$

Since $[2x] - 2[x]$ equals 0 or 1, we obtain the left side of (1.5):

$$n\log(2) \leq \sum_{p \leq 2n} \left(\sum_{m=1}^{\left[\frac{\log(2n)}{\log(p)}\right]} 1\right)\log(p) \leq \sum_{p \leq 2n} \log(2n) = \pi(2n)\log(2n).$$

Hence we obtain

$$\pi(2n) \geq \frac{n\log(2)}{\log(2n)} = \frac{\log(2)}{2}\frac{2n}{\log(2n)} > \frac{1}{4}\frac{2n}{\log(2n)}, \tag{1.7}$$

since $\log(2) > \frac{1}{2}$. Since $\frac{2n}{2n+1} \geq \frac{2}{3}$ we have

$$\begin{aligned}\pi(2n+1) \geq \pi(2n) &> \frac{1}{4}\frac{2n}{\log(2n)} > \frac{1}{4} \cdot \frac{2n}{2n+1} \cdot \frac{2n+1}{\log(2n+1)} \\ &\geq \frac{1}{6}\frac{2n+1}{\log(2n+1)}.\end{aligned}$$

Together with (1.7) we obtain, for $n \geq 2$,

$$\pi(n) > \frac{1}{6}\frac{n}{\log(n)}.$$

This shows the first inequality. For the second one we will estimate the right side of (1.6) by the term with $m = 1$. Since the terms for $m > 1$ are non-negative, we have

$$\begin{aligned}\log(2n!) - 2\log(n!) &\geq \sum_{p \leq 2n} \left(\left[\frac{2n}{p}\right] - 2\left[\frac{n}{p}\right] \right) \log(p) \\ &\geq \sum_{n<p\leq 2n} \log(p).\end{aligned}$$

In the last step we have used that $[2n/p] - 2[n/p] = 1$ for all prime numbers p in the interval $n < p \leq 2n$. Writing $\theta(x) = \sum_{p \leq x} \log(p)$, the right side of (1.5) gives

$$\theta(2n) - \theta(n) = \sum_{n<p\leq 2n} \log(p) < n\log(4).$$

For $n = 2^r$ this yields

$$\theta(2^{r+1}) - \theta(2^r) < 2^r \log(4) = 2^{r+1}\log(2).$$

Summing up for $r = 0, 1, \ldots, k$ we have a telescoping sum on the left side, so that

$$\theta(2^{k+1}) < 2^{k+2}\log(2).$$

Choosing k in such a way that $2^k \leq n < 2^{k+1}$, we obtain

$$\theta(n) \leq \theta(2^{k+1}) < 2^{k+2}\log(2) \leq 4n\log(2).$$

Then, with $0 < \alpha < 1$,

$$(\pi(n) - \pi(n^\alpha)) \cdot \log(n^\alpha) < \sum_{n^\alpha < p \le n} \log(p) \le \theta(n) < 4n\log(2),$$

and hence

$$\begin{aligned}\pi(n) &< \frac{4n\log(2)}{\alpha\log(n)} + \pi(n^\alpha) < \frac{4n\log(2)}{\alpha\log(n)} + n^\alpha \\ &= \frac{n}{\log(n)}\left(\frac{4\log(2)}{\alpha} + \frac{\log(n)}{n^{1-\alpha}}\right).\end{aligned}$$

For $\alpha = \frac{2}{3}$ this implies, considering $\frac{\log(n)}{n^{1/3}} \le \frac{3}{e}$,

$$\pi(n) < \frac{n}{\log(n)}\left(6\log(2) + \frac{3}{e}\right) < 6\frac{n}{\log(n)}.$$

Here we have used that for $c > 0$ and $x \ge 1$ the function $f(x) = x^{-c}\log(x)$ takes its maximum value at $x = e^{1/c}$,i.e.,

$$\frac{\log(n)}{n^c} \le \frac{1}{ce}.$$

$\square$

One can obtain much better estimates of this type for $\pi(x)$, but this requires also much more work, and the estimates will not hold for all $x \ge 1$, but only for all $x \ge x_0$ with a certain value of x_0. For example, we have

$$\frac{x}{\log(x)} < \pi(x)$$

for all $x \ge 17$, and

$$\pi(x) < \frac{219}{200} \cdot \frac{x}{\log(x)}$$

for all $x \ge 284860$. In this context we mention the results by Dusart [**5**]:

THEOREM 1.6.5 (Dusart). *We have the following inequalities:*

$$\pi(x) \ge \frac{x}{\log(x)}\left(1 + \frac{1}{\log(x)} + \frac{1.8}{\log^2(x)}\right), \quad x \ge 32299,$$

$$\pi(x) \le \frac{x}{\log(x)}\left(1 + \frac{1}{\log(x)} + \frac{2.51}{\log^2(x)}\right) < 1.094 \cdot \frac{x}{\log(x)}, \quad x \ge 355991,$$

$$\pi(x) \le \frac{x}{\log(x)}\left(1 + \frac{1.2762}{\log(x)}\right), \quad x > 1.$$

A nice application of such estimates on $\pi(x)$ is the proof of *Bertrand's postulate*: for every $n \geq 1$, the interval $(n, 2n]$ contains at least one prime. This has first been proved by Chebyshev in 1852.

REMARK 1.6.6. Chebyshev's explicit inequalities, see [**4**],

$$c_1 \frac{x}{\log(x)} < \pi(x) < c_2 \frac{x}{\log(x)},$$

$$\begin{aligned} c_1 &= \log(2^{1/2} 3^{1/3} 5^{1/5} 30^{-1/30}) \approx 0.921292022934, \\ c_2 &= \frac{6}{5} c_1 \approx 1.10555042752 \end{aligned}$$

are often cited to hold for all $x \geq 30$, which is wrong. In fact, they only hold for all $x \geq 96098$, see [**7**], [**8**]. This was also used in Khare's work on Serre's modularity conjecture [**11**], and had to be corrected.

Such inequalities also yield estimates for the n-th prime p_n, see [**6**]:

THEOREM 1.6.7 (Dusart). *The n-th prime p_n satisfies*

$$\begin{aligned} n(\log(n) + \log(\log(n)) - 1) &< p_n, \quad n \geq 1, \\ n(\log(n) + \log(\log(n))) &> p_n, \quad n \geq 6. \end{aligned}$$

In the proof of Proposition 1.6.4 we have used the Chebyshev θ-function:

$$\theta(x) = \sum_{p \leq x} \log(p).$$

It is related to the prime number theorem as follows.

THEOREM 1.6.8. *The prime number theorem $\pi(x) \sim \frac{x}{\log(x)}$ is equivalent to the statement $\theta(x) \sim x$, for $x \to \infty$.*

PROOF. With

$$a(n) = \begin{cases} 1 & \text{if } n \in \mathbb{P}, \\ 0 & \text{otherwise,} \end{cases}$$

we have

$$\sum_{1 < n \leq x} a(n) = \sum_{p \leq x} 1 = \pi(x), \qquad \sum_{1 < n \leq x} a(n) \log(n) = \sum_{p \leq x} \log(p) = \theta(x).$$

By Abel's summation formula, with $f(x) = \log(x)$ and $y = 1$ we obtain, for $x \geq 2$,

$$\theta(x) = \sum_{1 < n \leq x} a(n) \log(n) = \pi(x) \log(x) - \pi(1) \log(1) - \int_1^x \frac{\pi(t)}{t} \, dt.$$

Since $\pi(t) = 0$ for $t < 2$, it follows that

$$\theta(x) = \pi(x)\log(x) - \int_2^x \frac{\pi(t)}{t}\,dt. \tag{1.8}$$

With $b(n) = a(n)\log(n)$ we have

$$\sum_{\frac{3}{2}<n\le x} \frac{b(n)}{\log(n)} = \pi(x), \quad \sum_{n\le x} b(n) = \theta(x).$$

With $f(x) = \frac{1}{\log(x)}$ and $y = \frac{3}{2}$ we obtain from Proposition 1.4.2

$$\pi(x) = \frac{\theta(x)}{\log(x)} - \frac{\theta(\frac{3}{2})}{\log(\frac{3}{2})} + \int_{\frac{3}{2}}^x \frac{\theta(t)}{t\log^2(t)}\,dt.$$

Since $\theta(t) = 0$ for $t < 2$, we obtain

$$\pi(x) = \frac{\theta(x)}{\log(x)} + \int_2^x \frac{\theta(t)}{t\log^2(t)}\,dt. \tag{1.9}$$

Now we assume that $\pi(x) \sim \frac{x}{\log(x)}$ for $x \to \infty$. Then we rewrite (1.8) and (1.9) as follows:

$$\frac{\theta(x)}{x} - \frac{\pi(x)\log(x)}{x} = -\frac{1}{x}\int_2^x \frac{\pi(t)}{t}\,dt \tag{1.10}$$

$$\frac{\pi(x)\log(x)}{x} - \frac{\theta(x)}{x} = \frac{\log(x)}{x}\int_2^x \frac{\theta(t)}{t\log^2(t)}\,dt. \tag{1.11}$$

To show $\theta(x) \sim x$, we need to show, by (1.10), that

$$\lim_{x\to\infty} \frac{1}{x}\int_2^x \frac{\pi(t)}{t}\,dt = 0.$$

By assumption we have $\pi(t)/t = O(1/\log(t))$ for $t \ge 2$, so that

$$\frac{1}{x}\int_2^x \frac{\pi(t)}{t}\,dt = O\left(\frac{1}{x}\int_2^x \frac{dt}{\log(t)}\right).$$

On the other hand we have

$$\begin{aligned}\int_2^x \frac{dt}{\log(t)} &= \int_2^{\sqrt{x}} \frac{dt}{\log(t)} + \int_{\sqrt{x}}^x \frac{dt}{\log(t)} \le \frac{1}{\log(2)}\int_2^{\sqrt{x}} dt + \frac{1}{\log(\sqrt{x})}\int_{\sqrt{x}}^x dt \\ &\le \frac{\sqrt{x}}{\log(2)} + \frac{x-\sqrt{x}}{\log(\sqrt{x})}.\end{aligned}$$

This shows the claim.

The proof of the converse direction goes the same way, by using (1.11). The assumption is that $\lim_{x\to\infty} \theta(x)/x = 1$. To conclude the prime number theorem from it, one has to show that

$$\lim_{x\to\infty} \frac{\log(x)}{x} \int_2^x \frac{\theta(t)dt}{t\log^2(t)} = 0.$$

But by assumption we have $\theta(t) = O(t)$, so that

$$\frac{\log(x)}{x} \int_2^x \frac{\theta(t)dt}{t\log^2(t)} = O\left(\frac{\log(x)}{x} \int_2^x \frac{dt}{\log^2(t)}\right)$$

Because of

$$\int_2^x \frac{dt}{\log^2(t)} \le \frac{\sqrt{x}}{\log^2(2)} + \frac{x-\sqrt{x}}{\log^2(\sqrt{x})},$$

we obtain the claim. □

Remark 1.6.9. Another statement which is equivalent to the prime number theorem, is $\psi(x) \sim x$, where

$$\psi(x) = \sum_{n\le x} \Lambda(x).$$

CHAPTER 2

Periodic arithmetic functions and Gauß sums

Many important arithmetic functions are periodic. One example is given by Dirichlet characters.

Definition 2.0.1. Let k be a positive integer. An arithmetic function f is called *periodic* with period k, if

$$f(n+k) = f(n)$$

for all $n \in \mathbb{N}$. The smallest positive period of f is called *fundamental period*.

If k is a period of f, so is mk for all $m \in \mathbb{N}$. To give an example, $f(n) = \gcd(n,k)$ is periodic, because of $\gcd(n+k,k) = \gcd(n,k)$. Furthermore the function

$$f(n) = \zeta_k^n = \exp\left(\frac{2\pi i n}{k}\right)$$

is a periodic arithmetic function.

2.1. Dirichlet characters

The general definition of a character on a group is as follows.

Definition 2.1.1. A *character* f on a group G is a map $f\colon G \to \mathbb{C}$ satisfying $f(ab) = fa)f(b)$ for all $a, b \in G$.

We also require that there exists a $c \in G$ with $f(c) \neq 0$. Then we can conclude from $f(c)f(e) = f(c)$ that $f(e) = 1$, where e is the neutral element of G. If G is finite, then $a^n = e$ for all $a \in G$, and hence

$$f(a)^n = f(a^n) = f(e) = 1.$$

Hence, for $|G| = n$, the values of a character are n-th roots of unity. How many different characters has a finite group? We always have the *principal character*, given by $f(a) = 1$ for all $a \in G$. In case that G is *abelian*, the result is as follows, see Theorem 6.8 in [**1**]:

Theorem 2.1.2. *A finite abelian group of order n has exactly n distinct characters.*

In this case it is not difficult to show that the n characters f_i on G form a group $\widehat{G}$, with multiplication

$$(f_i f_j)(a) = f_i(a) f_j(a)$$

and neutral element $f_1(a) = 1$. The inverse of f_i is the reciprocal $\frac{1}{f_i}$. We have

$$f(a)\overline{f(a)} = |f(a)| = 1,$$
$$\overline{f(a)} = \frac{1}{f(a)} = f(a^{-1}),$$

so the inverse is the complex conjugate, and this is again a character on G. The group $\widehat{G}$ is isomorphic to G. We have the following orthogonality relations for characters, see Theorem 6.12 in [**1**]:

Theorem 2.1.3. *Let $G = \{a_1, \dots, a_n\}$ be a finite abelian group and $\widehat{G} = \{f_1, \dots, f_n\}$. Then we have*

$$\sum_{\ell=1}^{n} \overline{f}_\ell(a_i) f_\ell(a_j) = n\delta_{ij} = \begin{cases} n & \text{if } a_i = a_j, \\ 0 & \text{otherwise.} \end{cases}$$

A particular interesting finite abelian group in this context is the group of units of the ring $\mathbb{Z}/k\mathbb{Z}$, i.e., the group $G = (\mathbb{Z}/k\mathbb{Z})^\times$. It has $\varphi(k)$ elements. Let us denote the class of an integer n in G by $\widehat{n}$.

Definition 2.1.4 (Dirichlet character). Let f be a character on $G = (\mathbb{Z}/k\mathbb{Z})^\times$. Define an arithmetic function $\chi = \chi_f$ on $\mathbb{Z}$ by

$$\chi(n) = \begin{cases} f(\widehat{n}) & \text{if } \gcd(n, k) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

This function χ is called a *Dirichlet character modulo k*.

The principal character χ_1 is defined by $\chi_1(n) = 1$ for n with $\gcd(n, k) = 1$, and $\chi_1(n) = 0$ otherwise.

Proposition 2.1.5. *A Dirichlet character modulo k is strongly multiplicative and periodic with period k. For all n, m we have*

$$\chi(nm) = \chi(n)\chi(m)$$
$$\chi(n + k) = \chi(n).$$

There are $\varphi(k)$ distinct Dirichlet characters modulo k. Conversely, every arithmetic function χ, satisfying these two conditions, and $\chi(n) = 0$ for n with $\gcd(n, k) > 1$, is a Dirichlet character modulo k.

Proof. Let $G = (\mathbb{Z}/k\mathbb{Z})^\times$. Since $\widehat{G}$ is isomorphic to G, we have exactly $\varphi(k)$ characters f for G and $\varphi(k)$ Dirichlet characters χ_f modulo k. For $\gcd(n, k) = \gcd(m, k) = 1$ we have

$$\chi(nm) = f(\widehat{n}\widehat{m}) = f(\widehat{n})f(\widehat{m}) = \chi(n)\chi(m).$$

In case of $\gcd(n, k) > 1$ or $\gcd(m, k) > 1$ we have $\gcd(nm, k) > 1$. Then $\chi(nm) = 0 = \chi(n)\chi(m)$. The character is periodic, since $a \equiv b(k)$ implies that $\gcd(a, k) = \gcd(b, k)$. The converse statement is true, because any such function f with $f(\widehat{n}) = \chi(n)$ for $\gcd(n, k) = 1$ must be a character on G, and hence a Dirichlet character modulo k. $\square$

For $k = 1, 2$ we only have the principal character $\chi(1)$ with $\chi(1) = 1$. In fact, because of $\varphi(k) = 1$ the group is trivial.

EXAMPLE 2.1.6. *For $k = 3$ there are exactly two Dirichlet characters modulo k, namely*

n	1	2	3
$\chi_1(n)$	1	1	0
$\chi_2(n)$	1	-1	0

Indeed, we have $\chi(n)^{\varphi(k)} = 1$ for $\gcd(n, k) = 1$. Since $k = 3$, this means $\chi(1)^2 = \chi(2)^2 = 1$. We have $\chi(1) = 1$ and $\chi(2) = \pm 1$. The first choice of the sign leads to χ_1, the second one to χ_2.

For $k = 4$ we have again two Dirichlet characters χ_1, χ_2. Both have value 1 at $n = 1$, value 0 at $n = 2, 4$, and $\chi_1(3) = 1$, $\chi_2(3) = -1$.

EXAMPLE 2.1.7. *For $k = 5$ we have four Dirichlet characters modulo k, namely*

n	1	2	3	4	5
$\chi_1(n)$	1	1	1	1	0
$\chi_2(n)$	1	-1	-1	1	0
$\chi_3(n)$	1	i	$-i$	-1	0
$\chi_4(n)$	1	$-i$	i	-1	0

Since $\varphi(5) = 4$, the possible values of $\chi(n)$ are $1, -1, i, -i$, for $\gcd(n, 5) = 1$. Hence there are four possibilities for $\chi(2)$. They determine the table, because $\chi(3) = \overline{\chi(2)}$, because of $\chi(2)\chi(3) = \chi(6) = \chi(1) = 1$ and $\chi(4) = \chi(2)^2$, $\chi(5) = 0$.

THEOREM 2.1.8. *Let $\chi_1, \chi_2, \ldots, \chi_{\varphi(k)}$ be the $\varphi(k)$ distinct Dirichlet characters modulo k, and $\gcd(n, k) = 1$. Then we have for all $m \in \mathbb{Z}$:*

$$\sum_{r=1}^{\varphi(k)} \chi_r(m)\overline{\chi}_r(n) = \begin{cases} \varphi(k) & \text{if } n \equiv m \mod k, \\ 0 & \text{otherwise.} \end{cases}$$

PROOF. For $\gcd(m, k) = 1$ this follows from Theorem 2.1.3 with $G = (\mathbb{Z}/k\mathbb{Z})^\times$. For $\gcd(m, k) > 1$, every term in the sum is zero, with $n \not\equiv m \mod k$. $\square$

We will need certain estimates on sums involving Dirichlet characters, e.g., for Dirichlet's theorem on primes in arithmetic progressions.

THEOREM 2.1.9. *Let χ be a Dirichlet character modulo k with $\chi \neq \chi_1$, and f be a non-negative real function, which has a continuous, negative derivation $f'(x)$ for all $x \geq x_0$. Then we have, for all $y \geq x \geq x_0$,*

$$\sum_{x < n \leq y} \chi(n) f(n) = O(f(x)). \tag{2.1}$$

If, in addition $\lim_{x\to\infty} f(x) = 0$*, then the infinite series* $\sum_{n=1}^{\infty} \chi(n)f(n)$ *converges, and for* $x \geq x_0$*, we have*

$$\sum_{n\leq x} \chi(n)f(n) = \sum_{n=1}^{\infty} \chi(n)f(n) + O(f(x)). \tag{2.2}$$

Proof. Let $A(x) = \sum_{n\leq x} \chi(n)$. Since $\chi \neq \chi_1$, there exists an ℓ with $\gcd(\ell, k) = 1$ and $\chi(\ell) \neq 1$. Hence we have

$$\sum_{n=1}^{k} \chi(n) = \sum_{n=1}^{k} \chi(\ell n) = \chi(\ell) \sum_{n=1}^{k} \chi(n),$$

and $(\chi(\ell) - 1)A(k) = 0$, i.e., $A(k) = 0$:

$$\sum_{n=1}^{k} \chi(n) = 0, \quad \chi \neq \chi_1. \tag{2.3}$$

Since χ is periodic, $A(k) = A(2k) = A(3k) = \cdots = 0$. Furthermore

$$|A(x)| = \left| \sum_{n\leq x} \chi(n) \right| < \sum_{\substack{j=1 \\ (n,k)=1}}^{k} 1 = \varphi(k),$$

where we have used $|\chi(n)| \leq 1$ and $\chi(n) = 0$ for $\gcd(n, k) > 1$. Thus we have $A(x) = O(1)$. This we will use for applying Abel's summation formula 1.4.2, with x and y interchanged:

$$\begin{aligned}
\sum_{x<n\leq y} \chi(n)f(n) &= A(y)f(y) - A(x)f(x) - \int_x^y A(t)f'(t)dt \\
&= O(f(y)) + O(f(x)) + O\left(\int_x^y -f'(t)dt\right) \\
&= O(f(x)).
\end{aligned}$$

This shows (2.1). If $f(x) \to 0$ for $x \to \infty$, then this implies that the series

$$\sum_{n=1}^{\infty} \chi(n)f(n)$$

converges by Cauchy's criterion. To show (2.2) we consider

$$\sum_{n=1}^{\infty} \chi(n)f(n) = \sum_{n\leq x} \chi(n)f(n) + \lim_{y\to\infty} \left(\sum_{x<n\leq y} \chi(n)f(n) \right).$$

The limit on the right side is independent of y, and tends to $O(f(x))$, as we have shown. $\square$

Corollary 2.1.10. *Let $\chi \neq \chi_1$ be a Dirichlet character modulo k and $x \geq 1$. Then we have*

$$\sum_{n\leq x} \frac{\chi(n)}{n} = \sum_{n=1}^{\infty} \frac{\chi(n)}{n} + O\left(\frac{1}{x}\right), \tag{2.4}$$

$$\sum_{n\leq x} \frac{\chi(n)\log(n)}{n} = \sum_{n=1}^{\infty} \frac{\chi(n)\log(n)}{n} + O\left(\frac{\log(x)}{x}\right), \tag{2.5}$$

$$\sum_{n\leq x} \frac{\chi(n)}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{\chi(n)}{\sqrt{n}} + O\left(\frac{1}{\sqrt{x}}\right). \tag{2.6}$$

Proof. We apply the above theorem with $f(x) = 1/x$, respectively with $f(x) = \log(x)/x$ and $f(x) = 1/\sqrt{x}$. Then the claims follow by (2.2). $\square$

Definition 2.1.11. The *L-series* of χ is defined by

$$L(s, \chi) = \sum_{n=1}^{\infty} \chi(n) n^{-s}, \quad \mathrm{Re}(s) = \sigma > 1.$$

This series converges for $\sigma > 1$. The above corollary then shows that the following series also converge for $s = 1$:

$$L(1, \chi) = \sum_{n=1}^{\infty} \chi(n) n^{-1}, \quad \chi \neq \chi_1,$$

$$L'(1, \chi) = -\sum_{n=1}^{\infty} \chi(n) \log(n) n^{-1}, \quad \chi \neq \chi_1.$$

Remark 2.1.12. We will show later that $L(1, \chi) \neq 0$ for $\chi \neq \chi_1$. This is a crucial result in the proof of Dirichlet's theorem on primes in arithmetic progressions.

2.2. Finite Fourier series

Let $\zeta_k = \exp\left(\frac{2\pi i}{k}\right)$ be a primitive k-th root of unity. Then, for $m \in \mathbb{Z}$, the function $f(n) = \zeta_k^{nm}$ is periodic modulo k. Every finite linear combination

$$\sum_{m\in\mathbb{Z}} c(m) e^{\frac{2\pi i n m}{k}}$$

is again periodic modulo k. Such sums are called *finite Fourier series*. An important special case are geometric sums.

LEMMA 2.2.1. *Let $k \geq 1$ be fixed. Then we have*

$$\sum_{m=0}^{k-1} e^{\frac{2\pi i n m}{k}} = \begin{cases} 0 & \text{if } k \nmid n, \\ k & \text{if } k \mid n. \end{cases}$$

PROOF. With $x = \exp\left(\frac{2\pi i n}{k}\right)$ we have

$$\sum_{m=0}^{k-1} e^{\frac{2\pi i n m}{k}} = \sum_{m=0}^{k-1} x^m = \begin{cases} \frac{x^k-1}{x-1} & \text{if } x \neq 1, \\ k & \text{if } x = 1. \end{cases}$$

Because of $x^k = 1$ this gives the claim, since $x = 1$ if and only if $k \mid n$. □

Our aim is to show that every periodic arithmetic function has a finite Fourier expansion. For this we need the Lagrange interpolation theorem.

THEOREM 2.2.2 (Lagrange's interpolation theorem). *Let $z_0, z_1, \ldots, z_{k-1}$ be distinct complex numbers and $w_0, w_1, \ldots, w_{k-1}$ arbitrary complex numbers. Then there exists a unique polynomial $P(z)$ of degree $r \leq k-1$ such that $P(z_m) = w_m$ for $m = 0, 1, \ldots, k-1$.*

PROOF. Let $A(z) = (z - z_0)(z - z_1)\cdots(z - z_{k-1})$ and

$$A_m(z) = \frac{A(z)}{z - z_m} = (z - z_0)\cdots\widehat{(z - z_m)}\cdots(z - z_{k-1}).$$

Then $A_m(z)$ is a polynomial of degree $k-1$, with $A_m(z_m) \neq 0$, but $A_m(z_j) = 0$ for $j \neq m$. Hence $(A_m(z_m))^{-1}A_m(z)$ is a polynomial of degree $k-1$ in z, which is zero for each $z = z_j$, $j \neq m$, and one for $z = z_m$. Now let

$$P(z) = \sum_{m=0}^{k-1} w_m \frac{A_m(z)}{A_m(z_m)}.$$

Its degree, as a linear combination of polynomials of degree $k-1$, has a degree $r \leq k-1$ and satisfies $P(z_j) = w_j$ for all j. We claim that this polynomial is unique. If $Q(z)$ is another such polynomial, then $P(z) - Q(z)$ has degree $r \leq k-1$, but k distinct zeros $z_0, z_1, \ldots, z_{k-1}$. Consequently, $P(z) - Q(z)$ is the zero polynomial and hence $P(z) = Q(z)$. □

For the next result we choose the z_j as k-th roots of unity.

PROPOSITION 2.2.3. *For given complex numbers $w_0, w_1, \ldots, w_{k-1}$ there exist k uniquely determined complex numbers $a_0, a_1, \ldots, a_{k-1}$ with*

$$w_m = \sum_{n=0}^{k-1} a_n e^{2\pi inm/k}, \quad m = 0, 1, \ldots, k-1. \tag{2.7}$$

The coefficients a_n here are given by

$$a_n = \frac{1}{k} \sum_{m=0}^{k-1} w_m e^{-2\pi inm/k}, \quad n = 0, 1, \ldots, k-1. \tag{2.8}$$

PROOF. Let $z_m = \exp(2\pi im/k)$. Then the numbers $z_0, z_1, \ldots z_{k-1}$ are distinct, so that by Theorem 2.2.2 there is a uniquely determined polynomial

$$P(z) = \sum_{n=0}^{k-1} a_n z^n$$

with $P(z_m) = w_m$ for $m = 0, 1, \ldots, k-1$. This also determines the a_n, and (2.7) follows. To prove (2.8), multiply (2.7) by $\exp(-2\pi imr/k)$, $0 \le m, r < k$ and sum up over $m = 0$ to $k-1$:

$$\begin{aligned}\sum_{m=0}^{k-1} w_m e^{-2\pi imr/k} &= \sum_{n=0}^{k-1} a_n \sum_{m=0}^{k-1} e^{2\pi i(n-r)m/k} \\ &= ka_r.\end{aligned}$$

By Lemma 2.2.1 the sum on the right is zero for $k \nmid (n-r)$. For $k \mid (n-r)$ we have $n = r$, because $|n-r| \le k-1$, and the sum is equal to k. This implies (2.8). □

PROPOSITION 2.2.4. *Let f be an arithmetic function, which is periodic modulo k. Then there exists a uniquely determined arithmetic function g, which is periodic modulo k, with*

$$f(m) = \sum_{n=0}^{k-1} g(n) e^{2\pi imn/k}.$$

Here g is given by

$$g(n) = \frac{1}{k} \sum_{m=0}^{k-1} f(m) e^{-2\pi imn/k}.$$

PROOF. This follows from Proposition 2.2.3 with $w_m = f(m)$ and $g(m) = a_m$. One extends g to $\mathbb{N}$ by periodic continuation modulo k. □

REMARK 2.2.5. Since f and g are periodic, the summation need not be from 0 to $k-1$, but can be extended to an arbitrary complete residue system modulo k. We denote this by $\sum_{n \bmod k}$.

2.3. Ramanujan sums

Let us also write (m,n) for $\gcd(n,m)$, if no confusion arises.

DEFINITION 2.3.1. Let $k,n \in \mathbb{N}$. Then a sum of the form

$$c_k(n) = \sum_{\substack{m \bmod k \\ (m,k)=1}} e^{2\pi i mn/k}$$

is called a *Ramanujan sum.*

We need a lemma:

LEMMA 2.3.2. *Let f be a function, which is defined on $\mathbb{Q} \cap [0,1]$, and let*

$$F(n) = \sum_{m=1}^{n} f\left(\frac{m}{n}\right), \quad G(n) = \sum_{\substack{m=1 \\ (m,n)=1}}^{n} f\left(\frac{m}{n}\right).$$

Then we have $\mu * F = G$.

PROOF. We know that $\sum_{d|(m,n)} \mu(d) = 0$ for $(m,n) > 1$. Hence we have

$$\begin{aligned}(\mu * F)(n) &= \sum_{d|n} \mu(d) F\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d) \left(f\left(\frac{d}{n}\right) + f\left(\frac{2d}{n}\right) + \cdots + f\left(\frac{n}{n}\right)\right) \\ &= \sum_{m=1}^{n} \left(f\left(\frac{m}{n}\right) \sum_{d|(m,n)} \mu(d)\right) = \sum_{\substack{m=1 \\ (m,n)=1}}^{n} f\left(\frac{m}{n}\right) = G(n).\end{aligned}$$

□

PROPOSITION 2.3.3. *Let $k \in \mathbb{N}$ and p be a prime number. Then we have*

$$c_k(1) = \sum_{\substack{m=1 \\ (m,k)=1}}^{k} \left(e^{2\pi i/k}\right)^m = \mu(k),$$

$$c_p(1) = \sum_{m=1}^{p} \left(e^{2\pi i/p}\right)^m = \mu(p) = -1.$$

PROOF. Let $\zeta_k = \exp\left(\frac{2\pi i}{k}\right)$ be a primitive k-th root of unity. Then all powers ζ_k^m for $(m,k) = 1$ are primitive roots of unity. By Lemma 2.2.1, the sum of *all* k-th roots of unity equals zero,

$$\sum_{m=1}^{k} \zeta_k^m = 1 + \zeta_k + \zeta_k^2 + \cdots + \zeta_k^{k-1} = 0, \quad k > 1,$$

and 1 for $k=1$. Choosing $f(m/k)=\zeta_k^m$ in Lemma 2.3.2 we obtain

$$F(k)=\sum_{m=1}^{k}\zeta_k^m = I(k)=\left\lfloor \frac{1}{k}\right\rfloor,$$

$$G(k)=\sum_{\substack{m=1\\(m,k)=1}}^{k}\zeta_k^m=(\mu * F)(k)=(\mu * I)(k)=\mu(k).$$

□

Hence for $n=1$ the Ramanujan sum coincides with the μ-function. For $k\mid n$ it coincides with the φ-function. Indeed, each term of the sum equals one, and we have $\varphi(k)$ terms:

$$c_k(n)=\varphi(k),\quad k\mid n.$$

Ramanujan showed that all $c_k(n)$ are integral, see Corollary 2.3.7.

Definition 2.3.4. Let f and g be arithmetic functions, and $k,n\in\mathbb{N}$. Then define

$$s_k(n)=\sum_{d\mid(n,k)} f(d)g\left(\frac{k}{d}\right).$$

This sum looks similar to the Dirichlet product $f*g$, but one sums over only certain divisors d of n. Since n is only a factor in (n,k), we have $s_k(n+k)=s_k(n)$. Hence the $s_k(n)$ are periodic functions modulo k, which thus have a finite Fourier expansion.

Proposition 2.3.5. *The sums $s_k(n)$ have a finite Fourier expansion*

$$s_k(n)=\sum_{m \bmod k} a_k(m)e^{2\pi imn/k},\quad \textit{with} \tag{2.9}$$

$$a_k(m)=\sum_{d\mid(m,k)} g(d)f\left(\frac{k}{d}\right)\frac{d}{k}. \tag{2.10}$$

Proof. By Proposition 2.2.4 the $a_k(m)$ are given by

$$\begin{aligned}a_k(m)&=\frac{1}{k}\sum_{n \bmod k} s_k(n)e^{-2\pi inm/k}=\frac{1}{k}\sum_{n=1}^{k}\sum_{\substack{d\mid n\\ d\mid k}} f(d)g\left(\frac{k}{d}\right)e^{-2\pi inm/k}\\&=\frac{1}{k}\sum_{d\mid k} f(d)g\left(\frac{k}{d}\right)\sum_{c=1}^{k/d}e^{-2\pi icdm/k}\\&=\frac{1}{k}\sum_{d\mid k} f\left(\frac{k}{d}\right)g(d)\sum_{c=1}^{d}e^{-2\pi icm/d}.\end{aligned}$$

Here we have written $n = cd$. Then the index c, for fixed d, runs from 1 to k/d. Thereafter one can replace in the sum over c, the d by k/d. By Lemma 2.2.1 the sum equals zero, of not $d \mid m$, where it equals d:

$$a_k(m) = \frac{1}{k} \sum_{\substack{d \mid k \\ d \mid m}} f\left(\frac{k}{d}\right) g(d) d = \frac{1}{k} \sum_{d \mid (m,k)} g(d) f\left(\frac{k}{d}\right) d.$$

□

With 2.3.5 we can derive the following formula for $c_k(n)$.

PROPOSITION 2.3.6. *We have*

$$c_k(n) = \sum_{d \mid (n,k)} \mu\left(\frac{k}{d}\right) d.$$

PROOF. We choose $f(k) = k$ and $g(k) = \mu(k)$ in Proposition 2.3.5. Then (2.9) says:

$$\begin{aligned}
\sum_{d \mid (n,k)} d\mu\left(\frac{k}{d}\right) &= \sum_{m \bmod k} a_k(m) e^{2\pi i m n/k} \\
&= \sum_{m \bmod k} \sum_{d \mid (m,k)} \mu(d) e^{2\pi i m n/k} \\
&= \sum_{\substack{m \bmod k \\ (m,k)=1}} e^{2\pi i m n/k} = c_k(n).
\end{aligned}$$

In the last line we have used (2.10), which says

$$a_k(m) = \sum_{d \mid (m,k)} \mu(d) = \begin{cases} 1 & \text{if } (m,k) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

□

For $n = 1$ we again obtain $c_k(1) = \mu(k)$. The result also shows that the sums $c_k(n)$ arise as a special case form the $s_k(n)$, namely for $f = \mathrm{id}$ and $g = \mu$. We note two further consequences. The first one is immediate, the second one follows from the proposition for $k \mid n$.

COROLLARY 2.3.7. *We have* $c_k(n) \in \mathbb{Z}$, *and*

$$\sum_{d \mid k} d\mu\left(\frac{k}{d}\right) = \varphi(k).$$

The sums $s_k(n)$, and hence also the sums $c_k(n)$, have the following multiplicative properties. For a proof see [**1**]:

PROPOSITION 2.3.8. *We have the following identities:*

$$\begin{aligned}
s_{mk}(ab) &= s_m(a) s_k(b), \quad & (a,k) = (b,m) = (m,k) = 1, \\
s_m(ab) &= s_m(a), & (b,m) = 1, \\
s_{mk}(a) &= s_m(a) g(k), & (a,k) = (m,k) = 1.
\end{aligned}$$

We will prove the following result for the sums $s_k(n)$.

Proposition 2.3.9. *Let f, g, h be arithmetic functions with $g(k) = \mu(k)h(k)$, f be strongly multiplicative, and h be multiplicative. Assume that $f(p) \neq 0$ and $f(p) - h(p) \neq 0$ for all primes p. Let $F = f * g$. Then we have*

$$s_k(n) = \frac{F(k) g\left(\frac{k}{(n,k)}\right)}{F\left(\frac{k}{(n,k)}\right)}.$$

Proof. Let $N = k/(n,k)$. Then we have

$$\begin{aligned} F(k) &= \sum_{d|k} f(d)\mu\left(\frac{k}{d}\right) h\left(\frac{k}{d}\right) = \sum_{d|k} f\left(\frac{k}{d}\right)\mu(d)h(d) \\ &= f(k)\sum_{d|k}\mu(d)\frac{h(d)}{f(d)} \\ &= f(k)\prod_{p|k}\left(1 - \frac{h(p)}{f(p)}\right). \end{aligned}$$

In the last step we have used Proposition 1.3.12. By assumption the factors in the product are nonzero. Writing $a = (n,k)$, we have $k = aN$ and

$$\begin{aligned} s_k(n) &= \sum_{d|a} f(d)\mu\left(\frac{k}{d}\right) h\left(\frac{k}{d}\right) = \sum_{d|a} f(d)\mu\left(\frac{aN}{d}\right) h\left(\frac{aN}{d}\right) \\ &= \sum_{d|a} f\left(\frac{a}{d}\right)\mu(Nd)h(Nd) \\ &= \mu(N)h(N) \sum_{\substack{d|a \\ (N,d)=1}} f\left(\frac{a}{d}\right)\mu(d)h(d) \\ &= f(a)\mu(N)h(N) \sum_{\substack{d|a \\ (N,d)=1}} \mu(d)\frac{h(d)}{f(d)} \end{aligned}$$

$$\begin{aligned} s_k(n) &= f(a)\mu(N)h(N)\prod_{\substack{p|a \\ p\nmid N}}\left(1 - \frac{h(p)}{f(p)}\right) \\ &= f(a)\mu(N)h(N)\frac{\prod_{p|aN}\left(1 - \frac{h(p)}{f(p)}\right)}{\prod_{p|N}\left(1 - \frac{h(p)}{f(p)}\right)}. \end{aligned}$$

Here we have used Proposition 1.3.12, and that $\mu(Nd) = \mu(N)\mu(d)$ for $(N, d) = 1$, and $\mu(Nd) = 0$ otherwise. Substituting the above formula for $F(k)$ in it, and noting that $f(a)f(N) = f(k)$, $\mu h = g$, we obtain

$$\begin{aligned} s_k(n) &= f(a)\mu(N)h(N)\frac{F(k)}{f(k)}\frac{f(N)}{F(N)} \\ &= \frac{F(k)\mu(N)h(N)}{F(N)} = \frac{F(k)g(N)}{F(N)}. \end{aligned}$$

$\square$

Corollary 2.3.10. *We have*

$$c_k(n) = \frac{\varphi(k)\mu\left(\frac{k}{(n,k)}\right)}{\varphi\left(\frac{k}{(n,k)}\right)}.$$

Proof. Choose $f(n) = n$ and $h(n) = 1$. Then we have $g(n) = \mu(n)\varepsilon(n) = \mu(n)$ and $F = f * g = \mathrm{id} * \mu = \varphi$. For this special case it follows that $s_k(n) = c_k(n)$, and the assumptions of the above proposition are satisfied. $\square$

Remark 2.3.11. The Dirichlet series to $c_k(n)$ is given by

$$\sum_{n=1}^{\infty} \frac{c_k(n)}{n^s} = \zeta(s) \sum_{d \mid k} \mu\left(\frac{k}{d}\right) d^{1-s},$$

valid for all $k > 1$, and all complex $s \neq 1$ with $\sigma > 0$.

2.4. Gauß sums

DEFINITION 2.4.1. Let χ be a Dirichlet character modulo k. Then

$$G(n,\chi) = \sum_{m=1}^{k} \chi(m) e^{2\pi inm/k}$$

is called a *Gauß sum* associated to the character χ.

For the principal character $\chi = \chi_1$, the Gauß sum reduces to the Ramanujan sum. Indeed, because of $\chi_1(m) = 1$ for $(m,k) = 1$ and $\chi_1(m) = 0$ otherwise, we have

$$G(n,\chi_1) = \sum_{\substack{m=1 \\ (m,k)=1}}^{k} e^{2\pi inm/k} = c_k(n).$$

PROPOSITION 2.4.2. *For $(n,k) = 1$ we have $G(n,\chi) = \overline{\chi}(n)G(1,\chi)$.*

PROOF. If r runs through a complete residue system modulo k, so does nr because of $(n,k) = 1$. We have $\overline{\chi}(n)\chi(n) = |\chi(n)|^2 = 1$ and hence

$$\chi(r) = \overline{\chi}(n)\chi(n)\chi(r) = \overline{\chi}(n)\chi(nr).$$

Substituting this into $G(n,\chi)$, we obtain with $m = nr$

$$\begin{aligned} G(n,\chi) &= \sum_{r \bmod k} \chi(r) e^{2\pi inr/k} = \overline{\chi}(n) \sum_{r \bmod k} \chi(nr) e^{2\pi inr/k} \\ &= \overline{\chi}(n) \sum_{m \bmod k} \chi(m) e^{2\pi im/k} \\ &= \overline{\chi}(n) G(1,\chi). \end{aligned}$$

□

DEFINITION 2.4.3. The Gauß sum $G(n,\chi)$ is called *separable*, if

$$G(n,\chi) = \overline{\chi}(n)G(1,\chi). \tag{2.11}$$

By Theorem 2.4.2, $G(n,\chi)$ is separable for all n with $(n,k) = 1$.

PROPOSITION 2.4.4. *For positive integers n with $(n,k) > 1$ the Gauß sum $G(n,\chi)$ is separable if and only if $G(n,\chi) = 0$.*

PROOF. If $(n,k) = 1$, then $G(n,\chi)$ is separable. For $(n,k) > 1$ we have $\overline{\chi}(n) = 0$, hence $G(n,\chi) = 0$ by (2.11). □

PROPOSITION 2.4.5. *Let χ be a Dirichlet character modulo k, and $G(n,\chi)$ be separable for all $n \geq 1$. Then*

$$|G(1,\chi)|^2 = k. \tag{2.12}$$

PROOF. With (2.11) we have

$$\begin{aligned}
|G(1,\chi)|^2 &= G(1,\chi)\overline{G(1,\chi)} = G(1,\chi)\sum_{m=1}^{k}\overline{\chi}(m)e^{-2\pi i m/k}\\
&= \sum_{m=1}^{k} G(m,\chi)e^{-2\pi i m/k} = \sum_{m=1}^{k}\sum_{r=1}^{k}\chi(r)e^{2\pi i m r/k}e^{-2\pi i m/k}\\
&= \sum_{r=1}^{k}\chi(r)\sum_{m=1}^{k} e^{2\pi i m(r-1)/k} = k\chi(1) = k.
\end{aligned}$$

Here we have used Lemma 2.2.1. The last sum, from $m=1$ to k, equals k for $r=1$, and zero otherwise. We have $k \nmid (r-1)$, if not $r=1$. $\square$

If $(n,k)>1$, we want to know whether or not $G(n,\chi)$ is zero.

PROPOSITION 2.4.6. *Assume that $G(n,\chi)\neq 0$ for some n with $(n,k)>1$. Then there exists a divisor $d\mid k$, $d<k$, so that for all a with $(a,k)=1$ and $a\equiv 1(d)$, we have $\chi(a)=1$.*

PROOF. Let $q=(n,k)$ and $d=k/q$. Then we have $d\mid k$ and $d\neq k$, because $q\neq 1$. Choose some a with $(a,k)=1$ and $a\equiv 1(d)$. We want to show that $\chi(a)=1$. Because of $(a,k)=1$ we can replace in the defining sum for $G(n,\chi)$, the index m by am, i.e.,

$$\begin{aligned}
G(n,\chi) &= \sum_{m \bmod k}\chi(m)e^{2\pi i n m/k} = \sum_{m \bmod k}\chi(am)e^{2\pi i a n m/k}\\
&= \chi(a)\sum_{m \bmod k}\chi(m)e^{2\pi i a n m/k}.
\end{aligned}$$

Because of $a\equiv 1(d)$ and $d=k/q$ we can write $a=1+bk/q$ with $b\in\mathbb{Z}$. Since $q\mid n$, we have

$$\frac{anm}{k} = \frac{nm}{k} + \frac{bknm}{qk} = \frac{nm}{k} + c$$

for some $c\in\mathbb{Z}$. This implies $e^{2\pi i a m n/k} = e^{2\pi i n m/k}$ and thus

$$G(n,\chi) = \chi(a)\sum_{m \bmod k}\chi(m)e^{2\pi i n m/k} = \chi(a)G(n,\chi).$$

Because of $G(n,\chi)\neq 0$ it follows that $\chi(a)=1$. $\square$

The above result motivates the following definition.

DEFINITION 2.4.7. Let χ be a Dirichlet character modulo k. A positive divisor $d\mid k$ is called *induced modulus* for χ, if $\chi(a)=1$ for all a with $(a,k)=1$ and $a\equiv 1(d)$.

Note that $d=k$ itself is an induced modulus for χ.

PROPOSITION 2.4.8. *For a Dirichlet character χ modulo k, $d = 1$ is an induced modulus if and only if $\chi = \chi_1$.*

PROOF. If $\chi = \chi_1$, then $\chi(a) = 1$ holds for all a with $(a, k) = 1$, and the condition $a \equiv 1(1)$ is satisfied for all a. Hence $d = 1$ is an induced modulus for χ_1. Conversely, if $d = 1$ is an induced modulus, then $\chi(a) = 1$ for all a with $(a, k) = 1$, and $\chi(a) = 0$ for $(a, k) > 1$. This means that $\chi = \chi_1$. □

Dirichlet characters having only $d = 1$ as induced modulus, have a special name.

DEFINITION 2.4.9. A Dirichlet character modulo k is called *primitive* modulo k, if it has no induced modulus $d < k$.

Note that χ is primitive modulo k, if for each divisor $d \mid k$, $0 < d < k$ there exists an $a \in \mathbb{N}$ with $a \equiv 1(d)$ and $(a, k) = 1$, such that $\chi(a) \neq 1$.

PROPOSITION 2.4.10. *Let χ be a Dirichlet character modulo p, with p prime and $\chi \neq \chi_1$. Then χ is primitive modulo p.*

PROOF. An induced modulus is a divisor of p, so $d = 1$ or $d = p$. Because of $\chi \neq \chi_1$ and Proposition 2.4.8, $d = 1$ is not an induced modulus. Hence χ has no induced modulus $d < p$. □

We may reformulate the Propositions 2.4.4, 2.4.5, 2.4.6 as follows.

THEOREM 2.4.11. *Let χ be a primitive Dirichlet character modulo k. Then we have*

(*a*) *$G(n, \chi) = 0$ for all n with $(n, k) > 1$.*
(*b*) *$G(n, \chi)$ is separable for all n.*
(*c*) *$|G(1, \chi)|^2 = k$.*

PROOF. Assume that there is an n with $(n, k) > 1$ and $G(n, \chi) \neq 0$. Then χ is not primitive by Proposition 2.4.6, a contradiction. This implies (a).

For $(n, k) = 1$ we know that $G(n, \chi)$ is separable. Because of Proposition 2.4.4 and part (a), $G(n, \chi)$ is also separable for $(n, k) > 1$. This implies (b). Part (c) follows from (b) and Proposition 2.4.5. □

PROPOSITION 2.4.12. *Let d be a positive divisor of k and χ be a Dirichlet character modulo k. Then d is an induced modulus for χ if and only if*

$$\chi(a) = \chi(b) \quad \forall a, b \;\; mit \;\; (a, k) = (b, k) = 1, \; a \equiv b(d). \tag{2.13}$$

PROOF. By (2.13) with $b = 1$ it follows that d is an induced modulus for χ. Conversely, let d be an induced modulus. Let a, b be given with $(a, k) = (b, k) = 1$ and $a \equiv b(d)$. Because of $(a, k) = 1$ there exists an a' with $aa' \equiv 1(k)$, and hence with $aa' \equiv 1(d)$, since $d \mid k$. By assumption it follows that $\chi(aa') = 1$. Because of $a \equiv b(d)$ we have $aa' \equiv ba' \equiv 1(d)$, and thus

$$1 = \chi(aa') = \chi(ba') = \chi(a)\chi(a') = \chi(b)\chi(a').$$

In particular we have $\chi(a') \neq 0$, and hence $\chi(a')^{-1} = \chi(a) = \chi(b)$. □

EXAMPLE 2.4.13. *The following table describes a character modulo 9, where $d = 3$ is an induced modulus.*

n	1	2	3	4	5	6	7	8	9
$\chi(n)$	1	-1	0	1	-1	0	1	-1	0

Here (2.13) is satisfied with $d = 3$, and χ is periodic modulo 3. This character is an extension of the character ψ with $\psi(1) = 1, \psi(2) = -1$ and $\psi(3) = 0$. It is primitive by Proposition 2.4.10.

EXAMPLE 2.4.14. *Let χ be the following Dirichlet character modulo 6:*

n	1	2	3	4	5	6
$\chi(n)$	1	0	0	0	-1	0

The numbers n coprime to to 6 are 1 and 5. Only $n = 1$ is congruent $1 \mod 3$. Hence $d = 3$ is an induced modulus for χ. However, χ is not an extension of some character modulo 3, because none of them satisfies $\chi(2) = 0$.

PROPOSITION 2.4.15. *Let χ be a Dirichlet character modulo k and $d \mid k$, $d > 0$. Then the following statements are equivalent.*

(a) *d is an induced modulus for χ.*
(b) *There is a character ψ modulo d with $\chi(n) = \psi(n)\chi_1(n)$.*

PROOF. Assume (b). Choose a n with $(n, k) = 1$ and $n \equiv 1(d)$. Then $\chi_1(n) = \psi(n) = 1$ by definition. Hence $\chi(n) = 1$ for such n, and d is an induced modulus.
Assume (a). We want to construct a suitable ψ for (b). For $(n, d) > 1$ we put $\psi(n) = 0$. Then $(n, k) > 1$ and $\chi(n) = 0 = \psi(n)\chi_1(n)$. Assume now that $(n, d) = 1$. It is easy to show the following claim:

$$\text{There is some } m \in \mathbb{Z} \text{ with } m \equiv n(d),\ (m, k) = 1. \tag{2.14}$$

Fix such an m, which is unique modulo d and define $\psi(n) = \chi(m)$. This is well-defined, since $\chi(a) = \chi(b)$ for $a \equiv b(d)$ and $(a, k) = (b, k) = 1$. Hence ψ is a character modulo d. It remains to show the above property for $(n, k) = 1$, hence for $(n, d) = 1$. We have $\psi(n) = \chi(m)$ for $m \equiv n(d)$. By (a), $\chi_1(n) = 1$, and by Proposition 2.4.7 we have

$$\chi(n) = \chi(m) = \psi(n) = \psi(n)\chi_1(n).$$

□

DEFINITION 2.4.16. Let χ be a Dirichlet character modulo k. The smallest induced modulus d for χ is called the *conductor* of χ.

PROPOSITION 2.4.17. *Every Dirichlet character χ modulo k can be expressed as a product $\chi(n) = \psi(n)\chi_1(n)$, where ψ is a primitive character modulo d, and d is the conductor of χ.*

PROOF. By Proposition 2.4.15 we have that $\chi(n) = \psi(n)\chi_1(n)$. We only need to show that ψ is primitive. Assume that it is not primitive. Then there is a divisor $q \mid d$ with $q < d$, hence with $q \mid k$, which is an induced modulus for ψ. For n with $n \equiv 1(q)$ and $(n, k) = 1$ we have

$$1 = \psi(1) = \psi(n) = \psi(n)\chi_1(n) = \chi(n),$$

because q is an induced modulus for ψ. Hence q is also an induced modulus for χ. This is a contradiction, since d is the conductor of χ, and hence the *smallest* induced modulus. □

THEOREM 2.4.18. *Let χ be a Dirichlet character modulo k. Then χ is primitive modulo k if and only if the Gauß sum $G(n,\chi)$ is separable for all n.*

PROOF. If χ is primitive, then $G(n,\chi)$ is separable for all n by Theorem 2.4.11. Conversely, by Propositions 2.4.2 and 2.4.4, it is enough to show the following: if χ is not primitive, then we have $G(r,\chi)\neq 0$ for some r with $(r,k)>1$ (for $(r,k)=1$, $G(r,\chi)$ is always separable, and for $(r,k)>1$ it is separable if and only if $G(r,\chi)=0$). So assume that χ is not primitive modulo k, so $k>1$. Denote by $d<k$ the conductor of χ with $r=k/d$. We have $(r,k)>1$, because of $d\mid k$. By Proposition 2.4.17 there exists a primitive character ψ modulo d with $\chi(n)=\psi(n)\chi_1(n)$. Hence we have

$$\begin{aligned}
G(r,\chi) &= \sum_{m \bmod k} \psi(m)\chi_1(m)e^{2\pi irm/k} = \sum_{\substack{m \bmod k\\ (m,k)=1}} \psi(m)e^{2\pi irm/k} \\
&= \sum_{\substack{m \bmod k\\ (m,k)=1}} \psi(m)e^{2\pi im/d} \\
&= \frac{\varphi(k)}{\varphi(d)} \sum_{\substack{m \bmod d\\ (m,d)=1}} \psi(m)e^{2\pi im/d} \\
&= \frac{\varphi(k)}{\varphi(d)} G(1,\psi)\neq 0.
\end{aligned}$$

By Theorem 2.4.11 we have $|G(1,\psi)|^2=d$, since ψ is primitive. Hence $G(r,\chi)\neq 0$, and we are done. For the second last equation we have used the following fact, see [**1**] Theorem 5.33:

Let S be a reduced residue system modulo k and $d\mid k$ a positive divisor (such an S is a set of $\varphi(k)$ numbers coprime to k, which are pairwise incongruent modulo k). Then S is the union of $\varphi(k)/\varphi(d)$ disjoint sets, which are all reduced residue systems modulo d. $\square$

2.5. Pólyas inequality for Dirichlet characters

Since each Dirichlet character χ modulo k is periodic modulo k, it has a finite Fourier expansion

$$\chi(m) = \sum_{n=1}^{k} a_k(n) e^{2\pi imn/k}, \tag{2.15}$$

where the coefficients by Proposition 2.2.4 are given by

$$a_k(n) = \frac{1}{k}\sum_{m=1}^{k} \chi(m) e^{-2\pi imn/k} = \frac{1}{k} G(-n,\chi). \tag{2.16}$$

If χ is primitive, the Fourier expansion (2.15) can be expressed as follows.

PROPOSITION 2.5.1. *The finite Fourier expansion of a primitive Dirichlet character χ modulo k has the form*

$$\chi(n) = \frac{\tau_k(\chi)}{\sqrt{k}} \sum_{m=1}^{k} \overline{\chi}(m) e^{-2\pi imn/k}, \quad \textit{where} \tag{2.17}$$

$$\tau_k(\chi) = \frac{G(1,\chi)}{\sqrt{k}}, \quad |\tau_k(\chi)| = 1. \tag{2.18}$$

PROOF. Since χ is primitive, $G(-n,\chi)$ is separable by Theorem 2.4.18. So we have $G(-n,\chi) = \overline{\chi}(-n)G(1,\chi)$ and (2.16) implies that $a_k(n) = \frac{1}{k}\overline{\chi}(-n)G(1,\chi)$. So we can rewrite (2.15) as follows, with n and m interchanged,

$$\begin{aligned} \chi(n) &= \frac{G(1,\chi)}{k} \sum_{m=1}^{k} \overline{\chi}(-m) e^{2\pi imn/k} \\ &= \frac{G(1,\chi)}{k} \sum_{m=1}^{k} \overline{\chi}(m) e^{-2\pi imn/k}. \end{aligned}$$

This is equivalent to (2.17), and $\tau_k(\chi)$ is normalized in a way that

$$|\tau_k(\chi)| = \frac{|G(1,\chi)|}{\sqrt{k}} = \frac{\sqrt{k}}{\sqrt{k}} = 1.$$

□

We want to prove an inequality by Pólya. For χ modulo k and $x \geq 1$ let

$$A_\chi(x) = \sum_{n \leq x} \chi(n).$$

We have shown in Theorem 2.1.9 that $|A_\chi(x)| < \varphi(k)$ for $\chi \neq \chi_1$. For $\chi = \chi_1$ we have $A_{\chi_1}(k) = \varphi(k)$, hence such an inequality is impossible. If χ is primitive, one can improve the inequality as follows.

THEOREM 2.5.2 (Pólya). *Let χ be a primitive Dirichlet character modulo k and $x \geq 1$. Then we have*

$$|A_\chi(x)| < \sqrt{k}\log(k).$$

PROOF. We write $\chi(n)$ as in (2.17) and sum over all $n \leq x$. Then, with $\chi(k) = 0$,

$$\sum_{n \leq x} \chi(n) = \frac{\tau_k(\chi)}{\sqrt{k}} \sum_{m=1}^{k-1} \overline{\chi}(m) \sum_{n \leq x} e^{-2\pi i n m/k}$$

$$= \frac{\tau_k(\chi)}{\sqrt{k}} \sum_{m=1}^{k-1} \overline{\chi}(m) f(m),$$

where we have written $f(m) = \sum_{n \leq x} e^{-2\pi i n m/k}$. Multiplying this equation with $\sqrt{k}$ and taking absolute values, and using $|\tau_k(\chi)| = 1$, we obtain

$$\sqrt{k} \cdot \left| \sum_{n \leq x} \chi(n) \right| \leq \sum_{m=1}^{k-1} |f(m)|. \tag{2.19}$$

Because of $f(k-m) = \overline{f(m)}$ we have $|f(k-m)| = |f(m)|$. Hence (2.19) implies

$$\sqrt{k} \cdot |A_\chi(x)| \leq \begin{cases} 2\sum_{m<k/2} |f(m)| & \text{if } k \equiv 1(2), \\ 2\sum_{m<k/2} |f(m)| + |f(\frac{k}{2})| & \text{if } k \equiv 0(2). \end{cases} \tag{2.20}$$

Now $f(m)$ is a geometric sum of the form $f(m) = \sum_{n=1}^{r} y^n$ with $r = [x], y = e^{-2\pi i m/k}$. Because of $1 \leq m \leq k-1$ we have $y \neq 1$. With $z = e^{-\pi i m/k}$ we have $y = z^2$ and $z^2 \neq 1$. Hence

$$f(m) = y \cdot \frac{y^r - 1}{y - 1} = z^2 \cdot \frac{z^{2r} - 1}{z^2 - 1} = z^{r+1} \cdot \frac{z^r - z^{-r}}{z - z^{-1}}.$$

Now by $|z| = 1$ we obtain

$$|f(m)| = \left| \frac{z^r - z^{-r}}{z - z^{-1}} \right| = \frac{\left|\sin\left(\frac{\pi r m}{k}\right)\right|}{\left|\sin\left(\frac{\pi m}{k}\right)\right|} \leq \frac{1}{\sin\left(\frac{\pi m}{k}\right)}.$$

Since $\sin(t) \geq 2t/\pi$ for $0 \leq t \leq \pi/2$, it follows, with $t = \pi m/k$,

$$|f(m)| \leq \frac{1}{\frac{2}{\pi}\frac{\pi m}{k}} = \frac{k}{2m}.$$

We use this to study (2.20) for even and odd k. For $k \equiv 1(2)$ we have

$$\sqrt{k} \cdot |A_\chi(x)| \leq k \cdot \sum_{m<k/2} \frac{1}{m} < k \log(k),$$

and for $k \equiv 0(2)$ we have

$$\sqrt{k} \cdot |A_\chi(x)| \leq k \cdot \left(\sum_{m<k/2} \frac{1}{m} + \frac{1}{k} \right) < k \log(k).$$

□

Remark 2.5.3. It is also possible to show that

$$|A_\chi(x)| < \sqrt{k} + \frac{2}{\pi}\sqrt{k}\log(k),$$

which improves the main term by a factor of $2/\pi$.

2.6. Quadratic Gauß sums

We recall the definition of quadratic (non)-residues.

Definition 2.6.1. Let $p \in \mathbb{P}$ and $n \in \mathbb{N}$ with $p \nmid n$. If the congruence $x^2 \equiv n \pmod p$ is solvable, then n is called a *quadratic residue modulo* p. Otherwise n is called a *quadratic non-residue modulo* p.

Example 2.6.2. *The QR's modulo* 11 *are given by* $\{1, 3, 4, 5, 9\}$*, and the QNR's modulo* 11 *are given by* $\{2, 6, 7, 8, 10\}$.

To see this, we can take a residue system modulo 11, e.g., $\{0, 1, \dots, 10\}$. Then we compute all squares modulo 11. This yields exactly the residues ($1^2 \equiv 1, 2^2 \equiv 4, 3^2 \equiv 9, 4^2 \equiv 5$ etc.). Because of $11 \nmid n$, we have to exclude $n = 0$. Both sets have the same cardinality. This is no coincidence.

Proposition 2.6.3. *Let $p > 2$ be prime. Then each reduced residue system modulo p contains exactly $(p-1)/2$ quadratic residues, and exactly $(p-1)/2$ quadratic non-residues.*

Proof. Let $S = \{1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2\}$. These numbers are all incongruent modulo p. In fact, if $x^2 \equiv y^2 \pmod p$ with $1 \le x, y \le (p-1)/2$, then $(x-y)(x+y) \equiv 0 \pmod p$. Because of $1 < x+y < p$, we have $x - y \equiv 0 \pmod p$ and hence $x = y$. Every quadratic residue modulo p is congruent to exactly one of the numbers in S because of $(p-k)^2 \equiv k^2 \pmod p$. And the set S has $\frac{p-1}{2}$ elements, so we are done. □

Definition 2.6.4. Let $p > 2$ be a prime, and $n \in \mathbb{Z}$. The *Legendre symbol* is defined as

$$\left(\frac{n}{p}\right) = \begin{cases} +1 & \text{if } n \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } n \text{ is a quadratic non-residue modulo } p, \\ 0 & \text{if } n \equiv 0 \bmod p. \end{cases}$$

Theorem 2.6.5 (Euler). *Let $p > 2$ be a prime. Then we have for all $n \ge 1$*

$$\left(\frac{n}{p}\right) \equiv n^{(p-1)/2} \pmod p.$$

For a proof see [**1**], Theorem 9.2.

Corollary 2.6.6. *The Legendre symbol $\chi(n) = (n \mid p)$ is a strongly multiplicative function.*

Proof. For $p \mid m$ or $p \mid n$ we have $p \mid nm$, so that both sides are zero. For $p \nmid n, p \nmid m$ we have

$$\left(\frac{nm}{p}\right) \equiv (nm)^{(p-1)/2} = n^{(p-1)/2} m^{(p-1)/2} \equiv \left(\frac{n}{p}\right)\left(\frac{m}{p}\right) \pmod p.$$

The Legendre symbols can only have the values 1 or -1. Thus the difference

$$\left(\frac{nm}{p}\right) - \left(\frac{n}{p}\right)\left(\frac{m}{p}\right)$$

equals 0, 2 or -2. Since it is divisible by $p > 2$, it is zero. □

REMARK 2.6.7. Since the Legendre symbol is strongly multiplicative, periodic modulo p and vanishes for $p \mid n$, it is a Dirichlet character χ modulo p, satisfying $\chi^2 = \chi_1$. Such a character is called *quadratic*.

Gauß proved in 1796 the following famous result.

THEOREM 2.6.8 (Quadratic reciprocity law). *For odd distinct primes p and q we have*

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2},$$

$$\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8},$$

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.$$

So we have $(-1 \mid p) = 1$, if $p \equiv 1(4)$, and $(2 \mid p) = 1$, if $p \equiv \pm 1(8)$. The third part says that we have $(p \mid q) = (q \mid p)$, except for $p \equiv q \equiv 3(4)$, in which case we have $(p \mid q) = -(q \mid p)$. This yields an recursive algorithm to determine, whether or not a prime p is a QR or not.

EXAMPLE 2.6.9. $p = 1997$ *is a quadratic non-residue modulo* $q = 1999$.

Note that 1997 and 1999 are twin primes. As such they are never both congruent to 3 modulo 4. Hence $(1997 \mid 1999) = (1999 \mid 1997) = (2 \mid 1997)$. For twin primes $p, p+2$ we have in general

$$\left(\frac{p}{p+2}\right) = \left(\frac{p+2}{p}\right) = \left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}.$$

In our case, $\left(\frac{2}{1997}\right) = -1$, since $1997 \equiv -3 \pmod 8$.

There are many proofs for the quadratic reciprocity law. Some of them use *quadratic Gauß sums*

$$G(n,\chi) = \sum_{r \bmod p} \chi(r) e^{2\pi i n r/p}, \quad \chi(r) = \left(\frac{r}{p}\right).$$

Since p is prime, the quadratic character $\chi(r) = (r \mid p)$ is primitive. Hence the Gauß sum is separable, i.e., $G(n,\chi) = (n \mid p) G(1,\chi)$.

DEFINITION 2.6.10. Let χ be the Legendre character modulo p. Then we denote the associated quadratic Gauß sum by

$$g(\chi) = G(1,\chi) = \sum_{r=1}^{p} \left(\frac{r}{p}\right) e^{2\pi i r/p}.$$

We already know that $|g(\chi)| = \sqrt{p}$ for $p > 2$. Now we show a better result.

PROPOSITION 2.6.11. *Let χ be the Legendre character modulo p. Then we have*

$$g(\chi)^2 = \left(\frac{-1}{p}\right) p = (-1)^{(p-1)/2} p.$$

PROOF. We have

$$g(\chi)^2 = \sum_{r=1}^{p-1} \sum_{s=1}^{p-1} \left(\frac{r}{p}\right) \left(\frac{s}{p}\right) e^{2\pi i (r+s)/p}.$$

We can rewrite this. For each pair r, s there is a unique t mod p with $s \equiv tr(p)$. Hence we have

$$\left(\frac{r}{p}\right)\left(\frac{s}{p}\right) = \left(\frac{r}{p}\right)\left(\frac{tr}{p}\right) = \left(\frac{r^2}{p}\right)\left(\frac{t}{p}\right) = \left(\frac{t}{p}\right),$$

so that

$$\begin{aligned} g(\chi)^2 &= \sum_{t=1}^{p-1} \sum_{r=1}^{p-1} \left(\frac{t}{p}\right) e^{2\pi i r(1+t)/p} \\ &= \sum_{t=1}^{p-1} \left(\frac{t}{p}\right) \sum_{r=1}^{p-1} e^{2\pi i r(1+t)/p} \\ &= -\sum_{t=1}^{p-1} \left(\frac{t}{p}\right) + p\left(\frac{p-1}{p}\right) \\ &= \left(\frac{-1}{p}\right) p. \end{aligned}$$

Here we have used that the sum from $r = 1$ to $p-1$ equals -1, if $p \nmid (1+t)$, and p otherwise. The Legendre symbol $(t \mid p)$ is $\frac{p-1}{2}$-times $+1$ and $\frac{p-1}{2}$-times -1, so that we have $\sum_{t=1}^{p-1} \left(\frac{t}{p}\right) = 0$. $\square$

So we know that the value of $g(\chi)$ is $\pm\sqrt{p}$, if $p \equiv 1(4)$, and $\pm i\sqrt{p}$, if $p \equiv 3(4)$. However, the sign is surprisingly hard to determine. Gauß wrote in 1801 in his diary, that the correct sign must be $+1$, independently of p. But he needed four years to come up with a proof. On September 3-rd of 1805 he wrote: *Wie der Blitz einschlägt, hat sich das Rätsel gelöst ! – Finally, two days ago, I succeeded – not on account of my hard efforts, but by the grace of the Lord. Like a sudden flash of lightning, the riddle was solved.*

THEOREM 2.6.12 (Gauß). *Let $\chi(r) = \left(\frac{r}{p}\right)$. Then we have*

$$g(\chi) = \begin{cases} \sqrt{p} & \text{if } p \equiv 1(4), \\ i\sqrt{p} & \text{if } p \equiv 3(4). \end{cases}$$

PROOF. Let $\zeta = e^{2\pi i/p}$ be a p-th root of unity. Then $1, \zeta, \zeta^2, \ldots, \zeta^{p-1}$ are the roots of the polynomial $x^p - 1$ over $\mathbb{C}$. First we will prove the following identity:

$$\prod_{k=1}^{(p-1)/2} \left(\zeta^{2k-1} - \zeta^{-(2k-1)}\right)^2 = (-1)^{(p-1)/2} p. \tag{2.21}$$

Indeed, we have

$$x^p - 1 = (x-1)\prod_{j=1}^{p-1}(x - \zeta^j),$$
$$\frac{x^p - 1}{x - 1} = 1 + x + x^2 + \cdots x^{p-1} = \prod_{j=1}^{p-1}(x - \zeta^j).$$

In the second equation we may take $x = 1$, which gives

$$p = \prod_{\substack{r \bmod p \\ r \neq 0}} (1 - \zeta^r),$$

where the product may run over any complete residue system modulo p without zero. Choosing the numbers $\pm(4k-2)$ for $k = 1, 2, \ldots \frac{p-1}{2}$, we obtain

$$p = \prod_{k=1}^{(p-1)/2} \left(1 - \zeta^{4k-2}\right) \prod_{k=1}^{(p-1)/2} \left(1 - \zeta^{-(4k-2)}\right)$$
$$= \prod_{k=1}^{(p-1)/2} \left(\zeta^{-(2k-1)} - \zeta^{2k-1}\right) \prod_{k=1}^{(p-1)/2} \left(\zeta^{2k-1} - \zeta^{-(2k-1)}\right)$$
$$= (-1)^{(p-1)/2} \prod_{k=1}^{(p-1)/2} \left(\zeta^{2k-1} - \zeta^{-(2k-1)}\right)^2.$$

This shows (2.21). Taking the square root on both sides yields:

$$\prod_{k=1}^{(p-1)/2} \left(\zeta^{2k-1} - \zeta^{-(2k-1)}\right) = \begin{cases} \sqrt{p} & \text{falls } p \equiv 1(4), \\ i\sqrt{p} & \text{falls } p \equiv 3(4). \end{cases} \tag{2.22}$$

Here we will show that on the left side each factor has the form α or $i\alpha$, with $\alpha > 0$. Hence the positive sign on the right side is correct. We have

$$\begin{aligned}\prod_{k=1}^{(p-1)/2}\left(\zeta^{2k-1}-\zeta^{-(2k-1)}\right) &= \prod_{k=1}^{(p-1)/2} 2i\left(\frac{1}{2i}\left(e^{2\pi i/p}\right)^{2k-1}-\frac{1}{2i}\left(e^{-2\pi i/p}\right)^{2k-1}\right)\\ &= (i)^{(p-1)/2}\prod_{k=1}^{(p-1)/2} 2\sin\left(\frac{(4k-2)\pi}{p}\right).\end{aligned}$$

Here we have $\sin((4k-2)\pi/p) < 0$ für $(p+2)/4 < k \le (p-1)/2$. So the sine product has exactly

$$\frac{p-1}{2}-\left[\frac{p+2}{4}\right] = \begin{cases}\frac{p-1}{4} & \text{if } p \equiv 1(4),\\ \frac{p-3}{4} & \text{if } p \equiv 3(4)\end{cases}$$

negative terms, hence the factor $(-1)^{(p-1)/4}$ or $(-1)^{(p-3)/4}$. On the other hand we have

$$i^{(p-1)/2} = \begin{cases}(-1)^{(p-1)/4} & \text{if } p \equiv 1(4),\\ i(-1)^{(p-3)/4} & \text{if } p \equiv 3(4).\end{cases}$$

Together we obtain, for both cases of p modulo 4, an *even* power of -1 for the product in (2.22), hence always the positive sign. Hence (2.21), (2.22) and Proposition 2.6.11 yield

$$g(\chi)^2 = (-1)^{(p-1)/2}p = \prod_{k=1}^{(p-1)/2}\left(\zeta^{2k-1}-\zeta^{-(2k-1)}\right)^2,$$

$$g(\chi) = \varepsilon\prod_{k=1}^{(p-1)/2}\left(\zeta^{2k-1}-\zeta^{-(2k-1)}\right) = \begin{cases}\varepsilon\sqrt{p} & \text{if } p \equiv 1(4),\\ \varepsilon i\sqrt{p} & \text{if } p \equiv 3(4).\end{cases}$$

with $\varepsilon = \pm 1$.

We need to show that $\varepsilon = 1$. For this, consider the polynomial

$$f(x) = \sum_{j=1}^{p-1}\chi(j)x^j - \varepsilon\prod_{k=1}^{(p-1)/2}\left(x^{2k-1}-x^{p-(2k-1)}\right) \in \mathbb{Z}[x].$$

We have $f(\zeta) = g(\chi) - g(\chi) = 0$ and $f(1) = \sum_{j=1}^{p-1}\chi(j) = 0$. The field extension $\mathbb{Q}(\zeta)/\mathbb{Q}$ has the degree $p-1$, and the minimal polynomial of ζ is $1 + x + \cdots + x^{p-1}$. It divides $f(x)$ because of $f(\zeta) = 0$:

$$1 + x + \cdots + x^p \mid f(x).$$

Because of $f(1) = 0$ we have $x - 1 \mid f(x)$. Since $x - 1$ and $1 + x + \cdots + x^{p-1}$ are coprime, also their product $x^p - 1$ is a divisor of f:

$$x^p - 1 \mid f(x).$$

So we may write $f(x) = (x^p - 1)h(x)$ with some $h \in \mathbb{Z}[x]$. For $x = e^z$ we obtain

$$\begin{aligned} f(e^z) &= (e^{pz} - 1)h(e^z) \\ &= \sum_{j=1}^{p-1} \chi(j)e^{jz} - \varepsilon \prod_{k=1}^{(p-1)/2} \left(e^{(2k-1)z} - e^{(p-(2k-1))z}\right). \end{aligned}$$

Substituting $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$ and comparing the coefficients of $z^{(p-1)/2}$ on both sides, we obtain

$$\frac{pA}{B} = \sum_{j=1}^{p-1} \frac{\chi(j)j^{(p-1)/2}}{\left(\frac{p-1}{2}\right)!} - \varepsilon \prod_{k=1}^{(p-1)/2} (4k - p - 2);$$

here the left side is the coefficient of $z^{(p-1)/2}$ in $(e^{pz} - 1)h(e^z)$, and the right side the coefficient of $z^{(p-1)/2}$ of the other side. We have $A, B \in \mathbb{Z}$ with $p \nmid B$, see [**10**], page 78. Multiplying the above equation by $B\left(\frac{p-1}{2}\right)!$ and considering it modulo p shows that the left side then is congruent to zero. Because of $p \nmid B$ we can cancel the B and obtain

$$\begin{aligned} \sum_{j=1}^{p-1} \chi(j)j^{(p-1)/2} &\equiv \varepsilon \left(\frac{p-1}{2}\right)! \prod_{k=1}^{(p-1)/2} (4k-2) \\ &\equiv \varepsilon \cdot 2 \cdot 4 \cdot 6 \cdots (p-1) \prod_{k=1}^{(p-1)/2} (2k-1) \\ &\equiv \varepsilon(p-1)! \\ &\equiv -\varepsilon. \end{aligned}$$

In the last step we have used Wilson's Theorem. On the other hand we have $j^{(p-1)/2} \equiv \chi(j) \pmod p$ by Theorem 2.6.5, hence we have

$$-\varepsilon \equiv \sum_{j=1}^{p-1} \chi(j)j^{(p-1)/2} \equiv \sum_{j=1}^{p-1} \chi(j)^2 \equiv p - 1 \pmod p,$$

so that $\varepsilon \equiv 1 \pmod p$. Because of $\varepsilon = \pm 1$ we have $\varepsilon = 1$, and we are done. $\square$

2.7. Cubic Gauß sums

Let $\omega = e^{2\pi i/3} = \frac{\sqrt{-3}-1}{2}$ be a primitive third root of unity. Then ω is a root of $x^3 - 1 = (x-1)(x^2+x+1)$ and hence satisfies

$$0 = 1 + \omega + \omega^2 \tag{2.23}$$

$$\overline{\omega} = \omega^{-1} = \omega^2. \tag{2.24}$$

The set $D = \mathbb{Z}[\omega] = \{a + b\omega \mid a, b \in \mathbb{Z}\}$ is a subring of $\mathbb{C}$, since it is closed under multiplication. Indeed, with (2.23) we have

$$\begin{aligned}(a+b\omega)(c+d\omega) &= ac + (ad+bc)\omega + bd\omega^2 \\ &= (ac-bd) + (ad+bc-bd)\omega\end{aligned}$$

This ring also is closed under complex conjugation:

$$\overline{a+b\omega} = a + b\overline{\omega} = a + b\omega^2 = (a-b) - b\omega.$$

Definition 2.7.1. An integral domain R is called *Euclidean*, if there is a function $\lambda \colon R\backslash 0 \to \mathbb{N} \cup 0$ such that for all $a, b \in R$, $b \neq 0$ there are elements $c, d \in R$ with $a = cb + d$, and either $d = 0$ or $\lambda(d) < \lambda(b)$.

Example 2.7.2. *The rings $\mathbb{Z}$, $K[x]$, $\mathbb{Z}[i]$ and the rings of integers $\mathcal{O}_d$ in $\mathbb{Q}(\sqrt{d})$ for $d = -1, -2, -3, -7, -11$ are Euclidean, see section* 1.1.

Indeed, $\mathcal{O}_d = \mathbb{Z}[i]$ for $d = -1$ and $\mathcal{O}_d = \mathbb{Z}[\omega]$ for $d = -3$. Every Euclidean ring is a PID, and every PID is a UFD. The converse implications are not true in general. For example, $\mathbb{Z}[x]$ is a UFD but not a PID. And $\mathcal{O}_{-19}$ is a PID, but not a norm Euclidean ring.

Proposition 2.7.3. *The ring $D = \mathbb{Z}[\omega]$ is Euclidean.*

Proof. Let $\alpha = a + b\omega \in \mathbb{Z}[\omega]$. The norm $N(\alpha)$ is defined by

$$N(\alpha) = \alpha\overline{\alpha} = (a+b\omega)(a+b\overline{\omega}) = a^2 - ab + b^2.$$

Define $\lambda(\alpha) = N(\alpha)$. So let $\alpha, \beta \in \mathbb{Z}[\omega]$ with $\beta \neq 0$. Then we have $\beta\overline{\beta} > 0$ and $\alpha\overline{\beta} \in \mathbb{Z}[\omega]$, and

$$\frac{\alpha}{\beta} = \frac{\alpha\overline{\beta}}{\beta\overline{\beta}} = r + s\omega,$$

with $r, s \in \mathbb{Q}$. Choose $m, n \in \mathbb{Z}$ with $|r-m| \leq 1/2$ and $|s-n| \leq 1/2$. Let $\gamma = m + n\omega$. Then

$$\begin{aligned}\lambda\left(\frac{\alpha}{\beta} - \gamma\right) &= N(r - m + (s-n)\omega) \\ &= (r-m)^2 - (r-m)(s-n) + (s-n)^2 \\ &\leq \frac{1}{4} + \frac{1}{4} + \frac{1}{4} < 1.\end{aligned}$$

With $\delta = \alpha - \gamma\beta$ we have either $\delta = 0$ or

$$\lambda(\delta) = \lambda\left(\beta\left(\frac{\alpha}{\beta} - \gamma\right)\right) = \lambda(\beta)\lambda\left(\frac{\alpha}{\beta} - \gamma\right) < \lambda(\beta).$$

□

Proposition 2.7.4. *The units in the ring $D = \mathbb{Z}[\omega]$ are given by*

$$1, -1, \omega, -\omega, \omega^2, -\omega^2.$$

Proof. We have that $\alpha \in D$ is a unit if and only if $N(\alpha) = 1$. Indeed, if $N(\alpha) = 1$, we have $\alpha\overline{\alpha} = 1$ with $\overline{\alpha} \in D$. Hence α is a unit. Conversely, if α is a unit, then there is a $\beta \in D$ with $\alpha\beta = 1$, hence with $N(\alpha)N(\beta) = N(1) = 1$. This equation has only the solution $N(\alpha) = N(\beta) = 1$ in positive integers. Now let $\alpha = a + b\omega$ be a unit, i.e.,

$$1 = N(\alpha) = a^2 - ab + b^2 = \frac{3b^2 + (2a-b)^2}{4}.$$

The only integer solutions then are $2a - b = \pm 1, b = \pm 1$, or $2a - b = \pm 2, b = 0$. This gives 6 possibilities: $1, -1, \omega, -\omega, -1-\omega, 1+\omega$. Because of $-1-\omega = \omega^2$ and $1+\omega = -\omega^2$ this gives the claim. The unit group is cyclic of order 6. □

Concerning the prime elements of D, note that a rational prime p need not stay prime in D. For $p = 7$ we have

$$7 = 6 - \omega - \omega^2 = (3+\omega)(2-\omega).$$

We have the following result.

Lemma 2.7.5. *Let π be a prime element in D. Then there is a $p \in \mathbb{P}$ with $N(\pi) = p$ or $N(\pi) = p^2$. In the latter case, π and p are associated. In the first case, they are not. If conversely $N(\pi) = p$ with $p \in \mathbb{P}$, then π is prim in D.*

Proof. Since π is not a unit, we have $\pi\overline{\pi} = n > 1$. Hence there is a $p \in \mathbb{P}$ with $p \mid n$ and we have $\pi \mid p$. Writing $p = \pi\gamma$ with $\gamma \in D$ gives $p^2 = N(p) = N(\pi)N(\gamma)$. This equation has only two solutions. Either $N(\pi) = p^2, N(\gamma) = 1$, so that γ is a unit, or $N(\pi) = N(\gamma) = p$. Then π is not associated to any rational prime q. Assume that we have $\pi = \varepsilon q$, with a unit ε. Then

$$p = N(\pi) = N(\varepsilon)N(q) = q^2,$$

which is a contradiction.
Finally suppose that $\pi = \rho\gamma$ is not prime. Then $N(\rho), N(\gamma) > 1$, and $p = N(\pi) = N(\rho)N(\gamma)$. Hence p is not a prime, a contradiction. □

Proposition 2.7.6. *Let p and q be rational primes. Then we have:*

(1) *If $q \equiv 2(3)$, then q is prime in D.*
(2) *If $p \equiv 1(3)$, then $p = N(\pi) = \pi\overline{\pi}$ with a prime element $\pi \in D$.*
(3) *We have $3 = -\omega^2(1-\omega)^2$, where $1-\omega$ is prime in D.*

Proof. To (1): Assume that q is not prime. Then $q = \pi\gamma$ with $N(\pi), N(\gamma) > 1$, and hence $q^2 = N(q) = N(\pi)N(\gamma)$, so that $N(\pi) = q$. Let $\pi = a + b\omega$. Then

$$\begin{aligned} q &= N(\pi) = a^2 - ab + b^2, \\ 4q &= (2a-b)^2 + 3b^2. \end{aligned}$$

This implies $q \equiv (2a-b)^2 \equiv 2 \pmod 3$, which is a contradiction, since 2 is not a quadratic residue modulo 3.

To (2): Using the quadratic reciprocity law we have

$$\begin{aligned}\left(\frac{-3}{p}\right) &= \left(\frac{-1}{p}\right)\left(\frac{3}{p}\right) = (-1)^{(p-1)/2}(-1)^{\frac{p-1}{2}\frac{3-1}{2}}\left(\frac{p}{3}\right)\\ &= \left(\frac{p}{3}\right) = \left(\frac{1}{3}\right) = 1,\end{aligned}$$

since $p \equiv 1(3)$. Hence there exists an $a \in \mathbb{Z}$ with $a^2 \equiv -3(p)$, so that

$$pb = a^2 + 3 = (a + 1 + 2\omega)(a - 1 - 2\omega)$$

for some $b \in \mathbb{Z}$. Since D is a UFD, the following holds. If p is prime in D, then p divides one of the following two factors

$$p \mid (a + 1 + 2\omega) \ \text{ or } \ p \mid (a - 1 - 2\omega).$$

We may assume that $\gamma p = a + 1 + 2\omega$. Then there are $c, d \in \mathbb{Z}$ with $\gamma = c + d\omega$ and

$$cp - a - 1 + \omega(dp - 2) = 0,$$

hence with $dp = 2$. Since $p > 2$, this is a contradiction. Hence p is not prime in D, so that $p = \pi\gamma$ with $N(\pi), N(\gamma) > 1$. Hence we have $p^2 = N(\pi)N(\gamma)$ and $p = N(\pi) = \pi\overline{\pi}$.

To (3): Because of $x^3 - 1 = (x - 1)(x - \omega)(x - \omega^2)$ we have

$$\begin{aligned}x^2 + x + 1 &= (x - \omega)(x - \omega^2),\\ 3 &= (1 - \omega)(1 - \omega^2) = (1 + \omega)(1 - \omega)^2 = -\omega^2(1 - \omega)^2\\ 9 &= N(-\omega^2)N((1 - \omega)^2) = (N(1 - \omega))^2.\end{aligned}$$

This implies $N(1 - \omega) = 3$, so that $1 - \omega$ is prime by Lemma 2.7.5. ☐

Let $\gamma \neq 0$, not a unit. Consider the *residue class ring* $D/\gamma D$. The equivalence relation is given by

$$\alpha \equiv \beta(\gamma) \Longleftrightarrow \gamma \mid (\alpha - \beta).$$

For the following result, see a text on algebraic number theory.

THEOREM 2.7.7. *Let $\mathcal{O}$ be the ring of integers of a number field, and I an ideal in $\mathcal{O}$. Then $\mathcal{O}/I$ is a finite ring with $N(I)$ elements.*

If I is prime then $\mathcal{O}/I$ is a finite field.

COROLLARY 2.7.8. *Let $\pi \in D$ be a prime element. Then $D/\pi D$ is a finite field with $N(\pi)$ elements.*

For a prime element π in D, the multiplicative group of $D/\pi D$ has $N(\pi) - 1$ elements. By Euler's Theorem we have, for $\overline{\alpha} = \alpha \bmod \pi \in D/\pi D$, that

$$\overline{\alpha}^{N(\pi)-1} = \overline{1}.$$

We will show the following result.

LEMMA 2.7.9. *For $\pi \in D$ prime and $\alpha \in D$ with $\pi \nmid \alpha$ we have*

$$\alpha^{N(\pi)-1} \equiv 1 \pmod{\pi}.$$

Moreover, $N(\pi) = 3$ or $N(\pi) \equiv 1(3)$.

PROOF. Let $N(\pi) \neq 3$. Then $\overline{1}, \overline{\omega}, \overline{\omega^2}$ are distinct in $D/\pi D$. Indeed, assume that $\overline{1} = \overline{\omega}$, then $\pi \mid (1-\omega)$. By Proposition 2.7.6, $1-\omega$ is prime, and hence π and $1-\omega$ are associated. Then $N(\pi) = N(1-\omega) = 3$, a contradiction. Hence $U = \{\overline{1}, \overline{\omega}, \overline{\omega^2}\}$ is a subgroup of order 3 in $((D/\pi D)^*, \cdot)$. By Lagrange we have $3 \mid N(\pi) - 1$, hence $N(\pi) \equiv 1(3)$. ☐

LEMMA 2.7.10. *Let $\pi \in D$ be prime, $N(\pi) \neq 3$ and $\alpha \in D$ with $\pi \nmid \alpha$. Then there is a unique $m \in \{0, 1, 2\}$ with*

$$\alpha^{\frac{N(\pi)-1}{3}} \equiv \omega^m \pmod{\pi}. \tag{2.25}$$

PROOF. By Lemma 2.7.9 we have $\pi \mid \alpha^{N(\pi)-1} - 1$. We have

$$\alpha^{N(\pi)-1} - 1 = \left(\alpha^{\frac{N(\pi)-1}{3}} - 1\right)\left(\alpha^{\frac{N(\pi)-1}{3}} - \omega\right)\left(\alpha^{\frac{N(\pi)-1}{3}} - \omega^2\right).$$

Since π is prime and divides the left side, π must divide a factor on the right side. It is easy to see that π divides *exactly* one of the three factors. Assume that π divides two factors, say the first two, then it divides the difference $\pi \mid 1-\omega$. This is a contradiction to $N(\pi) \neq 3$. ☐

Now we are able to define the cubic character $\chi_\pi(\alpha) = \left(\frac{\alpha}{\pi}\right)_3$.

DEFINITION 2.7.11. Let $\pi \in D$ be prime and $N(\pi) \neq 3$. The *cubic character* of α modulo π is defined by

$$\left(\frac{\alpha}{\pi}\right)_3 = 0, \qquad \text{if } \pi \mid \alpha,$$

$$\left(\frac{\alpha}{\pi}\right)_3 \equiv \alpha^{\frac{N(\pi)-1}{3}} \pmod{\pi}, \qquad \text{if } \pi \nmid \alpha,$$

where $\left(\frac{\alpha}{\pi}\right)_3 \in \{1, \omega, \omega^2\}$ in the second case is uniquely determined by 2.7.10.

PROPOSITION 2.7.12. *The cubic character satisfies the following properties:*

(1) $\left(\frac{\alpha}{\pi}\right)_3 = 1$ *if and only if* $x^3 \equiv \alpha \pmod{\pi}$ *is solvable.*

(2) $\left(\frac{\alpha\beta}{\pi}\right)_3 = \left(\frac{\alpha}{\pi}\right)_3 \left(\frac{\beta}{\pi}\right)_3$.

(3) $\alpha \equiv \beta \pmod{\pi}$ *implies* $\left(\frac{\alpha}{\pi}\right)_3 = \left(\frac{\beta}{\pi}\right)_3$.

PROOF. To (1): For a finite field F and an $\alpha \in F^\times$, $x^n = \alpha$ has a solution in $F^\times$ if and only if $\alpha^{(q-1)/d} = 1$, where $|F^\times| = q-1$ and $d = (n, q-1)$. With $F = D/\pi D$, $q = N(\pi)$ and $n = 3$ this says that $x^3 \equiv \alpha \pmod{\pi}$ is solvable if and only if

$$\left(\frac{\alpha}{\pi}\right)_3 = \alpha^{(q-1)/d} = \alpha^{(N(\pi)-1)/3} \equiv 1 \pmod{\pi}$$

where $d = (3, N(\pi)-1) = 3$.

To (2):

$$\left(\frac{\alpha\beta}{\pi}\right)_3 \equiv (\alpha\beta)^{(N(\pi)-1)/3} \equiv \alpha^{(N(\pi)-1)/3}\beta^{(N(\pi)-1)/3} \equiv \left(\frac{\alpha}{\pi}\right)_3 \left(\frac{\beta}{\pi}\right)_3.$$

To (3): $\alpha \equiv \beta \pmod{\pi}$ implies

$$\left(\frac{\alpha}{\pi}\right)_3 \equiv \alpha^{(N(\pi)-1)/3} \equiv \beta^{(N(\pi)-1)/3} \equiv \left(\frac{\beta}{\pi}\right)_3,$$

and hence the claim. □

Note that every element $\alpha \in D^\times$ has six associates. We eliminate the ambiguity as follows.

DEFINITION 2.7.13. Let $\pi \in D$ be prime. The π is called *primary*, if $\pi \equiv 2 \pmod{3}$.

If $\pi = q$ is rational, then π is automatically primary, see Proposition 2.7.6. Otherwise, for $\pi = a + b\omega$ it means that $3 \mid (a - 2 + b\omega)$, i.e.,

$$a \equiv 2(3), \quad b \equiv 0(3).$$

PROPOSITION 2.7.14. *Suppose that for $\pi \in D$ we have $N(\pi) = p \equiv 1(3)$ with a rational prime p. Then π is prime in D and exactly one of it its six associates is primary.*

For a proof see [**10**], Proposition 9.3.5.

EXAMPLE 2.7.15. *The element $\pi = 3 + \omega$ is prime in D, and $-\omega^2(3+\omega) = 2 + 3\omega$ is primary.*

We have $N(\pi) = N(3+\omega) = 7$, hence π is prime. Obviously the five associates $\pi, \omega\pi, -\pi, -\omega\pi, \omega^2\pi$ are not primary, and $-\omega^2(3+\omega)$ is primary.

There also exists a *cubic reciprocity law*, see [**10**], page 114:

THEOREM 2.7.16. *Let π_1, π_2 be primary elements in D with $N(\pi_1) \neq N(\pi_2)$, and both norms different from 3. Then we have*

$$\left(\frac{\pi_2}{\pi_1}\right)_3 = \left(\frac{\pi_1}{\pi_2}\right)_3.$$

The proof involves Jacobi sums associated to a character $\chi_\pi(\alpha) = \left(\frac{\alpha}{\pi}\right)_3$.

DEFINITION 2.7.17. Let χ, ψ be Dirichlet characters modulo $p \in \mathbb{P}$. The *Jacobi sum* associated to χ and ψ is defined by

$$J(\chi, \psi) = \sum_{a \bmod p} \chi(a)\psi(1-a).$$

Jacobi sums can be expressed by Gauß sums in certain cases. If χ, ψ and $\chi\psi$ are non-trivial, then we have

$$J(\chi, \psi) = \frac{g(\chi)g(\psi)}{g(\chi\psi)}.$$

We need the following lemma, see [**10**], page 115.

LEMMA 2.7.18. *Let $\pi \in D$ be primary. Then we have $J(\chi_\pi, \chi_\pi) = \pi$.*

2.8. A conjecture by Kummer

Let χ be a Dirichlet character modulo $p \in \mathbb{P}$ of order n. Since the character group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$, there exists an element of order n if and only if $n \mid p-1$. So fix a $n \geq 2$ and a prime p with $p \equiv 1(n)$. Since the Gauß sum $g(\chi)$ has the absolute value $\sqrt{p}$ for all n, we have

$$|g(\chi)^n| = |g(\chi)|^n = p^{n/2}, \quad n \geq 2. \tag{2.26}$$

On the other hand is the complex number $g(\chi)^n$ an element in $\mathbb{Q}[\zeta_n]$, and the number $g(\chi)$ is an element in $\mathbb{Q}[\zeta_n]\mathbb{Q}[\zeta_p]$, which is n-fold determined. It is not difficult to show that

$$g(\chi)^n = \begin{cases} \chi(-1)pJ(\chi,\chi)J(\chi,\chi^2)\cdots J(\chi,\chi^{n-2}) & \text{if } n \geq 3, \\ \chi(-1)p & \text{if } n = 2. \end{cases} \tag{2.27}$$

Hence the complex number $g(\chi)$ lies on the circle $\{z \in \mathbb{C} \mid |z| = \sqrt{p}\}$ and is determined up to an n-th root of unity.

Kummer asked, which argument this number has, depending on p, running through the primes $p \equiv 1(n)$, for fixed $n \geq 2$. For $n = 2$ we already know the answer. (2.26) and (2.27) show that the number $g(\chi)$ is one of the roots of the polynomial $x^2 - (-1)^{(p-1)/2}p$. For $p \equiv 1(4)$ we have the choices $\sqrt{p}$ or $-\sqrt{p}$. For $p \equiv 3(4)$ we have the choices $i\sqrt{p}$ or $-i\sqrt{p}$. By Theorem 2.6.12 of Gauß we always have the positive sign, independently of $p > 2$.

For $n = 3$, Kummer's question is more difficult. Let π be prime in $D = \mathbb{Z}[\omega]$, with $N(\pi) = p \equiv 1(3)$. Furthermore let $\chi_\pi(\alpha) = \left(\frac{\alpha}{\pi}\right)_3$ and

$$g(\pi) := g(\chi_\pi) = G(1, \chi_\pi)$$

the Gauß sum to χ_π. By (2.26) and (2.27) we have

$$g(\pi)^3 = p\pi, \quad |g(\pi)^3| = p^{3/2}. \tag{2.28}$$

Indeed, (2.27) says that $g(\pi)^3 = \chi_\pi(-1)pJ(\chi_\pi, \chi_\pi)$, and we have

$$\chi_\pi(-1) = \chi_\pi((-1)^3) = (\chi_\pi(-1))^3 = 1,$$

since χ_π is a cubic character. We have $J(\chi_\pi, \chi_\pi) = \pi$ by Lemma 2.7.18. We will normalize $g(\pi)$ as follows.

Definition 2.8.1. The Gauß sum

$$h(\pi) = \frac{g(\pi)}{\sqrt{p}} = \frac{g(\pi)}{\sqrt{N(\pi)}}$$

is called the *normalized cubic Gauß sum*. It lies on the unit circle in $\mathbb{C}$:

$$|h(\pi)^3| = 1. \tag{2.29}$$

We want to study the complex number $g(\pi)$, and the real number $\mathrm{Re}(g(\pi))$.

Lemma 2.8.2. *Let $\gamma = 2\mathrm{Re}(g(\pi)) = g(\pi) + \overline{g(\pi)}$. Then we have*

$$\gamma = \sum_{r=0}^{p-1} \cos\left(\frac{2\pi r^3}{p}\right).$$

PROOF. We have

$$\begin{aligned}\gamma &= g(\chi_\pi) + \chi_\pi(-1)\overline{g(\chi_\pi)} = g(\chi_\pi) + g(\overline{\chi_\pi}) \\ &= g(\chi_\pi) + g(\chi_\pi^2).\end{aligned}$$

We will use, for cubic characters $g(\chi)$, see [**10**],

$$\sum_{r=0}^{p-1} \zeta_p^{r^3} = g(\chi) + g(\chi^2). \tag{2.30}$$

Since γ is real, we obtain

$$\begin{aligned}\gamma = g(\chi_\pi) + g(\chi_\pi^2) &= \sum_{r=0}^{p-1} e^{2\pi i r^3/p} \\ &= \sum_{r=0}^{p-1} \cos\left(\frac{2\pi r^3}{p}\right) + \sum_{r=0}^{p-1} i\sin\left(\frac{2\pi r^3}{p}\right) \\ &= \sum_{r=0}^{p-1} \cos\left(\frac{2\pi r^3}{p}\right).\end{aligned}$$

□

Which possibilities are there for the real number $\gamma = 2\mathrm{Re}(g(\pi))$?

LEMMA 2.8.3. *Let $\pi = a + b\omega \in D$. Then γ is a real root of the polynomial $x^3 - 3px - (2a-b)p$.*

PROOF. By (2.28) we have

$$\begin{aligned}\gamma^3 &= (g(\pi) + \overline{g(\pi)})^3 = g(\pi)^3 + 3g(\pi)\overline{g(\pi)}\left(g(\pi) + \overline{g(\pi)}\right) + \left(\overline{g(\pi)}\right)^3 \\ &= p\pi + 3|g(\pi)|^2\gamma + p\overline{\pi} \\ &= p\pi + 3p\gamma + p\overline{\pi} \\ &= p(a + b\omega + a + b\overline{\omega}) + 3p\gamma \\ &= p(2a-b) + 3p\gamma.\end{aligned}$$

□

The complex number $g(\pi)$ is determined by (2.28) up to a cubic root $1, \omega, \omega^2$. Kummer asked, which cubic root to take, depending on the prime $\pi \in D$ with $N(\pi) = p \equiv 1(3)$. He didn't find any congruence rule on π to determine $g(\pi)$, like it was possible in the case of quadratic Gauß sums. Therefore Kummer computed $g(\pi)$ numerically for many π. For this, he put the primes π into *three classes* C_1, C_2, C_3, according to the conditions for the real number $\gamma = 2\mathrm{Re}(g(\pi))$:

$$\begin{aligned}
&C_1 : -2\sqrt{p} < \gamma < -\sqrt{p}\\
&C_2 : -\sqrt{p} < \gamma < \sqrt{p}\\
&C_3 : \sqrt{p} < \gamma < 2\sqrt{p}
\end{aligned}$$

By Lemma 2.8.3, exactly one of these three cases must occur. The roots of $x^3 - 3px - (2a - b)p$ are $\gamma = 2\mathrm{Re}(g(\pi))$, $2\mathrm{Re}(\omega g(\pi))$ and $2\mathrm{Re}(\omega^2 g(\pi))$. Using $|g(\pi)| = \sqrt{p}$, one sees that each interval $(-2\sqrt{p}, -\sqrt{p})$, $(-\sqrt{p}, \sqrt{p})$ and $(\sqrt{p}, 2\sqrt{p})$ contains exactly one of the three roots of $x^3 - 3px - (2a - b)p$.

Kummer considered, for $j = 1, 2, 3$ the numbers

$$N_j(x) = \#\{\pi \in D, N(\pi) = p \equiv 1(3) \mid N(\pi) \leq x,\ \gamma \in C_j\}.$$

He computed for $x = 500$ the following values:

$$N_1(500) = 7,\ N_2(500) = 14,\ N_3(500) = 24.$$

The ratios $7 : 14 : 24$ are roughly $1 : 2 : 3$. Kummer conjectured, that this will be the asymptotic distribution, for $x \to \infty$. Hasse called this the *Kummer conjecture*. However, in 1953, J. von Neumann and H. H. Goldstine 1953 found the following distribution:

$$N_1(10000) = 138,\ N_2(10000) = 201,\ N_3(10000) = 272.$$

This corresponds more or less to $2 : 3 : 4$. Later E. Lehmer found a relation of $3 : 4 : 5$, so that one begun to suspect, that the distribution is *uniform*. This was proved by Heath-Brown and Patterson in 1978, see [**9**].

THEOREM 2.8.4. *For $\pi \in D$ prime with $\pi \equiv 1(3)$, the numbers $h(\pi)$ are uniformly distributed on the unit circle. There exists an absolute constant $A > 0$, such that we have for all angles $0 \leq \theta_1 \leq \theta_2 \leq 2\pi$*

$$\sum_{\substack{N(\pi)\leq x\\ \theta_1 < \arg(h(\pi)) \leq \theta_2}} 1 = \frac{\theta_2 - \theta_1}{2\pi}\,\mathrm{li}(x) + O\left(x \exp\left(-A\sqrt{\log(x)}\right)\right).$$

Note that the prime number theorem says that

$$\pi(x) = \sum_{p \leq x} 1 = \mathrm{li}(x) + O\left(x \exp\left(-A\sqrt{\log(x)}\right)\right)$$

for some $A > 0$.

REMARK 2.8.5. One also can find an *explicit* formula for the Gauß sum $g(\pi)$ in terms of products of division values of an elliptic function, involving the Weierstrass $\wp$ function, see [**12**].

2.9. Prime numbers in arithmetic progressions

In this section we want to prove Dirichlet's Theorem. It says that each arithmetic progression $a(n) = kn + h$ with $(k, h) = 1$ contains infinitely many prime numbers. For, say, $a(n) = 2n + 1$ this is obvious. For $a(n) = 4n \pm 1$ it is a bit less obvious. However, there still is an elementary proof.

EXAMPLE 2.9.1. *There are infinitely many primes of the form $4n - 1$, respectively $4n + 1$.*

PROOF. Suppose that there are only finitely many primes $\{p_1, \dots, p_r\}$ of the form $4n - 1$. Then consider $N = 4p_1 \cdots p_r - 1$. Not all of the prime factors of N can be congruent 1 modulo 4, because otherwise N would be as well, because of

$$(4a + 1)(4b + 1) = 4(4ab + a + b) + 1.$$

Hence there exists a prime divisor $p \mid N$ with $p \equiv -1(4)$. Obviously, p is different from every p_i, by construction of N. This is a contradiction and we are done.

Suppose that there are only finitely many primes $\{p_1, \dots, p_r\}$ of the form $4n+1$. The consider $N = 4(p_1 \cdots p_r)^2 + 1$. Assume that $p \mid N$. Then

$$4(p_1 \cdots p_r)^2 + 1 \equiv 0(p).$$

Hence -1 is a quadratic residue modulo p, so that

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2} = 1,$$

and hence $p \equiv 1(4)$. This is a contradiction because of $p \neq p_i$. □

It is known that such an elementary proof does not always work. In fact, it only works for $a(n) = kn+h$ if $h^2 \equiv 1(k)$. So it doesn't work for $a(n) = 7n+2$, to give an example. Dirichlet's proof works in all cases:

THEOREM 2.9.2 (Dirichlet). *Let $k > 0$ and $(h, k) = 1$. Then, for all $x > 1$,*

$$\sum_{\substack{p \le x \\ p \equiv h(k)}} \frac{\log(p)}{p} = \frac{1}{\varphi(k)} \log(x) + O(1). \tag{2.31}$$

In particular, there are infinitely many primes $p \equiv h(k)$.

The last statement just follows from (2.31) for $x \to \infty$. The proof is split up into several lemmas. We also recall Proposition 1.3.23. Let f, g be two arithmetic functions. For $h = f * g$ let $H(x) = \sum_{n \le x} h(n)$, $F(x) = \sum_{n \le x} f(n)$ and $G(x) = \sum_{n \le x} g(n)$. Then we have

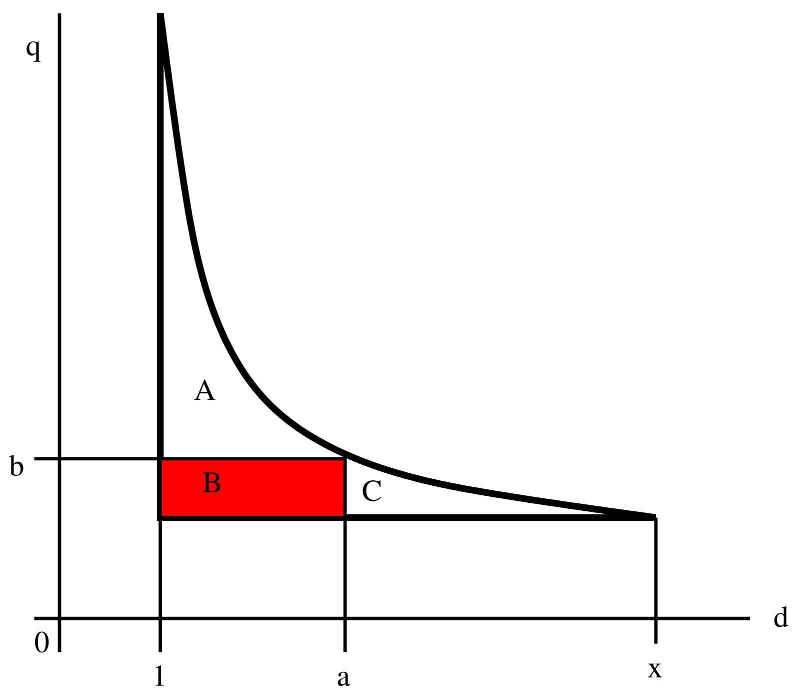
$$\begin{aligned} H(x) &= \sum_{n \le x} f(n) G\left(\frac{x}{n}\right) = \sum_{n \le x} g(n) F\left(\frac{x}{n}\right) \\ &= \sum_{n \le x} \sum_{d \mid n} f(d) g\left(\frac{n}{d}\right) = \sum_{\substack{q,d \\ qd \le x}} f(d) g(q). \end{aligned}$$

With these notations we can generalize the proposition.

PROPOSITION 2.9.3. *For positive real numbers a, b with $ab = x$ we have*

$$\sum_{\substack{q,d\\qd\le x}} f(d)g(q) = \sum_{n\le a} f(n)G\left(\frac{x}{n}\right) + \sum_{n\le b} g(n)F\left(\frac{x}{n}\right) - F(a)G(b).$$

PROOF. Consider the lattice points (q, d) in the picture below:



On the left side of the equation we sum over all lattice points under the hyperbola $qd = x$, i.e., in the region $A \cup B \cup C$. On the right side we split the sum into two parts, namely into a vertical one and a horizontal one. Thus, we sum up there over the lattice points in $A \cup B$, and then over the lattice points in $B \cup C$, and the subtract the sum of those in B, because we have counted them twice:

$$\begin{aligned} H(x) &= \sum_{d\le a}\sum_{q\le x/d} f(d)g(q) + \sum_{q\le b}\sum_{d\le x/q} f(d)g(q) - \sum_{d\le a}\sum_{q\le b} f(d)g(q) \\ &= \sum_{n\le a} f(n)G\left(\frac{x}{n}\right) + \sum_{n\le b} g(n)F\left(\frac{x}{n}\right) - F(a)G(b). \end{aligned}$$

$\square$

LEMMA 2.9.4. *Let χ be a real valued Dirichlet character modulo k, and $A(n) = \sum_{d|n} \chi(d)$. Then we have $A(n) \ge 0$ for all n and $A(n) \ge 1$, if $n = m^2$ is a perfect square.*

PROOF. For prime powers p^α we have

$$A(p^\alpha) = \sum_{t=0}^{\alpha} \chi(p^t) = 1 + \sum_{t=1}^{\alpha} (\chi(p))^t.$$

Since χ is real valued, $\chi(p)$ is equal to $0, 1, -1$. For $\chi(p) = 0$ we have $A(p^\alpha) = 1$; for $\chi(p) = 1$ we have $A(p^\alpha) = \alpha + 1$, and for $\chi(p) = -1$ we have

$$A(p^\alpha) = \begin{cases} 1 & \text{falls } \alpha \equiv 0(2), \\ 0 & \text{falls } \alpha \equiv 1(2). \end{cases}$$

So for α even we always have $A(p^\alpha) \geq 1$. Now let $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$. Since A is multiplicative, we have $A(n) = A(p_1^{\alpha_1}) \cdots A(p_r^{\alpha_r})$. Each factor is non-negative, hence also $A(n) \geq 0$. If n is a square, then all α_i are even, and hence $A(p_i^{\alpha_i}) \geq 1$ for all i, and thus $A(n) \geq 1$. □

PROPOSITION 2.9.5. *Let $\chi \neq \chi_1$ be a real valued character modulo k and*

$$A(n) = \sum_{d|n} \chi(d), \quad B(x) = \sum_{n \leq x} \frac{A(n)}{\sqrt{n}}.$$

Then we have:

(1) $B(x) \to \infty$ *for* $x \to \infty$.

(2) $B(x) = 2\sqrt{x}L(1,\chi) + O(1)$ *for all* $x \geq 1$.

(3) $L(1,\chi) \neq 0$.

PROOF. To (3): If $L(1,\chi) = 0$, then (1) and (2) are contradictory.

Zu (1): By Lemma 2.9.3 we have

$$B(x) = \sum_{n \leq x} \frac{A(n)}{\sqrt{n}} \geq \sum_{\substack{n \leq x \\ n = m^2}} \frac{1}{\sqrt{n}} = \sum_{m \leq x} \frac{1}{m}.$$

This implies

$$\lim_{x \to \infty} B(x) = \sum_{m=1}^{\infty} \frac{1}{m} = \infty.$$

To (2): Applying Proposition 2.9.3 with $a = b = \sqrt{x}$, so $ab = x$, and $f(n) = \frac{\chi(n)}{\sqrt{n}}$, $g(n) = \frac{1}{\sqrt{n}}$ yields:

$$B(x) = \sum_{qd \le x} \frac{\chi(d)}{\sqrt{qd}} = \sum_{n \le \sqrt{x}} \frac{\chi(n)}{\sqrt{n}} G\left(\frac{x}{n}\right) + \sum_{n \le \sqrt{x}} \frac{1}{\sqrt{n}} F\left(\frac{x}{n}\right) - F(\sqrt{x})G(\sqrt{x}).$$

Using Proposition 1.4.4 with $\sigma = 1/2$ and $A = \zeta(1/2)$ yields:

$$\begin{aligned} G(x) = \sum_{n \le x} g(n) = \sum_{n \le x} \frac{1}{\sqrt{n}} &= \frac{x^{1/2}}{1/2} + \zeta\left(\frac{1}{2}\right) + O\left(\frac{1}{\sqrt{x}}\right) \\ &= 2\sqrt{x} + A + O\left(\frac{1}{\sqrt{x}}\right). \end{aligned}$$

Then by (2.6) and $B = \sum_{n=1}^{\infty} \frac{\chi(n)}{\sqrt{n}}$ we have

$$\begin{aligned} F(x) = \sum_{n \le x} \frac{\chi(n)}{\sqrt{n}} &= \sum_{n=1}^{\infty} \frac{\chi(n)}{\sqrt{n}} + O\left(\frac{1}{\sqrt{x}}\right) \\ &= B + O\left(\frac{1}{\sqrt{x}}\right). \end{aligned}$$

Together we obtain $F(\sqrt{x})G(\sqrt{x}) = 2B\sqrt[4]{x} + O(1)$. Substituting this into the above formula for $B(x)$, we obtain

$$\begin{aligned} B(x) = &\sum_{n \le \sqrt{x}} \frac{\chi(n)}{\sqrt{n}} \left(2\sqrt{\frac{x}{n}} + A + O\left(\sqrt{\frac{n}{x}}\right)\right) + \sum_{n \le \sqrt{x}} \frac{1}{\sqrt{n}} \left(B + O\left(\sqrt{\frac{n}{x}}\right)\right) \\ &- 2B\sqrt[4]{x} + O(1). \\ &= 2\sqrt{x} \sum_{n \le \sqrt{x}} \frac{\chi(n)}{n} + A \sum_{n \le \sqrt{x}} \frac{\chi(n)}{\sqrt{n}} + O\left(\frac{1}{\sqrt{x}} \sum_{n \le \sqrt{x}} |\chi(n)|\right) + B \sum_{n \le \sqrt{x}} \frac{1}{\sqrt{n}} \\ &+ O\left(\frac{1}{\sqrt{x}} \sum_{n \le \sqrt{x}} 1\right) - 2Bx^{1/4} + O(1). \end{aligned}$$

Here we have $\sum_{n \le \sqrt{x}} \frac{\chi(n)}{\sqrt{n}} = O(1)$ by Theorem 2.1.9, and

$$\sum_{n \le \sqrt{x}} \frac{1}{\sqrt{n}} = 2x^{1/4} + A + O\left(\frac{1}{x^{1/4}}\right)$$

by Proposition 1.4.4 as above, but with $x^{1/2}$ instead of x. Hence we have

$$B \sum_{n \le \sqrt{x}} \frac{1}{\sqrt{n}} - 2Bx^{1/4} = O(1).$$

Together we obtain

$$\begin{aligned} B(x) &= 2\sqrt{x}\left(\sum_{n\le x} \frac{\chi(n)}{n}\right) + O(1) \\ &= 2\sqrt{x}\left(L(1,\chi) + O\left(\frac{1}{x}\right)\right) + O(1) \\ &= 2\sqrt{x}L(1,\chi) + O(1). \end{aligned}$$

□

Now we can start the proof of Dirichlet's Theorem, i.e., of Theorem 2.9.2. We will need Lemma 2.9.6, Lemma 2.9.7, Lemma 2.9.8 and Theorem 2.9.9. The above theorem says that $L(1,\chi) \ne 0$ for all real valued Dirichlet characters $\chi \ne \chi_1$. The proof uses Lemma 2.9.10 and Lemma 2.9.11. Recall that $L(1,\chi)$ and $L'(1,\chi)$ are given by a convergent series as follows, for $\chi \ne \chi_1$:

$$L(1,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n},$$

$$L'(1,\chi) = -\sum_{n=1}^{\infty} \frac{\chi(n)\log(n)}{n}.$$

Proof of Theorem 2.9.2: Let $k \in \mathbb{N}$ and h coprime to k. Let $\chi_1, \dots, \chi_{\varphi(k)}$ be the Dirichlet characters modulo k. Let us start with the lemmas.

LEMMA 2.9.6. *For $x > 1$ we have*

$$\sum_{\substack{p\le x\\ p\equiv h(k)}} \frac{\log(p)}{p} = \frac{1}{\varphi(k)}\log(x) + \frac{1}{\varphi(k)}\sum_{r=2}^{\varphi(k)} \overline{\chi}_r(h)\sum_{p\le x}\frac{\chi_r(p)\log(p)}{p} + O(1).$$

This will imply Dirichlet's result, as soon as we can show that

$$\sum_{p\le x} \frac{\chi_r(p)\log(p)}{p} = O(1), \quad \chi \ne \chi_1. \tag{2.32}$$

LEMMA 2.9.7. *For $x > 1$ and $\chi \ne \chi_1$ we have*

$$\sum_{p\le x} \frac{\chi_r(p)\log(p)}{p} = -L'(1,\chi)\sum_{n\le x}\frac{\mu(n)\chi(n)}{n} + O(1). \tag{2.33}$$

This lemma will imply (2.32), as soon as we can show that

$$\sum_{n\le x}\frac{\mu(n)\chi(n)}{n}=O(1),\quad \chi\neq\chi_1. \tag{2.34}$$

LEMMA 2.9.8. *For $x>1$ and $\chi\neq\chi_1$ we have*

$$L(1,\chi)\sum_{n\le x}\frac{\mu(n)\chi(n)}{n}=O(1). \tag{2.35}$$

This lemma will imply (2.34), and hence (2.32) and Dirichlet's Theorem, as soon as we can show the following theorem.

THEOREM 2.9.9. *We have $L(1,\chi)\neq 0$ for all $\chi\neq\chi_1$.*

So this theorem certainly is a key result. Let us start proving the lemmas.

Proof of Lemma 2.9.6: We start with Proposition 1.5.2 by Mertens, i.e., with

$$\sum_{p\le x}\frac{\log(p)}{p}=\log(x)+O(1).$$

We want to extract on the left side the terms belonging to $p\equiv h(k)$. This we can do with Theorem 2.1.8, for $(n,k)=1$, namely

$$\sum_{r=1}^{\varphi(k)}\chi_r(m)\overline{\chi}_r(n)=\begin{cases}\varphi(k) & \text{falls } n\equiv m(k),\\ 0 & \text{sonst.}\end{cases}$$

We choose $m=p, n=h$. Noe that $(n,k)=(h,k)=1$. Multiplying the above equation with $\log(p)/p$, and summing over all $p\le x$ yields:

$$\begin{aligned}\varphi(k)\sum_{\substack{p\le x\\ p\equiv h(k)}}\frac{\log(p)}{p}&=\sum_{p\le x}\sum_{r=1}^{\varphi(k)}\chi_r(p)\overline{\chi}_r(h)\frac{\log(p)}{p}\\
&=\overline{\chi}_1(h)\sum_{p\le x}\frac{\chi_1(p)\log(p)}{p}+\sum_{r=2}^{\varphi(k)}\overline{\chi}_r(h)\frac{\chi_r(p)\log(p)}{p}\\
&=\sum_{p\le x}\frac{\log(p)}{p}+\sum_{r=2}^{\varphi(k)}\overline{\chi}_r(h)\frac{\chi_r(p)\log(p)}{p}+O(1)\\
&=\log(x)+\sum_{r=2}^{\varphi(k)}\overline{\chi}_r(h)\frac{\chi_r(p)\log(p)}{p}+O(1).\end{aligned}$$

In the last step we have used Proposition 1.5.2. For the third equation we have used that $\overline{\chi}_1(h) = 1$ and $\chi_1(p) = 1$, if $(p,k) = 1$ and $\chi_1(p) = 0$ otherwise. This gives

$$\begin{aligned}\sum_{p\le x} \frac{\chi_1(p)\log(p)}{p} &= \sum_{\substack{p\le x\\(p,k)=1}} \frac{\log(p)}{p} = \sum_{p\le x}\frac{\log(p)}{p} - \sum_{\substack{p\le x\\p\mid k}}\frac{\log(p)}{p}\\ &= \sum_{p\le x}\frac{\log(p)}{p} + O(1),\end{aligned}$$

because the sum over $p \le x, p \mid k$ is bounded for all $x > 1$. Dividing by $\varphi(k)$, the lemma follows. $\square$

Proof of Lemma 2.9.7: By definition of $\Lambda(n)$ we have

$$\begin{aligned}\sum_{n\le x}\frac{\chi(n)\Lambda(n)}{n} &= \sum_{a=1}^{\infty}\sum_{\substack{p\le x\\p^a\le x}}\frac{\chi(p^a)\log(p)}{p^a}\\ &= \sum_{p\le x}\frac{\chi(p)\log(p)}{p} + \sum_{a=2}^{\infty}\sum_{\substack{p\le x\\p^a\le x}}\frac{\chi(p^a)\log(p)}{p^a}\\ &= \sum_{p\le x}\frac{\chi(p)\log(p)}{p} + O(1),\end{aligned}$$

because the last double sum is bounded, see Remark 1.4.7, by

$$\sum_p \log(p)\sum_{a=2}^{\infty}p^{-a} = \sum_p\frac{\log(p)}{p(p-1)} < \sum_{n=2}^{\infty}\frac{\log(n)}{n(n-1)} \le \log(4).$$

Using $\Lambda = \mu * \log$ of Example 1.3.19, and writing $n = cd$ and noting that χ is multiplicative, we obtain

$$\begin{aligned}\sum_{n\le x}\frac{\chi(n)\Lambda(n)}{n} &= \sum_{n\le x}\frac{\chi(n)}{n}\sum_{d\mid n}\mu(d)\log\left(\frac{n}{d}\right)\\ &= \sum_{d\le x}\frac{\mu(d)\chi(d)}{d}\sum_{c\le x/d}\frac{\chi(c)\log(c)}{c}.\end{aligned}$$

By (2.5) we also have

$$\begin{aligned}\sum_{c\le x/d}\frac{\chi(c)\log(c)}{c} &= \sum_{c=1}^{\infty}\frac{\chi(c)\log(c)}{c}+O\left(\frac{\log(x/d)}{x/d}\right)\\ &= -L'(1,\chi)+O\left(\frac{\log(x/d)}{x/d}\right),\end{aligned}$$

and therefore in the above computation

$$\begin{aligned}\sum_{n\le x}\frac{\chi(n)\Lambda(n)}{n} &= -L'(1,\chi)\sum_{d\le x}\frac{\mu(d)\chi(d)}{d}+O\left(\sum_{d\le x}\frac{1}{d}\frac{\log(x/d)}{x/d}\right)\\ &= -L'(1,\chi)\sum_{d\le x}\frac{\mu(d)\chi(d)}{d}+O(1).\end{aligned}$$

Here the O-term is just an $O(1)$, because we have by Proposition 1.4.5

$$\begin{aligned}\sum_{d\le x}\frac{1}{d}\frac{\log(x/d)}{x/d} &= \frac{1}{x}\sum_{d\le x}(\log(x)-\log(d)) = \frac{1}{x}\left([x]\log(x)-\sum_{d\le x}\log(d)\right)\\ &= O(1).\end{aligned}$$

Together we obtain

$$\begin{aligned}\sum_{p\le x}\frac{\chi(p)\log(p)}{p} &= \sum_{n\le x}\frac{\chi(n)\Lambda(n)}{n}+O(1)\\ &= -L'(1,\chi)\sum_{d\le x}\frac{\mu(d)\chi(d)}{d}+O(1).\end{aligned}$$

Proof of Lemma 2.9.8: We will apply the second Möbius inversion formula with $\alpha(n)=\chi(n)$, $F(x)=x$ and $G(x)=\sum_{n\le x}\chi(n)\frac{x}{n}$. Here we can write $G(x)$, by (2.4), as follows:

$$G(x)=x\sum_{n\le x}\frac{\chi(n)}{n}=xL(1,\chi)+O(1).$$

By Proposition 1.3.22 then we have

$$x = F(x) = \sum_{n \le x} \mu(n)\chi(n) G\left(\frac{x}{n}\right) = \sum_{n \le x} \mu(n)\chi(n)\left(\frac{x}{n} L(1,\chi) + O(1)\right)$$

$$= xL(1,\chi) \sum_{n \le x} \frac{\mu(n)\chi(n)}{n} + O(x).$$

Dividing by x yields the claim.

Proof of Theorem 2.9.9: Let $N(k)$ be the number of characters $\chi \neq \chi_1$ modulo k with $L(1,\chi) = 0$. Then we also have $L(1,\overline{\chi}) = 0$. If χ is real valued, the claim follows from Proposition 2.9.5. So let $\overline{\chi} \neq \chi$. Then we have $N(k) \equiv 0(2)$, because the characters χ with $L(1,\chi) = 0$ come in complex conjugated pairs. The following lemma now yields the proof of the theorem:

Lemma 2.9.10. *For $x > 1$ we have*

$$\sum_{\substack{p \le x \\ p \equiv 1(k)}} \frac{\log(p)}{p} = \frac{1 - N(k)}{\varphi(k)} \log(x) + O(1). \tag{2.36}$$

Assume that $N(k) \neq 0$. Then $N(k) \ge 2$, because of $N(k) \in 2\mathbb{N}$. Hence we have $\frac{1-N(k)}{\varphi(k)} < 0$, so that the right side of (2.36) tends to $-\infty$ for $x \to \infty$. This is impossible, since all terms on the left side are positive. So we conclude that $N(k) = 0$. $\square$

Proof of Lemma 2.9.10: Lemma 2.9.6 with $h = 1$ yields

$$\sum_{\substack{p \le x \\ p \equiv 1(k)}} \frac{\log(p)}{p} = \frac{1}{\varphi(k)} \log(x) + \frac{1}{\varphi(k)} \sum_{r=2}^{\varphi(k)} \sum_{p \le x} \frac{\chi_r(p)\log(p)}{p} + O(1). \tag{2.37}$$

Note that we don't know yet that $L(1,\chi) \neq 0$ in general. If $L(1,\chi_r) \neq 0$, then (2.35) implies that $\sum_{n \le x} \frac{\mu(n)\chi(n)}{n} = O(1)$. Then by (2.33) we have

$$\sum_{p \le x} \frac{\chi_r(p)\log(p)}{p} = -L'(1,\chi_r) \sum_{n \le x} \frac{\mu(n)\chi(n)}{n} + O(1)$$

$$= O(1).$$

However, if $L(1,\chi_r) = 0$, then we have

$$\sum_{p \le x} \frac{\chi_r(p)\log(p)}{p} = -\log(x) + O(1),$$

because of the following lemma:

LEMMA 2.9.11. *For each character $\chi \neq \chi_1$ with $L(1,\chi) = 0$ we have*

$$L'(1,\chi)\sum_{n\le x}\frac{\mu(n)\chi(n)}{n} = \log(x) + O(1).$$

Putting this together, (2.37) reads as follows:

$$\begin{aligned}\sum_{\substack{p\le x\\ p\equiv 1(k)}}\frac{\log(p)}{p} &= \frac{1}{\varphi(k)}\left(-N(k)\log(x) + O(1)\right) + \frac{1}{\varphi(k)}\log(x)\\ &= \frac{1-N(k)}{\varphi(k)}\log(x) + O(1).\end{aligned}$$

□

Proof of Lemma 2.9.11: we apply Proposition 1.3.22 with $\alpha(n) = \chi(n)$, $F(x) = x\log(x)$ and

$$\begin{aligned}G(x) &= \sum_{n\le x}\chi(n)\frac{x}{n}\log\left(\frac{x}{n}\right) = \sum_{n\le x}\frac{\chi(n)}{n}x(\log(x)-\log(n))\\ &= x\log(x)\sum_{n\le x}\frac{\chi(n)}{n} - x\sum_{n\le x}\frac{\chi(n)\log(n)}{n}\\ &= x\log(x)\left(L(1,\chi)+O(x^{-1})\right) + x\left(L'(1,\chi)+O\left(\frac{\log(x)}{x}\right)\right)\\ &= xL'(1,\chi) + O(\log(x)).\end{aligned}$$

Here we have used corollary 2.1.10. By Proposition 1.3.22 we obtain

$$\begin{aligned}x\log(x) &= \sum_{n\le x}\mu(n)\chi(n)\left(\frac{x}{n}L'(1,\chi) + O\left(\log\left(\frac{x}{n}\right)\right)\right)\\ &= xL'(1,\chi)\sum_{n\le x}\frac{\mu(n)\chi(n)}{n} + O\left(\sum_{n\le x}(\log(x)-\log(n))\right)\\ &= xL'(1,\chi)\sum_{n\le x}\frac{\mu(n)\chi(n)}{n} + O(x).\end{aligned}$$

Dividing by x yields the claim, and we finally have proven Dirichlet's Theorem. □

Analogous to $\pi(x)$ we consider the following prime counting function.

Definition 2.9.12. Let $k > 0$ and $(h, k) = 1$. For $x \geq 1$ define

$$\pi_{h,k}(x) = \sum_{\substack{p \leq x \\ p \equiv h(k)}} 1.$$

By Dirichlet's Theorem we have $\pi_{h,k}(x) \to \infty$ for $x \to \infty$. The main term in (2.31), namely $\log(x)/\varphi(k)$, doesn't depend on h. It seems that the primes are uniformly distributed among the $\varphi(k)$ residue classes k. In fact, there is the following prime number theorem for arithmetic progressions:

Theorem 2.9.13 (PNT2). *Let $k > 0$ and $(h, k) = 1$. Then we have*

$$\pi_{h,k}(x) \sim \frac{1}{\varphi(k)} \frac{x}{\log(x)}, \quad x \to \infty.$$

Remark 2.9.14. Theorem 2.9.9, or Abel's summation formula also implies the following result for $(h, k) = 1$, see Remark 1.1.14: There is a constant $C = C(h, k)$ such that, for all $x \geq 2$,

$$\sum_{\substack{p \leq x \\ p \equiv h(k)}} \frac{1}{p} = \frac{1}{\varphi(k)} \log(\log(x)) + C + O\left(\frac{1}{x}\right).$$

Dirichlet's Theorem follows from a more general result in algebraic number theory, namely from the *Density theorem by Chebotarev*.

Theorem 2.9.15 (Chebotarev). *Let K be a number field and L/K be a Galois extension with $G = Gal(L/K)$. For $\sigma \in G$ let C_σ be the conjugacy class of σ. Let S be the set of prime ideals $\mathfrak{p}$ of K, such that for each prime ideal $\mathfrak{P}$ of L lying over $\mathfrak{p}$ the Frobenius element of $\mathfrak{P}$ lies in C_σ. Then S has Dirichlet density $\frac{|C_\sigma|}{|G|}$.*

Dirichlet's Theorem is a special case as follows. Let $K = \mathbb{Q}$ and $L = \mathbb{Q}(\zeta_k)$ with a primitive k-th root of unity ζ_k. Then L is an abelian extension of $\mathbb{Q}$ with $G = (\mathbb{Z}/k\mathbb{Z})^*$, such that $C_\sigma = \{\sigma\}$ for all $\sigma \in G$. The Frobenius element of $\mathfrak{P}$ is the Artin symbol

$$\left(\frac{\mathbb{Q}(\zeta_k)/\mathbb{Q}}{p}\right) = \overline{p} \in (\mathbb{Z}/k\mathbb{Z})^\times,$$

for all $\mathfrak{P}$ over prime numbers p with $p \nmid k$. The conjugacy classes modulo k of primes not dividing k are therefore in a bijective correspondence to the elements of G. Hence the set S is of the form

$$S_h = \{p \in \mathbb{P} \mid p \equiv h(k)\}.$$

Because of $|C_\sigma| = 1$ and $|G| = \varphi(k)$, the set S_h has Dirichlet density $\frac{1}{\varphi(k)}$ for each $h \in (\mathbb{Z}/k\mathbb{Z})^\times$. This is exactly Dirichlet's Theorem.

CHAPTER 3

Dirichlet series, Riemann zeta function and L-series

The analytic properties of Dirichlet series, in particular of the Riemann zeta function, play an important role in number theory. We will start with general properties of Dirichlet series.

3.1. Dirichlet series

DEFINITION 3.1.1. A *Dirichlet series* is a series of the form

$$F(s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s},$$

where f is an arithmetic function and $s = \sigma + it \in \mathbb{C}$.

For $f(n) = 1$ we have $F(s) = \zeta(s)$, and for $f(n) = \chi(n)$ we have $F(s) = L(s, \chi)$. The set of points $s = \sigma + it$ with $\sigma > a$ is called a *half plane* in $\mathbb{C}$. For $\sigma \geq a$ we have $|n^s| = n^\sigma \geq n^a$, and therefore

$$\left| \frac{f(n)}{n^s} \right| \leq \frac{|f(n)|}{n^a}.$$

If a Dirichlet series $\sum_{n=1}^{\infty} \frac{f(n)}{n^s}$ converges absolutely for some $s = a + ib$, then it also converges absolutely for all s with $\sigma \geq a$, by the comparison test.

PROPOSITION 3.1.2. *Suppose that the series $\sum_{n=1}^{\infty} |f(n)n^{-s}|$ does not converge for all s, or does not diverge for all s. Then there exists a real number σ_a, called the abscissa of absolute convergence, such that the series $\sum_{n=1}^{\infty} f(n)n^{-s}$ converges absolutely for $\sigma > \sigma_a$, but does not converge absolutely for $\sigma < \sigma_a$.*

PROOF. Let D be the set of $\sigma \in \mathbb{R}$ such that $\sum_{n=1}^{\infty} |f(n)n^{-s}|$ diverges. By assumption (the first sentence of the statement) D is non-empty, and bounded from above. Hence D has a least upper bound $\sigma_a := \sup(D)$. For $\sigma < \sigma_a$ we have $\sigma \in D$, because otherwise σ would be an upper bound for D smaller than $\sup(D)$, which is impossible. For $\sigma > \sigma_a$ we obtain $\sigma \notin D$, since σ_a is an upper bound for D. $\square$

We define $\sigma_a = -\infty$, if $\sum_{n=1}^{\infty} |f(n)n^{-s}|$ converges everywhere, and $\sigma_a = \infty$, if $\sum_{n=1}^{\infty} |f(n)n^{-s}|$ converges nowhere.

EXAMPLE 3.1.3. *For $\zeta(s)$ and $L(s, \chi)$ we have $\sigma_a = 1$. For $\sum_{n=1}^{\infty} n^n n^{-s}$ we have $\sigma_a = \infty$, and for $\sum_{n=1}^{\infty} n^{-n} n^{-s}$ we have $\sigma_a = -\infty$.*

The series $\zeta(s)$ converges absolutely for $\sigma > 1$ and diverges at $s = 1$. Hence we have $\sigma_a = 1$. If f is bounded, i.e., if $|f(n)| \le C$, then $\sum_{n=1}^{\infty} f(n)n^{-s}$ converges absolutely for $\sigma > 1$. This holds for $L(s,\chi)$, since χ is bounded. The third series converges nowhere, the fourth everywhere in $\mathbb{C}$.

Proposition 3.1.4. *Let $F(s) = \sum_{n=1}^{\infty} f(n)n^{-s}$ for $\sigma > \sigma_a$. Then*

$$\lim_{\sigma\to\infty} F(\sigma + it) = f(1)$$

uniformly for $-\infty < t < \infty$.

Proof. Writing $F(s) = f(1) + \sum_{n=2}^{\infty} f(n)n^{-s}$, we only need to show that the series converges to zero for $\sigma \to \infty$. Choose a $c > \sigma_a$. For $\sigma \ge c$ we have

$$\begin{aligned}\left|\sum_{n=2}^{\infty} f(n)n^{-s}\right| &\le \sum_{n=2}^{\infty}|f(n)|n^{-\sigma} = \sum_{n=2}^{\infty}|f(n)|n^{-c}n^{-(\sigma-c)} \\ &\le 2^{-(\sigma-c)}\sum_{n=2}^{\infty}|f(n)|n^{-c} = 2^{-\sigma}A,\end{aligned}$$

where A is independent of σ and t. The claim follows by $\lim_{\sigma\to\infty} 2^{-\sigma}A = 0$. $\square$

Corollary 3.1.5. *We have $\lim_{\sigma\to\infty}\zeta(\sigma+it) = \lim_{\sigma\to\infty} L(\sigma+it,\chi) = 1$.*

We now prove that all the coefficients are uniquely determined by the sum function. For a proof, see Theorem 11.3 in [**1**].

Theorem 3.1.6 (Uniqueness). *Let $F(s) = \sum_{n=1}^{\infty} f(n)n^{-s}$ and $G(s) = \sum_{n=1}^{\infty} g(n)n^{-s}$ be two Dirichlet series, which converge absolutely for $\sigma > \sigma_a$. Let (s_k) be an infinite sequence of complex numbers with $\lim_{k\to\infty}\sigma_k = \infty$. If $F(s) = G(s)$ for all $s = s_k$, then $f(n) = g(n)$ for all $n \in \mathbb{N}$.*

The uniqueness implies the following result.

Proposition 3.1.7. *Let $F(s) = \sum_{n=1}^{\infty} f(n)n^{-s}$ be a Dirichlet series with $F(s) \ne 0$ for some s with $\sigma > \sigma_a$. Then there exists a half plane $\sigma > c \ge \sigma_a$, in which $F(s)$ never vanishes.*

Proof. Assume no such half plane exists. Then there exists a point s_k for every $k \in \mathbb{N}$ with $\sigma_k = \mathrm{Re}(s_k) > k$ with $F(s_k) = 0$. Because of $\lim_{k\to\infty}\sigma_k = \infty$ it follows that $f(n) = 0$ by the uniqueness result. Indeed, then $F(s) = 0$ everywhere, which is a contradiction. $\square$

Proposition 3.1.8. *Let $F(s)$ and $G(s)$ be two functions, given by the Dirichlet series*

$$\begin{aligned}F(s) &= \sum_{n=1}^{\infty} f(n)n^{-s}, \quad \sigma > a, \\ G(s) &= \sum_{n=1}^{\infty} g(n)n^{-s}, \quad \sigma > b.\end{aligned}$$

Then in the half plane, where both converge absolutely, we have

$$F(s)G(s) = \sum_{n=1}^{\infty} (f * g)(n) n^{-s},$$

where $(f * g)(n) = \sum_{d|n} f(d)g(n/d)$ *is the Dirichlet convolution of* f *and* g.

Proof. We have

$$\begin{aligned} F(s)G(s) &= \sum_{n=1}^{\infty} f(n)n^{-s} \sum_{m=1}^{\infty} g(m)m^{-s} = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} f(n)g(m)(nm)^{-s} \\ &= \sum_{k=1}^{\infty}\left(\sum_{mn=k} f(n)g(m)\right) k^{-s} = \sum_{k=1}^{\infty} (f * g)(n) k^{-s}. \end{aligned}$$

Here we may choose s in such a way that both series converge absolutely. Then we may rearrange the series. One collects the terms for which $mn = k$ is constant. This yields the claim. □

Example 3.1.9. *We have* $\zeta(s) \neq 0$ *for* $\sigma > 1$, *and*

$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}.$$

This follows from Proposition 3.1.8 with $F(s) = \zeta(s)$ and $G(s) = \sum_{n=1}^{\infty} \mu(n)n^{-s}$. These series converge absolutely for $\sigma > 1$. We obtain $(f * g)(n) = (\varepsilon * \mu)(n) = I(n)$, so that

$$\zeta(s) \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} = 1.$$

Example 3.1.10. *Let* χ *be a Dirichlet character modulo* k. *Then we have, for* $\sigma > 1$

$$\sum_{n=1}^{\infty} \frac{\mu(n)\chi(n)}{n^s} = \frac{1}{L(s,\chi)}.$$

If $F(s) = \sum_{n=1}^{\infty} f(n)n^{-s}$ converges absolutely for $\sigma > \sigma_a$, and if f is strongly multiplicative, then we have $f^{-1}(n) = \mu(n)f(n)$. Since $|f^{-1}(n)| \le |f(n)|$, the series $\sum_{n=1}^{\infty} \mu(n)f(n)n^{-s}$ converges absolutely for $\sigma > \sigma_a$, and we have

$$\sum_{n=1}^{\infty} \frac{\mu(n)f(n)}{n^s} = \frac{1}{F(s)}.$$

This yields the above claim for $f(n) = \chi(n)$.

Example 3.1.11. *For $\sigma > 2$ we have*

$$\sum_{n=1}^{\infty} \frac{\varphi(n)}{n^s} = \frac{\zeta(s-1)}{\zeta(s)}.$$

To see this, let $f(n) = 1$, $g(n) = \varphi(n)$ and $(f * g)(n) = \sum_{d|n} \varphi(d) = n$. Since $\varphi(n) \leq n$, the series $\sum_{n=1}^{\infty} \varphi(n) n^{-s}$ converges absolutely for $\sigma > 2$. By Proposition 3.1.8 it follows that

$$\zeta(s) \sum_{n=1}^{\infty} \frac{\varphi(n)}{n^s} = \sum_{n=1}^{\infty} \frac{n}{n^s} = \zeta(s-1).$$

More such identities can be shown in the same way, see also Remark 1.2.14:

$$\begin{aligned}
\sum_{n=1}^{\infty} \mu^2(n) n^{-s} &= \frac{\zeta(s)}{\zeta(2s)}, \quad \sigma > 1 \\
\sum_{n=1}^{\infty} 2^{\omega(n)} n^{-s} &= \frac{\zeta^2(s)}{\zeta(2s)}, \quad \sigma > 1 \\
\sum_{n=1}^{\infty} \tau(n^2) n^{-s} &= \frac{\zeta^3(s)}{\zeta(2s)}, \quad \sigma > 1 \\
\sum_{n=1}^{\infty} \tau^2(n) n^{-s} &= \frac{\zeta^4(s)}{\zeta(2s)}, \quad \sigma > 1.
\end{aligned}$$

3.2. Euler products

The Euler product gives a relation between Dirichlet series and the prime numbers.

PROPOSITION 3.2.1. *Let f be a multiplicative arithmetic function, such that the series $\sum_{n=1}^{\infty} f(n)$ converges absolutely. Then we have*

$$\sum_{n=1}^{\infty} f(n) = \prod_{p\in\mathbb{P}} \left(\sum_{k=0}^{\infty} f(p^k) \right),$$

where the infinite product converges absolutely. If f is strongly multiplicative, then

$$\sum_{n=1}^{\infty} f(n) = \prod_{p\in\mathbb{P}} \frac{1}{1-f(p)}.$$

PROOF. Consider the finite product

$$P(x) = \prod_{p\le x} (1 + f(p) + f(p^2) + \cdots).$$

This is a product of finitely many absolutely convergent series, So we can multiply the series and rearrange the terms in any fashion without altering the sum. A typical term of $P(x)$ is of the form

$$f(p_1^{\alpha_1}) \cdots f(p_r^{\alpha_r}) = f(p_1^{\alpha_1} \cdots p_r^{\alpha_r}).$$

By the fundamental theorem of arithmetic we have $P(x) = \sum_{n\in A} f(n)$, where A is the set of $n \in \mathbb{N}$, where all prime divisors $p \mid n$ satisfy $p \le x$. If B is the set of those n having at least a prime factor $p > x$, we have

$$\left| \sum_{n=1}^{\infty} f(n) - P(x) \right| \le \sum_{n\in B} |f(n)| \le \sum_{n>x} |f(n)|.$$

Since the series $\sum_{n=1}^{\infty} |f(n)|$ is convergent, the right side tends to zero for $x \to \infty$. Hence we have

$$\lim_{x\to\infty} \left(P(x) - \sum_{n=1}^{\infty} f(n) \right) = 0.$$

If f is strongly multiplicative, we have $f(p^n) = f(p)^n$ and hence $\sum_{k=0}^{\infty} (f(p)^k = (1-f(p))^{-1}$. Finally we show that the product converges absolutely. Indeed, a product $\prod_{n=1}^{\infty}(1+a_n)$ converges absolutely, if the series $\sum_{n=1}^{\infty} a_n$ converges absolutely. Since the series $\sum_p |f(p) + f(p^2) + \cdots|$ converges absolutely, because of

$$\sum_{p\le x} |f(p) + f(p^2) + \cdots| \le \sum_{p\le x} (|f(p)| + |f(p^2)| + \cdots) \le \sum_{n=2}^{\infty} |f(n)|,$$

we are done. □

For $g(n) = f(n)n^{-s}$ this yields

Proposition 3.2.2. *Let f be a multiplicative arithmetic function, such that the Dirichlet series $\sum_{n=1}^{\infty} f(n)n^{-s}$ converges absolutely for $\sigma > \sigma_a$. Then we have*

$$\sum_{n=1}^{\infty} f(n)n^{-s} = \prod_{p\in\mathbb{P}} \left(\sum_{k=0}^{\infty} f(p^k)p^{-ks} \right), \quad \sigma > \sigma_a.$$

If f is strongly multiplicative, we have

$$\sum_{n=1}^{\infty} f(n)n^{-s} = \prod_{p\in\mathbb{P}} \frac{1}{1 - f(p)p^{-s}}, \quad \sigma > \sigma_a.$$

For the arithmetic functions discussed this yields the following Euler products:

$$\sum_{n=1}^{\infty} n^{-s} = \zeta(s) = \prod_p \frac{1}{1-p^{-s}}, \quad \sigma > 1,$$

$$\sum_{n=1}^{\infty} \mu(n) n^{-s} = \frac{1}{\zeta(s)} = \prod_p (1-p^{-s}), \quad \sigma > 1,$$

$$\sum_{n=1}^{\infty} \varphi(n) n^{-s} = \frac{\zeta(s-1)}{\zeta(s)} = \prod_p \frac{1-p^{-s}}{1-p^{1-s}}, \quad \sigma > 2,$$

$$\sum_{n=1}^{\infty} \lambda(n) n^{-s} = \frac{\zeta(2s)}{\zeta(s)} = \prod_p \frac{1}{1+p^{-s}}, \quad \sigma > 1,$$

$$\sum_{n=1}^{\infty} \sigma_\alpha(n) n^{-s} = \zeta(s)\zeta(s-\alpha) = \prod_p \frac{1}{(1-p^{-s})(1-p^{\alpha-s})}, \quad \sigma > b,$$

$$\sum_{n=1}^{\infty} \tau(n) n^{-s} = \zeta(s)^2 = \prod_p (1-p^{-s})^{-2}, \quad \sigma > 1,$$

$$\sum_{n=1}^{\infty} \chi(n) n^{-s} = L(s,\chi) = \prod_p \frac{1}{1-\chi(p)p^{-s}}, \quad \sigma > 1,$$

$$\sum_{n=1}^{\infty} \mu^2(n) n^{-s} = \frac{\zeta(s)}{\zeta(2s)} = \prod_p (1+p^{-s}), \quad \sigma > 1,$$

where $b = \max\{1, 1+\mathrm{Re}(\alpha)\}$. Given the Euler products of $F(s)$, one can derive the representation by a zeta function. We did this without Euler products in 3.1.8.

3.3. Analytic continuation of the zeta function

The Riemann ζ-function is defined only for $\sigma > 1$. It is a holomorphic function in this half plane. We have seen in Proposition 1.4.4, that there is an analytic continuation to the half plane $\sigma > 0$. In fact, one uses Euler's summation formula to show for $\sigma > 1$ that

$$\zeta(s) = 1 - \frac{1}{1-s} - s\int_1^\infty \frac{t-[t]}{t^{s+1}}\,dt.$$

The right side converges uniformly and absolutely for $\mathrm{Re}(s) \geq \sigma > 0$. By the Weierstrass Theorem, this integral is a holomorphic function of s in the half plane $\sigma > 0$. Hence it gives an explicit analytic continuation of $\zeta(s)$ on the half plane $\sigma > 0$.

This idea can be generalized in such a way, that one obtains an analytic continuation to the whole complex plane. On the other hand, there is a different proof for this, which we want to present now.

THEOREM 3.3.1. *The function $\zeta(s)$ has an analytic continuation to a meromorphic function on the complex plane $\mathbb{C}$, with just one singularity, which is a simple pole at $s = 1$ with residue 1.*

PROOF. The Gamma function has the integral representation

$$\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}\,dt, \quad \sigma > 0.$$

It has an analytic continuation to a meromorphic function on $\mathbb{C}$, with simple poles at $s = 0, -1, -2, -3, \ldots$, and residue $\frac{(-1)^n}{n!}$ at $s = -n$. The Gamma function satisfies the functional equation

$$\Gamma(s+1) = s\Gamma(s)$$
$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}.$$

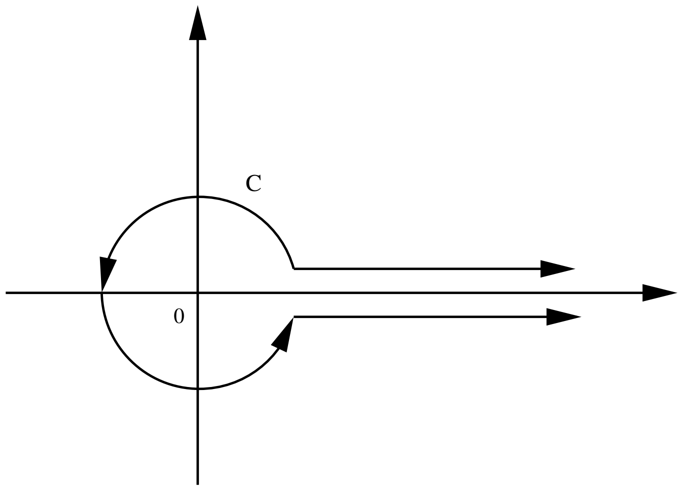
We start with the formula

$$\Gamma(s)n^{-s} = \int_0^\infty t^{s-1}e^{-nt}\,dt, \quad \sigma > 0.$$

Summing over $n \geq 1$, we obtain for $\sigma > 1$,

$$\Gamma(s)\zeta(s) = \sum_{n=1}^\infty \int_0^\infty t^{s-1}e^{-nt}\,dt = \int_0^\infty \frac{t^{s-1}}{e^t-1}dt.$$

Now we obtain an analytic continuation of this integral as follows. We replace the half line $[0,\infty]$ of the integration by the *Hankel path* $C = C_\rho$ in $\mathbb{C}$, where $0 < \rho < 2\pi$ is a real parameter:



The path C_ρ consists of three parts: the set of points with argument $0+$ on the real half line $[\rho,\infty]$, the circle $|z| = \rho$, and the set of points with argument $2\pi-$ on the half line $[\rho,\infty]$ in reversed orientation. The first and third part are the top and bottom edges of a slot along the positive half line. The path has distance ε from the real axis, and one has to imagine the limit of paths for $\varepsilon \to 0$. Since the function $z \mapsto z^{s-1}(e^z - 1)^{-1}$ is holomorphic in the horizontal strip $|\mathrm{Im}(z)| < 2\pi$ without the half line $[0,\infty]$, the integral

$$I(s) := \int_{C_\rho} z^{s-1}(e^z - 1)^{-1} dz$$

is independent of ρ in $(0, 2\pi)$. It is absolutely convergent for all $s \in \mathbb{C}$ and uniformly convergent on every compact set. Hence $I(s)$ defines an entire function. We have

$$I(s) = \int_{|z|=\rho} z^{s-1}(e^z - 1)^{-1} dz + (e^{2\pi i s} - 1) \int_\rho^\infty t^{s-1}(e^t - 1)^{-1} dt. \tag{3.1}$$

Now we let ρ tend to zero. For $|z| = \rho \le \pi$ we have $|z^{s-1}(e^z - 1)^{-1}| = O(\rho^{\sigma-2})$. This yields the formula

$$I(s) = (e^{2\pi i s} - 1)\Gamma(s)\zeta(s), \quad \sigma > 1.$$

Together with $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$ is follows that

$$\zeta(s) = \frac{1}{2\pi i} e^{-i\pi s}\Gamma(1-s)I(s). \tag{3.2}$$

This formula, originally defined only for $\sigma > 1$, now yields an explicit analytic continuation of $\zeta(s)$ on $\mathbb{C} \setminus 1$. For $\sigma \le 0$, the factor $\Gamma(1-s)$ is holomorphic. Hence $\zeta(s)$ has no further singularity except for the one at $s = 1$. □

Definition 3.3.2. Let $0 < a \le 1$. For $\sigma > 1$ define the *Hurwitz Zeta Function* $\zeta(s,a)$ by the Dirichlet series

$$\zeta(s,a) = \sum_{n=0}^{\infty} \frac{1}{(n+a)^s}.$$

We have $\zeta(s,1) = \zeta(s)$, and $L(s,\chi)$ can be expressed by $\zeta(s,a)$. One can also show, like in the previous theorem, that $\zeta(s,a)$ has an analytic continuation to $\mathbb{C} \setminus 1$, with a simple pole at $s = 1$ with residue 1. This implies that also $L(s,\chi)$, for $\chi \neq \chi_1$, has an analytic continuation on $\mathbb{C}$. For the principal character, $L(s,\chi_1)$ has an analytic continuation to $\mathbb{C} \setminus 1$, with a simple pole at $s = 1$ with residue $\varphi(k)/k$.

Let us state a consequence of (3.1). Let B_n be the n-th Bernoulli number, defined by the following Laurent series

$$\frac{1}{e^z - 1} = \sum_{m=0}^{\infty} \frac{1}{m!} B_m z^{m-1}.$$

It is easy to see that $B_{2m+1} = 0$ for $m \ge 1$ and $B_1 = -1/2$. Here are a few values:

m	2	4	6	8	10	12	14	16	18	20
B_m	$\frac{1}{6}$	$-\frac{1}{30}$	$\frac{1}{42}$	$-\frac{1}{30}$	$\frac{5}{66}$	$-\frac{691}{2730}$	$\frac{7}{6}$	$-\frac{3617}{510}$	$\frac{43867}{798}$	$-\frac{174611}{330}$

Proposition 3.3.3. *For all $n \ge 0$ we have*

$$\zeta(-n) = (-1)^n \frac{B_{n+1}}{n+1} = -\frac{B_{n+1}}{n+1}.$$

In particular we have $\zeta(-2n) = 0$ for all $n \ge 1$.

Proof. With (3.1) we have

$$I(-n) = \int_{|z|=\rho} z^{-n-1}(e^z - 1)^{-1} dz = 2\pi i \frac{B_{n+1}}{(n+1)!}.$$

Substituting this into (3.2) with $s = -n$ yields the claim, because we have $\Gamma(n+1) = n!$ and $\cos(\pi n) = (-1)^n$. Note that $B_{n+1} = 0$ for even n, and $(-1)^n = -1$ for odd n. So above we can replace $(-1)^n$ by -1. □

3.4. The functional equation for $\zeta(s)$ and $L(s,\chi)$

The duplication formula for the Gamma function is given as follows:

$$\Gamma(s)\Gamma\left(s+\frac{1}{2}\right) = 2^{1-2s}\sqrt{\pi}\Gamma(2s).$$

We may rewrite the functional equation for $\zeta(s)$ by introducing the notation $\Phi(s) := \pi^{-s/2}\Gamma(s/2)\zeta(s)$. Then the functional equation for $\zeta(s)$ is just equivalent to the functional equation $\Phi(s) = \Phi(1-s)$.

THEOREM 3.4.1. *For every $s \neq 1$ we have*

$$\zeta(s) = 2^s\pi^{s-1}\sin\left(\frac{1}{2}\pi s\right)\Gamma(1-s)\zeta(1-s),$$

or equivalently, $\Phi(s) = \Phi(1-s)$ with $s \neq 0, 1$.

PROOF. For $k \geq 1$ let H_k be the Hankel path with parameter $\rho_k = (2k+1)\pi$. Then $|e^z - 1|^{-1}$ is bounded for z on H_k, and we have

$$\left|\int_{H_k} z^{s-1}(e^z-1)^{-1}dz\right| = O(k^\sigma). \tag{3.3}$$

(3.4)

Let ρ be given with $0 < \rho < 2\pi$. The path $C_\rho - H_k$ goes around the poles $z = 2n\pi i$ for $n = \pm1, \pm2, \ldots, \pm k$ anticlockwise. The the residue theorem shows that

$$\begin{aligned} I(s) &= \int_{C_\rho} z^{s-1}(e^z-1)^{-1}dz \\ &= \int_{H_k} z^{s-1}(e^z-1)^{-1}dz - 2\pi i \sum_{1\leq|n|\leq k}(2n\pi i)^{s-1} \\ &= \int_{H_k} z^{s-1}(e^z-1)^{-1}dz - (2\pi i)^s(1-e^{i\pi s})\sum_{1\leq n\leq k} n^{s-1}. \end{aligned}$$

With (3.3) we obtain, for $k \to \infty$ and for every s in the half plane $\sigma < 0$,

$$I(s) = (2\pi i)^s(e^{i\pi s}-1)\zeta(1-s).$$

Putting this into (3.2) yields the functional equation for $\sigma < 0$. By analytic continuation we obtain it then for all s. $\square$

COROLLARY 3.4.2. *For all $n \geq 1$ we have*

$$\zeta(2n) = (-1)^{n-1} 2^{2n-1} \frac{B_{2n}}{(2n)!} \pi^{2n}.$$

PROOF. By Proposition 3.3.3 we have

$$\zeta(1-2n) = -\frac{B_{2n}}{2n}.$$

The functional equation for $s = 1 - 2n$ is equivalent to

$$\zeta(2n) = 2^{2n-1} \pi^{2n} (\sin(\pi/2 - \pi n))^{-1} \zeta(1-2n).$$

Substituting $\zeta(1-2n)$ in it, and using $\sin(\pi/2 - \pi n) = \cos(-\pi n) = (-1)^n$, one obtains the claim. $\square$

EXAMPLE 3.4.3. *For $n = 1, \ldots, 7$ we obtain*

$$\zeta(2) = \frac{\pi^2}{6}, \quad \zeta(4) = \frac{\pi^4}{90}, \quad \zeta(6) = \frac{\pi^6}{945}, \quad \zeta(8) = \frac{\pi^8}{9450}$$

$$\zeta(10) = \frac{\pi^{10}}{93555}, \quad \zeta(12) = \frac{691\pi^{12}}{638512875}, \quad \zeta(14) = \frac{2\pi^{14}}{18243225}.$$

The fractions grow exponentially. For example, $\zeta(50)$ is given by

$$\zeta(50) = \frac{39604576419286371856998202\pi^{50}}{285258771457546764463363635252374414183254365234375}.$$

It is still an open problem, to decide whether or not the values $\zeta(2n+1)$ are irrational. Apéry showed in 1978, that $\zeta(3)$ is irrational. He found the following formula for $\zeta(3)$:

$$\zeta(3) = \frac{5}{2} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3 \binom{2n}{n}}.$$

The conjecture is that all numbers

$$\zeta(3), \zeta(5), \zeta(7), \zeta(9), \ldots$$

are algebraically independent. W. Zudilin proved that at least one of the numbers $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational.

For real a and $\sigma > 1$ let

$$F(a,s) = \sum_{n=1}^{\infty} \frac{e^{2\pi i n a}}{n^s}.$$

This function is a periodic function of a with period 1 and $F(1,s) = \zeta(s)$. The series converges absolutely for $\sigma > 1$. For the proof of the following result, see [**1**]:

Proposition 3.4.4 (Hurwitz formula). *For $0 < a \leq 1$ and $\sigma > 1$ we have*

$$\zeta(1-s,a) = \frac{\Gamma(s)}{(2\pi)^s}\left(e^{-\pi i s/2}F(a,s) + e^{\pi i s/2}F(-a,s)\right).$$

For $a = 1$ we obtain again the functional equation of $\zeta(s)$, in the form

$$\zeta(1-s) = 2(2\pi)^{-s}\Gamma(s)\cos\left(\frac{\pi s}{2}\right)\zeta(s).$$

Similarly, Hurwitz formula also gives the functional equation for the Dirichlet L-function.

Proposition 3.4.5. *Let χ be a primitive Dirichlet character modulo k. Then for all s we have*

$$L(1-s,\chi) = \frac{k^{s-1}\Gamma(s)}{(2\pi)^s}\left(e^{-\pi i s/2} + \chi(-1)e^{\pi i s/2}\right)G(1,\chi)L(s,\overline{\chi}).$$

3.5. The zeros of the zeta function

Since the Euler product

$$\zeta(s) = \prod_p \frac{1}{1-p^{-s}}, \quad \sigma > 1$$

converges, $\zeta(s)$ does not vanish for $\sigma > 1$. We will show that $\zeta(s)$ also does not vanish on the line $\sigma = 1$. This is often used in the proof of the prime number theorem.

PROPOSITION 3.5.1. *Let $F(s) = \sum_{n=1}^\infty a_n n^{-s}$ be a Dirichlet series with non-negative coefficients, and with abscissa of absolute convergence σ_c. Then we have, for all $\sigma > \sigma_c$,*

$$3F(\sigma) + 4\mathrm{Re}F(\sigma + it) + \mathrm{Re}F(\sigma + 2it) \geq 0.$$

PROOF. For $\theta \in \mathbb{R}$ let $V(\theta) = 3 + 4\cos(\theta) + \cos(2\theta)$. Then we have

$$V(\theta) = 3 + 4\cos(\theta) + 2\cos^2(\theta) - 1 = 2(1 + \cos(\theta))^2.$$

This implies $V(\theta) \geq 0$. The left side of the claimed inequality is given by

$$\sum_{n=1}^\infty a_n n^{-\sigma} V(t \log(n)).$$

This gives the claim. □

COROLLARY 3.5.2. *For $\sigma > 1$ we have*

$$\zeta(\sigma)^3 |\zeta(\sigma + it)|^4 |\zeta(\sigma + 2it)| \geq 1.$$

PROOF. This follows by the above result applied to the function

$$F(s) = \log(\zeta(s)) = -\sum_p \log(1 - p^{-s}) = \sum_{n \geq 2} \frac{\Lambda(n)}{\log(n)} n^{-s}.$$

It converges for $\sigma > 1$. Taking exp yields the claim. □

Now we are able to prove the following result.

THEOREM 3.5.3. *The function $\zeta(s)$ has no zero in the half plane $\sigma \geq 1$.*

PROOF. We only need to prove the case $\sigma = 1$. Assume that $\zeta(1+it) = 0$. Then $t \neq 0$. By Corollary 3.5.2 we have for all $\sigma > 1$,

$$((\sigma-1)\zeta(\sigma))^3 \left|\frac{\zeta(\sigma+it)}{\sigma-1}\right|^4 |\zeta(\sigma+2it)| \geq \frac{1}{\sigma-1}.$$

Consider the limit $\sigma \to 1+$. The first factor tends to 1, because $\zeta(s)$ has residue 1 at the pole $s = 1$. The third factor tens to $|\zeta(1+2it)|$. Then the second one satisfies

$$\left|\frac{\zeta(\sigma+it)}{\sigma-1}\right|^4 = \left|\frac{\zeta(\sigma+it)-\zeta(1+it)}{\sigma-1}\right|^4 \to |\zeta'(1+it)|^4$$

for $\sigma \to 1+$. So the left side tends to $|\zeta'(1+it)|^4|\zeta(1+2it)|$, but the right side tends to ∞, for $\sigma \to 1+$. This is a contradiction. $\square$

The functional equation now implies the following corollary.

COROLLARY 3.5.4. *The only zeros of $\zeta(s)$ in the half plane $\sigma \leq 0$ are the trivial zeros at $s = -2n$, which are simple poles.*

It is easy to show that $\zeta(s)$ has, except for the trivial zeros, no other zeros on the real line.

LEMMA 3.5.5. *For $\sigma > 0$ we have*

$$(1-2^{1-s})\zeta(s) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s}.$$

This implies $\zeta(\sigma) < 0$ for $0 < \sigma < 1$. In particular, $\zeta(\sigma) = 0$ is impossible in this strip, and $\zeta(\sigma) > 0$ for $\sigma > 1$.

PROOF. For $\sigma > 1$ we have

$$(1-2^{1-s})\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} - 2\sum_{n=1}^{\infty} \frac{1}{(2n)^s} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s}.$$

The series on the right side converges also for $\sigma > 0$, so that the identity is valid for $\sigma > 0$, by analytic continuation. For real s, the alternating series has a positive sum. For $0 < \sigma < 1$, the factor $1 - 2^{1-\sigma}$ is negative, so that $\zeta(\sigma)$ then is negative, too. $\square$

The so-called *critical strip* is the set of $s \in \mathbb{C}$ with $0 < \sigma < 1$. There $\zeta(s)$ has infinitely many non-real zeros. In fact, let $N(T)$ be the number of zeros of $\zeta(s)$ in the region $0 < \sigma < 1$, $0 < t \leq T$. Then the *Riemann-von Mangoldt formula* says that

$$N(T) = \frac{T}{2\pi} \log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + O(\log(T)), \quad T \to \infty.$$

Hence $\zeta(s)$ has infinitely many zeros in the critical strip. They lie symmetrically with respect to the real axis and the *critical line* $\sigma = \frac{1}{2}$. The **Riemann hypothesis** says that all of these non-trivial zeros lie on the critical line. The *Generalized Riemann Hypothesis* predicts the same for $L(s, \chi)$, and for all automorphic L-functions.

3.6. The prime number theorem

Finally we want to give a proof of the *prime number theorem*, which says that

$$\pi(x) \sim \frac{x}{\log(x)} \sim \mathrm{li}(x).$$

This was first proven in 1896 independently by Hadamard and de la Vallee-Poussin, first by proving a result on the zeroes of the Riemann zeta function $\zeta(s)$, namely that none occur on the line $\Re(s) = 1$, and then by deducing the prime number theorem from this result, by a fairly complicated analysis. Over the decades this analysis was whittled down more and more until it reached a fairly definitive form, in the form of a succinct "Tauberian theorem" due to D. J. Newman in 1980. A streamlined account was written up in the American Mathematical Monthly by Don Zagier in 1997, see [**17**], who managed to produce a proof virtually from scratch in a little less than three pages (it requires just a bit of background in complex analysis, not much more than Cauchy's theorem).

Recall that

$$\theta(x) = \sum_{p \le x} \log(p).$$

The prime number theorem is equivalent to the asymptotic statement $\theta(x) \sim x$ by Theorem 1.6.8. Actually, let us give another short proof for only one direction of it.

Lemma 3.6.1. *The asymptotic statement $\theta(x) \sim x$ implies the prime number theorem.*

Proof. We will estimate $\frac{\pi(x)\log(x)}{x}$ from both sides. In one direction, we have an obvious inequality

$$\theta(x) = \sum_{p \le x} \log(p) \le \sum_{p \le x} \log(x) = \pi(x)\log(x).$$

Assuming that $\theta(x) \sim x$, this implies, given $\epsilon > 0$, the inequality

$$1 - \epsilon < \frac{\pi(x)\log(x)}{x}$$

holds for all sufficiently large x. In the other direction, for any $\epsilon > 0$ we have

$$\begin{aligned}
\theta(x) &\geq \sum_{x^{1-\epsilon}<p\leq x} \log(p) \\
&\geq \sum_{x^{1-\epsilon}<p\leq x} \log(x^{1-\epsilon}) \\
&= (1-\epsilon)\log(x)\cdot(\pi(x)-\pi(x^{1-\epsilon})) \\
&\geq (1-\epsilon)\log(x)\cdot(\pi(x)-x^{1-\epsilon}).
\end{aligned}$$

From the last line and the inequality above with $1-\varepsilon$ (which implies $\frac{\pi(x)\log(x)}{x}$ is bounded below by a positive constant), we obtain

$$\begin{aligned}
\frac{\theta(x)}{x} &\geq (1-\epsilon)\frac{\pi(x)\log(x)}{x} - x^{-\epsilon} \\
&\geq (1-2\epsilon)\frac{\pi(x)\log(x)}{x}
\end{aligned}$$

for sufficiently large x. Thus for any small $\epsilon > 0$ (say less than $1/3$) and the assumption $\theta(x) \sim x$ we have

$$1-\epsilon \leq \frac{\pi(x)\log(x)}{x} \leq (1-2\epsilon)^{-1}\frac{\theta(x)}{x} \leq (1-3\epsilon)^{-1}$$

or simply

$$1-\epsilon \leq \frac{\pi(x)\log(x)}{x} \leq (1-3\epsilon)^{-1}$$

for all sufficiently large x. □

LEMMA 3.6.2. *We have $\theta(x) = O(x)$.*

PROOF. This follows from the proof of Proposition 1.6.4, where we have shown that $\theta(x) \leq 4\log(2)x$ for sufficiently large x. □

The asymptotic result $\theta(x) \sim x$ will be established by using the following result:

THEOREM 3.6.3. *Suppose that the integral*

$$\int_1^\infty \frac{\theta(x)-x}{x^2}\,dx$$

converges. Then $\theta(x) \sim x$.

PROOF. Suppose that for some $\lambda > 1$ there exist arbitrarily large x such that $\lambda x \leq \theta(x)$. The monotonicity of θ implies the first inequality that follows:

$$\int_x^{\lambda x} \frac{\theta(t)-t}{t}\,\frac{dt}{t} \geq \int_x^{\lambda x} \frac{\lambda x - t}{t}\,\frac{dt}{t} = \int_1^\lambda \frac{\lambda x - tx}{tx}\,\frac{dt}{t} = \int_1^\lambda \frac{\lambda - t}{t^2}\,dt > 0.$$

This is a contradiction, because the convergence of $\int_1^\infty \frac{\theta(t)-t}{t^2}\,dt$ implies that $\int_x^{\lambda x} \frac{\theta(t)-t}{t^2}\,dt$ can be made arbitrarily small by taking x large. Similarly, if we suppose that for some $\lambda < 1$ there exist arbitrarily large x such that $\theta(x) \leq \lambda x$, then

$$\int_{\lambda x}^{x} \frac{\theta(t)-t}{t}\,\frac{dt}{t} \leq \int_{\lambda x}^{x} \frac{\lambda x - t}{t}\,\frac{dt}{t} = \int_\lambda^1 \frac{\lambda x - tx}{tx}\,\frac{dt}{t} = \int_\lambda^1 \frac{\lambda - t}{t^2}\,dt < 0$$

which again leads to a contradiction. □

The function $\theta(x)$ is related to the function $\Phi(s)$ by way of a Riemann-Stieltjes integral:

$$\Phi(s) = \sum_p \frac{\log(p)}{p^s} = \int_1^\infty \frac{d\theta(x)}{x^s}$$

for $\Re(s) > 1$. An integration by parts yields

$$\int_1^M \frac{d\theta(x)}{x^s} = \left[\frac{\theta(M)}{M^s} - \theta(1)\right] + \int_1^M \theta(x)\frac{s}{x^{s+1}}\, dx$$

where $\lim_{M\to\infty} \frac{\theta(M)}{M^s} = 0$ by Lemma 3.6.2. Thus

$$\Phi(s) = s\int_1^\infty \frac{\theta(x)}{x^{s+1}}\, dx = s\int_0^\infty e^{-st}\theta(e^t)\, dt. \tag{3.5}$$

We now study the convergence of

$$\int_1^\infty \frac{\theta(x) - x}{x}\frac{dx}{x} = \int_0^\infty \frac{\theta(e^t) - e^t}{e^t}\, dt \tag{3.6}$$

Recall that $\Phi(s) - \frac{1}{s-1}$ is holomorphic in the region $\Re(s) > 1$, and $\zeta(s)$ has no zeros in the half plane $\Re(s) \geq 1$ by Theorem 3.5.3. Thus, for $\Re(s) > 1$ we have from (3.5),

$$\begin{aligned} \frac{\Phi(s)}{s} - \frac{1}{s-1} &= \int_0^\infty e^{-st}\theta(e^t) - e^{(1-s)t}\, dt \\ &= \int_0^\infty e^{-st}(\theta(e^t) - e^t)\, dt \end{aligned}$$

or what is the same, for the region $\Re(s) > 0$,

$$\frac{\Phi(s+1)}{s+1} - \frac{1}{s} = \int_0^\infty e^{-(s+1)t}(\theta(e^t) - e^t)\, dt = \int_0^\infty e^{-st}\left(\frac{\theta(e^t) - e^t}{e^t}\right)\, dt \tag{3.7}$$

where the left side is in fact holomorphic over $\Re(s) \geq 0$, hence takes a finite value at $s = 0$, say L. By a Tauberian theorem (see below) we will be able to conclude that

$$L = \lim_{s\searrow 0}\int_0^\infty e^{-st}\left(\frac{\theta(e^t) - e^t}{e^t}\right)\, dt = \int_0^\infty \frac{\theta(e^t) - e^t}{e^t}\, dt$$

which in turn will yield the Prime Number Theorem in the following form:

Theorem 3.6.4. *The integral* $\int_0^\infty \frac{\theta(e^t) - e^t}{e^t}\, dt$ *converges.*

In fact, this implies the prime number theorem together with (3.6), Theorem 3.6.3 and Lemma 3.6.1. Here is the statement of the particular Tauberian theorem used.

Theorem 3.6.5. *Let $f(t)$ be a function that is bounded and locally integrable over the domain $t \geq 0$, and suppose that its Laplace transform, defined for $\Re(s) > 0$ by the equation*

$$g(s) = \int_0^\infty e^{-st} f(t)\, dt,$$

extends to a function g that is holomorphic over the region $\Re(s) \geq 0$. Then the integral $\int_0^\infty f(t)\, dt$ converges and equals $g(0)$.

Taking $f(t) = \frac{\theta(e^t)-e^t}{e^t}$, we see that f is bounded and locally integrable, and we have already observed that its Laplace transform, which is

$$g(s) = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}$$

by (3.7), is holomorphic in the region $\Re(s) \geq 0$. Thus all the hypotheses of the theorem are satisfied in this case, and its conclusion in this case is Theorem 3.6. So all that remains is to prove Theorem 3.6.5.

Proof. For $T \geq 0$, put

$$g_T(s) = \int_0^T e^{-st} f(t)\, dt.$$

Clearly g_T is an entire function. We are trying to show that $\lim_{T\to\infty} g_T(0) = g(0)$. A straightforward application of Cauchy's theorem yields

$$g_T(0) - g(0) = \frac{1}{2\pi i}\int_C (g_T(z) - g(z))\, e^{zT}\, \frac{dz}{z}$$

for contours C having 0 as an interior point, such that $g_T - g$ is holomorphic on C and its interior. We will choose C to be the boundary of the closed region $D = \{z : |z| \leq R \text{ and } Re(z) \geq -\delta\}$ where for a given R, we choose δ small enough that $g_T - g$ is holomorphic in D (this uses compactness of the interval $I = \{it : -R \leq t \leq R\}$ and holomorphicity of $g_T - g$ along I). The idea would be to estimate the contribution of the integral over C_+, the part of C where the real part is nonnegative, and over C_- where it is non-positive, hoping to show these contributions are small when R is large. This almost works, except that we don't have good control over the size of the contributions near the area where $Re(z) = 0$. To offset this, a mollifying factor $(1 + \frac{z^2}{R^2})$ is introduced in the integrand ("Carleman's formula"): this is close to zero near $\Re(z) = 0$ and has the desired mitigating effect. Thus we instead use

$$g_T(0) - g(0) = \frac{1}{2\pi i}\int_C (g_T(z) - g(z))\, e^{zT} \left(1 + \frac{z^2}{R^2}\right) \frac{dz}{z}$$

Suppose now that $|f(t)|$ is globally bounded by B. For $\Re(z) > 0$ we have

$$|g_T(z) - g(z)| \leq \int_T^\infty |f(t)e^{-zt}|\, dt \leq B\int_T^\infty e^{-Re(z)t}\, dt = B\frac{e^{-Re(z)T}}{Re(z)}$$

and we also have, over the circle $|z| = R$,

$$\begin{aligned} |e^{zT}(1 + z^2/R^2)(1/z)| &= e^{Re(z)T} \cdot |(R^2 z^{-1} + z)/R^2| \\ &= e^{Re(z)T} \cdot |(|z|^2 z^{-1} + z)/R^2| \\ &= e^{Re(z)T} \cdot |(\bar{z} + z)/R^2| \\ &= e^{Re(z)T} \cdot |2Re(z)|/R^2 \end{aligned}$$

so that putting the factors together, the integral over C_+ is bounded in absolute value by

$$2B/R^2 \int_{C_+} |dz| = 2\pi B/R$$

(and of course this hasn't taken into account the factor $\frac{1}{2\pi i}$ outside the integral). In any case, with the Carleman factor, the contribution from the C_+ part is B/R, which is small if R is large. Now we bound the contribution from C_- by considering g_T and g separately. First let us bound

$$\frac{1}{2\pi i}\int_{C_-} g_T(z)e^{zT}\left(1+\frac{z^2}{R^2}\right)\frac{dz}{z}.$$

For this, we can take advantage of the fact that g_T is entire, and (courtesy of Cauchy's theorem) replace C_- by the semicircle where $|z| = R, \Re(z) < 0$. The estimates are similar in form to those above, starting with

$$\begin{aligned}|g_T(z)| &\leq \int_0^T |f(t)|\cdot|e^{-zt}|\,dt\\ &\leq B\int_0^T e^{-Re(z)t}\,dt \leq B\int_{-\infty}^T e^{-Re(z)t}\,dt\\ &= B\frac{e^{-Re(z)T}}{-Re(z)}.\end{aligned}$$

Bear in mind here that we are now considering $\Re(z) < 0$ and $-\Re(z) = |\Re(z)|$; the bound produced tends to be very large. But, large as it may be, it is offset by the bound on the product of the remaining factors in the integrand, which is $e^{\Re(z)T}\cdot|2\Re(z)|/R^2$. We conclude as before that over C_- the overall contribution from g_T has upper bound B/R.

Finally, we bound

$$\frac{1}{2\pi i}\int_{C_-} g(z)e^{zT}\left(1+\frac{z^2}{R^2}\right)\frac{dz}{z}.$$

Holding R fixed, the expression $g(z)\cdot\left(1+\frac{z^2}{R^2}\right)\cdot\frac{1}{z}$ taken over C_- is bounded in absolute value by a number K. Whereas along $\Re(z) = -\delta$, by taking T very large we can make $|e^{zT}|$ as small as we like, so that considering g separately, the contribution over C_- can be discarded as vanishingly small (in other words, first pick R, then pick δ, and finally pick T, to get this contribution to be as small as we like). This completes the proof. $\square$

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